



Periodic Financing Request Report

Project Number: 43253-026
MFF Number: 0085
September 2019

India: Karnataka Integrated and Sustainable Water Resources Management Investment Program (Tranche 2)

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CURRENCY EQUIVALENTS

(as of 1 August 2019)

Currency unit	–	Indian rupee (₹)
₹1.00	=	\$0.014
\$1.00	=	₹69.19

ABBREVIATIONS

ACIWRM	–	Advanced Center for Integrated Water Resources Management
ADB	–	Asian Development Bank
EIA	–	environmental impact assessment
GAP	–	gender action plan
ha	–	hectare
IWRM	–	integrated water resources management
KNNL	–	Karnataka Neeravari Nigam Limited
m ³	–	cubic meter
O&M	–	operation and maintenance
PIO	–	project implementation office
PSC	–	project support consultant
SPS	–	Safeguard Policy Statement, 2009
TMC	–	thousand million cubic feet
VNC	–	Vijayanagara Channels
WRD	–	Water Resources Department
WUCS	–	water user cooperative societies

NOTES

- (i) The fiscal year (FY) of the Government of India and its agencies ends on 31 March. “FY” before a calendar year denotes the year in which the fiscal year ends, e.g., FY2019 ends on 31 March 2019.
- (ii) In this report, “\$” refers to United States dollars.

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TRANCHE AT A GLANCE

1. Basic Data		Project Number: 43253-026	
Project Name	Karnataka Integrated and Sustainable Water Resources Management Investment Program - Tranche 2	Department/Division	SARD/SAER
Country Borrower	India Government of India	Executing Agency	Advanced Centre for Integrated Water Resources Management, Karnataka Neeravari Nigam Limited
Country Economic Indicators	https://www.adb.org/Documents/LinkedDocs/?id=43253-026-CEI		
Portfolio at a Glance	https://www.adb.org/Documents/LinkedDocs/?id=43253-026-PortAtaGlance		
2. Sector		ADB Financing (\$ million)	
✔ Agriculture, natural resources and rural development	Subsector(s)		
	Irrigation		54.60
	Rural water policy, institutional and capacity development		18.20
	Water-based natural resources management		18.20
		Total	91.00
3. Operational Priorities		Climate Change Information	
✔ Addressing remaining poverty and reducing inequalities		Climate Change impact on the Project	Medium
✔ Accelerating progress in gender equality			
✔ Tackling climate change, building climate and disaster resilience, and enhancing environmental sustainability		ADB Financing	
✔ Promoting rural development and food security		Adaptation (\$ million)	13.80
✔ Strengthening governance and institutional capacity			
Sustainable Development Goals		Gender Equity and Mainstreaming	
SDG 1.1, 1.4, 1.5		Effective gender mainstreaming (EGM)	✔
SDG 2.4			
SDG 5.a			
SDG 6.4, 6.5, 6.b		Poverty Targeting	
SDG 10.1		General Intervention on Poverty	✔
SDG 12.2			
SDG 13.a			
4. Risk Categorization: Complex			
5. Safeguard Categorization Environment: A Involuntary Resettlement: C Indigenous Peoples: C			
6. Financing			
Modality and Sources		Amount (\$ million)	
ADB		91.00	
Sovereign MFF-Tranche (Regular Loan): Ordinary capital resources		91.00	
Cofinancing		0.00	
None		0.00	
Counterpart		39.00	
Government		39.00	
Total		130.00	
Currency of ADB Financing: US Dollar			

I. BACKGROUND

1. **Multitranche financing facility.** The investment program will improve water availability to meet competing water demands in selected river basins in the State of Karnataka by implementing integrated water resources management (IWRM). It will improve water use efficiency in irrigated agriculture to provide economic opportunities to improve rural incomes.

2. The state, in the south of India, has a population of 61.1 million and covers an area of 191,976 square kilometers (5.83% of the area of India). The contribution of agriculture to its gross domestic product is relatively small, at about 16% in fiscal year 2012, having steadily declined from about 30% in fiscal year 2001. However, agriculture remains the main source of employment for 55% of the population and about 75% of the rural population. Karnataka's urban population has increased from 34% in 2001 to 39% in 2011. The changing industrial structure and urbanization have had a significant impact on the water balance in the state. Karnataka has become water-stressed because of increasing demand for water from urban and industrial sectors. This is exacerbated by the uneven spatial and temporal distribution of water resources and the impacts of climate change. Although Karnataka has two major river basins (the Krishna and Cauvery), its water resources are limited (about 1,072 cubic meters [m³] per person per year in eastward flowing rivers, which is slightly above the threshold of 1,000 m³ per person per year for water scarcity). Droughts are frequent and 70% of the annual rainfall occurs during the June to September monsoon. The state's water demand (particularly for industry and household water) is projected to increase by 40% from 37,419 million m³ in 2000 to 52,366 million m³ in 2025. These demands will cause a decline in the agricultural proportion of total water from 84% in 2000 to 73% by 2025. Meeting the anticipated rise in demand is a major challenge, which, if unmet, may constrain sustainable economic growth.

3. About 3.1 million hectares (ha) of land are under irrigation in Karnataka, and more than 84% of the state's water resources are used for agriculture. The state's irrigation performance is functioning below potential. Current cropping intensities are only 110%–125%, which is below the national average of 136% and well below their potential of 180%–200%. Also, irrigation efficiency is about 40%, below the national target of 65%. The irrigation subsector suffers from deficient infrastructure (resulting from insufficient maintenance and lack of water control structures), management constraints (low service quality of operators), and limited promotion of water-saving crops and practices. Efficient distribution is impeded by inadequate implementation of command area development works and low participation of water user cooperative societies (WUCS) in irrigation management. Hence, Karnataka has the potential to improve its irrigation water productivity by (i) increasing water use efficiency, (ii) reducing the gap between the actual and potential area that can be irrigated, (iii) promoting water-efficient techniques and technologies such as drip irrigation, and (iv) improving agronomic practices. Greater efficiency will be required in water used for agriculture to meet the demands of other users, such as industry.

4. The projected impacts of climate change on the project area include increases in temperature and mean annual rainfall, and less but more variable monsoon rainfall. Overall, the investment program area is found to be vulnerable to more seasonal droughts. Thus, IWRM is a means to reconcile varied and changing water uses and demands. It provides greater flexibility and adaptive capacity in managing water deliveries to various users than do conventional, inflexible water resources management approaches.

5. An IWRM road map and investment plan was developed when preparing the investment program in consultation with senior management of Karnataka's Water Resources Department (WRD) and combines the Karnataka State Water Policy 2002; the state medium-term

development plan; the National Water Policy, 2012; and the Twelfth Five Year Plan, 2012–2017 to guide the strengthening of institutional and policy frameworks, capacity building, and the modernization of infrastructure. The road map, through physical and nonphysical investments, is supporting the state to meet its objective of improved water resources management with milestone performance targets.

6. The Government of India requested the Asian Development Bank (ADB) to contribute to the state's investment program using the multitranche financing facility because it is an effective modality to (i) provide a longer-term lending instrument suited to the IWRM process; (ii) afford flexibility in terms of project scope, timing, and size, which may be adjusted to meet emerging priorities of the road map; and (iii) provide sufficient time for the institutional strengthening in IWRM of key agencies and water users. The Karnataka Integrated and Sustainable Water Resources Management Investment Program was approved for \$150 million on 25 September 2014.¹ The initial availability period of the facility was until 30 September 2021 and has subsequently been extended until 24 September 2024 to allow sufficient time for completion of the remaining activities planned under the investment program. The first tranche of \$31 million was approved on 17 October 2014—\$5 million was partially cancelled on 6 May 2019 at the request of the government—and is due for closing on 31 December 2019. The investment program has three outputs:

- (i) **Output 1: State and basin institutions for integrated water resources management strengthened.** The investment program will strengthen the institutional capacities of the WRD and the Advanced Center for Integrated Water Resources Management (ACIWRM) to implement IWRM in selected river basins in Karnataka. This will include (a) the development and implementation of river basin plans to manage water resources better; (b) capacity building and certification of the WRD—and its associated agencies, such as Karnataka Neeravari Nigam Limited (KNNL)—in IWRM, with about 30% of trainees being women; and (c) the establishment of a state water resources information system. The activities to be undertaken are detailed in the IWRM road map.²
- (ii) **Output 2: Irrigation system infrastructure and management modernized.** This entails (a) modernizing irrigation infrastructure in three irrigation subprojects within the K-8 sub-basin: Gondi (4,600 ha), Vijayanagara Channels (VNC, 11,150 ha), and a portion of the Tungabhadra Left Bank Canal (144,000 ha); (b) strengthening asset management and main system operation and maintenance (O&M); and (c) building the capacity of more than 82 WUCS, with at least 30% women's representation on management boards for decision making.
- (iii) **Output 3: Program management systems operational.** This output will deliver the above program outputs on time and within budget. The main activities include (a) establishing a fully staffed program management unit and project implementation offices (PIOs) with an appointed program director; (b) procuring works and equipment packages; (c) recruiting and monitoring a project support consultant (PSC); (d) operationalizing a program management information system database; (e) meeting annual disbursement projections; and (f) conducting periodic program performance reviews.

¹ ADB. 2014. [*Report and Recommendation of the President to the Board of Directors: Proposed Multitranche Financing Facility to India for the Karnataka Integrated and Sustainable Water Resources Management Investment Program*](#). Manila.

² The IWRM road map is included in Updated Facility Administration Manual in Appendix 4.

7. **Tranches.** The investment program was originally envisaged as comprising two tranches.³ The first tranche served to prepare drafts of the State IWRM Strategy and Updated Water Resources Policy, and the K-8 Basin Plan, and to modernize the Gondi subproject (footnote 1). The second tranche was originally designed to prepare basin plans for the Upper and Middle Krishna Basins and modernize the VNC subproject as well as a portion of the Tungabhadra Left Bank Canal subproject. Now, tranche 2 will focus on preparing the Middle Krishna Basin Plan (sub-basins K-2 to K-4) and modernizing the VNC subproject. The remaining activities under the investment program will be considered for financing under a third tranche, tentatively scheduled for approval in 2020. The decision to limit the scope of the second tranche is based on its readiness criteria: the designs and bid documents for the VNC subproject have been prepared and contracting is proceeding as an advance action, while designs for the Tungabhadra Left Bank Canal subproject are still under preparation.

8. The investment program's executing agency for outputs 2 and 3 is KNNL, which has established separate PIOs for each of the modernization projects under output 2. ACIWRM is the executing agency for output 1.⁴

9. On 8 July 2019, ADB received the government's periodic financing request for tranche 2, totaling \$91 million, to continue the implementation of the investment program (as detailed in paras. 15–17). Tranche 2 is included in the ADB country operations business plan, 2018–2020 for India,⁵ and is aligned with several operational priorities of ADB's Strategy 2030: (i) tackling climate change, building climate and disaster resilience, and enhancing environmental sustainability; (ii) promoting rural development and food security; and (iii) strengthening governance and institutional capacity.

II. ASSESSMENT OF MULTITRANCHE FINANCING FACILITY IMPLEMENTATION

10. **Progress on road map and improvements in policy framework.** The road map covers six key IWRM themes: (i) integrated river basin management; (ii) institutions and policies for IWRM; (iii) data, information, and knowledge management; (iv) IWRM-based irrigation management; (v) improving water use efficiency and water productivity; and (vi) stakeholder participation. As part of the activities under output 1 of the tranche 1 project (Project 1, para. 7), progress is being made against all six themes and will continue throughout the investment program. In particular, the project team drafted the State IWRM Strategy and Updated Water Resources Policy. The policy document is now under consideration by the state cabinet.

11. **Progress of investment program.** As of 31 December 2018, the overall performance of the investment program is *on track*. Its undertakings compliance and tranche performance are both rated *on track*, but the timeliness is rated *at risk* because, as per the financing framework agreement, the last financing tranche was expected to be executed no later than 30 August 2015. Approval of tranche 2 project (Project 2) is in the 2019 pipeline (footnote 5). Advance procurement actions were undertaken, and the contract for one of the two Project 2 works packages was awarded in March 2019.

12. **Progress of Project 1.** Project 1 comprises one main works package to modernize the Gondi subproject; 228 community participation procurement packages to be implemented by the

³ The MFF is geographically sliced, and separate project completion reports will be completed for each tranche.

⁴ For tranche 1, KNNL is the executing agency and implementing agency for outputs 2 and 3, while ACIWRM is the implementing agency for output 1.

⁵ ADB. 2017. *Country Operations Business Plan: India, 2018–2020*. Manila. The government's submission of the periodic financing request for Project 2 was delayed until 2019.

11 WUCS in the Gondi irrigated area; one major goods package to install 100 telemetry flow-monitoring stations throughout the K-8 basin; one main consulting package for the PSC; and several individual consultant packages to support ACIWRM in its output 1 activities. The Gondi works and community participation procurement packages are to be completed by 31 December 2019. As for the telemetry stations, 82 are operational and 18 will be completed under Project 2. The project has trained 233 engineers under output 1 and will train another 150 by September 2019. As of 20 June 2019, cumulative contract awards were \$25.2 million, or 97% of the loan amount; and cumulative disbursements were \$16.8 million, or 65% of the loan amount. The overall physical progress is about 80%.

13. **Compliance with undertakings and loan covenants.** For tranche 1, all particular loan covenants are being complied with, including all safeguards-related covenants. The government is also in compliance with all facility-level undertakings set out in the framework facility agreement that are currently required to be complied with, including the requirement to implement the sector road map and other sector-related commitments.

III. PERIODIC FINANCING REQUEST

A. Impact and Outcome

14. Project 2 is aligned with the following impacts: (i) sustainable water security in selected river basins in Karnataka improved, and (ii) water use efficiency in irrigated agriculture improved.⁶ Project 2 will have the following outcome: IWRM successfully implemented in selected river basins in Karnataka.⁷

B. Outputs

15. **Output 1: State and basin institutions strengthened for integrated water resources management.** Project 2 will (i) prepare the middle Krishna Basin Plan (for sub-basins K2, K3, and K4); (ii) expand the state water resources information system; and (iii) continue strengthening WRD staff through IWRM training, with at least 30% of participants to be women.

16. **Output 2: Vijayanagara Channels irrigation system infrastructure and management modernized.** Project 2 will modernize the VNC subproject to (i) reduce VNC irrigation water consumption from 12.5 thousand million cubic feet (TMC) to 5.80 TMC, a saving of 6.7 TMC that will be reallocated to new irrigation developments within the river basin; and (ii) improve the water delivery service to allow farmers to increase cropping intensities. Modernization works include repairing 12 anicuts and intake structures; and about 442 kilometers of main, distributary, and minor canals. The project will establish and strengthen 30 WUCS in the project area, which will then be responsible for improving their farmer irrigation canals. Each WUCS will be governed by a board comprising at least 30% women. Project 2 will also provide KNNL staff and WUCS with gender sensitization workshops; farmers with extension services to improve water management and crop production; and will operationalize an asset management plan.

17. **Output 3: Program management systems operational.** This output ensures that investment program outputs are delivered on time and within budget. Project 2 will involve (i) establishing a fully staffed PIO at KNNL's Munirabad's office with an appointed project implementation officer; (ii) procuring works and equipment financed under Project 2;

⁶ This impact was formulated following the previous design and monitoring framework guidelines. However, it remains aligned with: NITI Aayog. 2017. *India Three Year Action Agenda 2017–2018 to 2019–2020*. New Delhi.

⁷ The design and monitoring framework is in Appendix 1.

(iii) coordinating with ACIWRM and managing the PSC; (iv) coordinating with other agencies responsible for command area development and on-farm improvement activities; (v) expanding the program management information system database that was established under Project 1 and including gender indicators; (vi) meeting annual contract award and disbursement projections; (vii) conducting periodic safeguard and performance monitoring reviews; (viii) preparing the technical, economic, and safeguard due diligence for the proposed subsequent project, and preparing the associated draft loan approval documents;⁸ and (ix) undertaking advanced procurement actions for the proposed subsequent project.

C. Summary Cost Estimates and Financing Plan

18. Project 2 is estimated to cost \$130 million (Table 1).

Table 1: Summary Cost Estimates
(\$ million)

Item	Amount ^a
A. Base Cost^b	
1. State and basin institutions strengthened for IWRM	8.5
2. VNC irrigation system infrastructure and management modernized	85.3
3. Program management systems operational	6.6
Subtotal (A)	100.4
B. Contingencies^c	23.0
C. Financial Charges During Implementation^d	6.6
Total (A+B+C)	130.0

IWRM = integrated water resources management, VNC = Vijayanagara Channels.

^a Includes taxes and duties of \$11.67 million. Such amount does not represent an excessive share of the project cost. The government will finance taxes and duties of \$11.67 million as cash contribution.

^b In 2019 prices as of 1 May 2019.

^c Physical contingencies computed at 10% for civil works; and 5%–10% for field research and development, training, surveys, and studies. Price contingencies computed at average of 1.5% on foreign exchange costs and 4.6% on local currency costs; includes provision for potential exchange rate fluctuation under the assumption of a purchasing power parity exchange rate.

^d Includes interest and commitment charges. Interest during construction for the ordinary capital resources loan has been computed at the 5-year United States dollar fixed swap rate plus an effective contractual spread of 0.5% and maturity premium of 0.0%. Commitment charges for the ordinary capital resources loan are 0.15% per year to be charged on the undisbursed loan amount.

Source: Asian Development Bank estimates.

19. The government has requested a regular loan of \$91 million from ADB's ordinary capital resources to help finance the project. The loan will have a 20-year term, including a grace period of 5 years; an annual interest rate determined in accordance with ADB's London interbank offered rate (LIBOR)-based lending facility; a commitment charge of 0.15% per year; and such other terms and conditions set forth in the draft loan and project agreements. Based on the straight-line method, the average maturity is 12.75 years, and the maturity premium payable to ADB is 0.00% per year.

20. The summary financing plan is in Table 2. ADB will finance the expenditures in relation to civil works, consulting services, trainings and workshops, equipment, and office operation.

⁸ The Sustainable Development and Climate Change Department may provide training and technical expertise on the assessment of water productivity using remote sensing and water accounting in support of preparing Project 3.

Table 2: Summary Financing Plan

Source	Amount (\$ million)	Share of Total (%)
Asian Development Bank		
Ordinary capital resources (regular loan)	91.0	70.0
Government	39.0	30.0
Total	130.0	100.0

Source: Asian Development Bank.

21. Climate adaptation is estimated to cost about \$18.4 million. ADB will finance 75% of this cost. Details are in the climate change assessment.⁹

D. Implementation Arrangements

22. The implementation arrangements are summarized in Table 3 and described in detail in the updated facility administration manual.¹⁰

Table 3: Implementation Arrangements for Project 2

Table 3: Implementation Arrangements for Project 2			
Aspects	Arrangements		
Implementation period	October 2019 to September 2023		
Estimated completion date	30 September 2023		
Estimated loan closing date	31 March 2024		
Management			
(i) Oversight body	State IWRM Steering Committee Chair: Chief Secretary of the State Members: Principal Secretary, Water Resources Department; Principal Secretary, Urban Development Department; Principal Secretary, Rural Development Department; Principal Secretary, Agriculture Department; Principal Secretary, Commerce and Industries Department; Principal Secretary, Forest, Ecology and Environment Department; Secretary, Minor Irrigation; Managing Director, KUIDFC; Managing Director, Karnataka Urban Water Supply and Drainage Board; and Chairman, Bangalore Water Supply and Sanitation Board.		
(ii) Executing agencies	KNNL, ACIWRM		
(iii) Key implementing agencies	ACIWRM will be responsible for implementing output 1. KNNL will be responsible for implementing outputs 2 and 3.		
(iv) Implementation unit	ACIWRM, Bangalore: 35 staff KNNL PMU, Bangalore: 42 staff KNNL PIO, Munirabad: 42 staff		
Procurement	OCB	3 contracts	\$84 million
	CPP	575 contracts	\$20 million
	RFQ	Multiple	\$2 million
Consulting services	ICS	167 person-months	\$3 million
	QCBS (PSC) ^a	16 person-months international and 1,103 person months national	\$6.4 million
Retroactive financing and/or advance contracting	Advance contracting will be used for civil works, consulting services, and procurement of goods. Retroactive financing will be considered for eligible expenditures not exceeding 20% of the loan amount incurred prior to loan effectiveness, but not earlier than 12 months prior to the signing of the loan agreement.		

⁹ Climate Change Assessment is in Appendix 18.

¹⁰ Updated Facility Administration Manual is in Appendix 4.

Aspects	Arrangements
Disbursement	The loan proceeds will be disbursed following ADB's <i>Loan Disbursement Handbook</i> (2017, as amended from time to time) and detailed arrangements agreed between the government and ADB.

ACIWRM = Advanced Center for Integrated Water Resources Management, ADB = Asian Development Bank, CPP = community participation in procurement, ICS = individual consultant selection, IWRM = integrated water resources management, KNNL = Karnataka Neeravari Nigam Limited, KUIDFC = Karnataka Urban Infrastructure Development and Finance Corporation, OCB = open competitive bidding (international and national advertising), PIO = project implementation office, PMU = program management unit, PSC = Program Support Consultant, QCBS = quality- and cost-based selection, RFQ = request for quotation.

^a The Program Support Consultant (Tranche 1) will initially continue to support the PMU and PIO in implementing Project 2 through a contract amendment while the Program Support Consultant (Tranche 2) is recruited.

Source: Asian Development Bank.

E. Project Readiness

23. Project readiness is high. The PSC was recruited under Project 1 and will continue to assist the government in implementing Project 2 until the Project 2 PSC has been recruited and mobilized. All designs and bid documents for Project 2 works packages have been completed. The first works package under Project 2, which is about 40% of the total project amount, was awarded on 8 March 2019. The second works package will be advertised in late-2019 and awarded in 2020. Environmental and social safeguards were assessed following ADB's Safeguard Policy Statement, 2009 (SPS). An environmental impact assessment (EIA) for Project 2 was disclosed on ADB's website on 4 July 2018. A social safeguards due diligence report was also prepared to confirm that no land acquisition is expected under Project 2.

IV. DUE DILIGENCE

A. Technical

24. The VNC subproject was initially constructed between 1336 and 1646, during the Vijayanagara Empire, to irrigate about 11,150 ha of arable land along both sides of the Tungrabhadra River near Hospet. The project will finance the modernization of the VNC's main irrigation infrastructure, construction of farmer irrigation canals, and on-farm extension services to assist farmers in improving their agricultural and water application practices. The designs take account of international best practices, the levels of familiarity with associated technologies, and local conditions, including architectural considerations within the proximity of heritage areas. The designs for the weirs, intake structures, and canals are based on crop water demands and scheduling, and take into account climate change risks and vulnerabilities. Climate change may lead to (i) temperature increases, resulting in more demand for crop water and less crop productivity; (ii) a change in the frequency and intensity of precipitation events, which may lead to more runoff, causing greater flood flows, and erosion and siltation of water courses; and (iii) a decrease in precipitation, which will result in higher soil moisture deficits and lead to an increase in irrigation demand and a reduction in crop yields. The design of the structures factored in the projected increases in evapotranspiration and floods, and the project includes on-farm extension support to help farmers manage changes in water availability and crop productivity.

B. Economic and Financial

25. The major project benefit is increased agricultural production from improved water supply, which leads to improvements in both cropping intensity and crop yields. A Monte Carlo simulation model was used to estimate the economic benefits of the project, deriving an economic internal

rate of return of 10.8% and an economic net present value of \$1.32 million. Based on ADB's requirement of a minimum 9% economic internal rate of return, the project is economically viable.

26. The government committed to finance the O&M and periodic replacement costs of the project infrastructure from its own resources. The project's incremental increase in O&M costs is less than 1% of KNNL's current annual O&M expenses and therefore does not present a burden on available resources. However, the financial sustainability risk is considered *substantial*.

C. Governance

27. **Financial management.** An updated financial management assessment of KNNL and ACIWRM was carried out in accordance with ADB guidelines.¹¹ The overall risk assessment for the project is *moderate*, and the assessment concludes that the implementing agencies have appropriate financial management capacity to implement the project.¹²

28. **Procurement.** The government requested that the procurement of goods and works follow ADB's Procurement Policy (2017, as amended from time to time). An updated procurement capacity assessment concluded that the overall procurement risk for the project is *moderate*.¹³

29. **Anticorruption.** ADB's Anticorruption Policy (1998, as amended to date) was explained to and discussed with the government, KNNL and ACIWRM. The specific policy requirements and supplementary measures are described in the updated facility administration manual.

D. Poverty, Social, and Gender

30. The project is a general poverty intervention aimed at increasing agricultural productivity in rural Karnataka. The project is designed to ensure that benefits are targeted and equally distributed to marginalized and poor people living in the project area. Direct project benefits include (i) capacity building and participation of women, vulnerable groups, and tail-end farmers in water governance and decision-making processes; (ii) capacity building in innovative and sustainable irrigation and agriculture practices, including drip and micro irrigation and crop diversification for cultivators and agricultural workers; (iii) improved availability of and access to water for small, marginal, and landless farmers; (iv) enhanced off-farm livelihood opportunities; and (v) reinforcement of agricultural infrastructure, research, and extension that considers the needs of marginalized groups working in agriculture.

31. Agriculture is the main livelihood activity of men and women in the project area. Nonetheless, women have limited access to productive resources (land, credit, inputs, transport, extension services, storage, and technical assistance), which prevents them from adopting new technologies or achieving economies of scale. Women farmers perceive that their opportunities in agriculture are decreasing because of the mechanization of the sector. The challenge is further exacerbated by women's triple burden and second-class status.¹⁴ Women in the project area are frequently excluded from agricultural governance bodies and continue to be paid less than their male counterparts for the same activities. Women's daily wages in the project area vary from ₹150 to ₹200, while men receive between ₹250 and ₹350 for the same agricultural work. The gender action plan (GAP) for Project 2 will build women's technical skills and leadership capacity to

¹¹ ADB. 2005. *Guidelines for the Financial Management and Analysis of Projects*. Manila.

¹² Updated Financial Management Assessment is in Appendix 15.

¹³ Updated Procurement Capacity Assessment is in Appendix 14.

¹⁴ Food and Agriculture Organization of the United Nations. 2015. *Running out of time: The reduction of women's work burden in agricultural production*. Rome.

increase their engagement in the agriculture sector, primarily through the WUCS. While the GAP activities for Project 1 are ongoing, monitoring data highlights that the project is on track to meet gender quotas for training and representation in key decision-making bodies. Furthermore, project interventions have raised women's income and livelihood standards through improved agricultural opportunities such as crop diversification. Project 2 will expand the scope of these activities.

32. The project is categorized *effective gender mainstreaming*. The GAP was designed to ensure that measurable targets and quotas for women's participation and representation in the project are included in the project design and monitoring framework. Women focus group discussions were held during project preparation to ascertain women's priorities and interests. Priority needs evolved around animal husbandry, skilled employment in farm and non-farm activities, and rural infrastructure strengthening. The GAP was designed to integrate women's needs into the project, and support women's capacity building, participation, and representation.

33. The increase in agricultural production as a result of the project is expected to create employment opportunities especially for the landless, women, and the poor, with access to skilled farm and off-farm jobs in subproject areas. The project will not incur additional costs to populations living in the project area. Improved market links and mechanisms for informed pricing will be established to prevent any exploitation of target populations and to increase the participation of socially excluded and marginal people, such as women, in project activities.

E. Safeguards

34. In compliance with ADB's SPS, the project's safeguard categories are as follows.¹⁵

35. **Environment (category A).** The project is classified category A for environmental impact in accordance with the SPS. The EIA and an environmental management plan were disclosed on ADB's website on 4 July 2018 and an update was disclosed on receipt.¹⁶ The environment assessment and review framework was updated to describe the grievance redress mechanism that had been established.¹⁷ A tranche report was also prepared.

36. The category A classification is triggered for two reasons. First, works will be undertaken within the Tungabhadra Otter Conservation Reserve, which is considered a critical habitat. The works will rehabilitate the anicuts, intake structures, and canals. The EIA includes a rapid biodiversity assessment of the reserve area to define the extent of possible impacts. Implementation and operation of the project will ensure that (i) no measurable adverse impacts on the critical habitat emerge that could impair its ability to function, (ii) no reduction occurs in the population of any recognized endangered or critically endangered species, and (iii) any lesser impacts are mitigated. Clearance from the State Board for Wildlife and the National Board for Wildlife as stated under the Wildlife (Protection) Act of 1972 will be obtained for any works to be undertaken within the reserve area.

37. Secondly, ADB's environmental safeguards policy principle no. 11 is triggered by the rehabilitation of several canals within the Hampi World Heritage Area. As specified in the SPS, these physical cultural resources must be protected from any construction damages. Moreover, the presence of heritage areas in the project area triggers additional management demand from the Hampi World Heritage Area Management Authority to ensure high standards of protection

¹⁵ ADB. Safeguard Categories. <https://www.adb.org/site/safeguards/safeguard-categories>.

¹⁶ Environmental Impact Assessment: VNC is in Appendix 9.

¹⁷ Environmental Assessment and Review Framework is in Appendix 8.

and compliance with the requirements of the world heritage processes. The authority has granted clearance to implement the works within the heritage area, along with conditions for the maintenance of the heritage architectures.

38. Most impacts occur during the anicut and canal rehabilitation works. The EIA revealed no adverse environmental impacts in the project locations that will be detrimental for the protected wildlife and physical cultural resources, while some impacts will likely occur during construction of campsites, material sourcing, and activities at the construction sites. Works will include dredging of accumulated sediments, which may be polluted from adjacent urban areas. KNNL have committed to undertake silt testing at potentially contaminated sites prior to commencing works. Management and planning measures for environmental impact mitigation in the Tungabhadra Otter Conservation Reserve, Hampi World Heritage Area, and the rest of the project sites are included in the EMP, including safe transport and disposal of toxic materials to reduce harm to the environment and people's health and safety.

39. The project's environmental categorization was also assessed against the Government of India's EIA Notification of 2006. This determines that modernization of irrigation projects does not require prior environmental clearance if it does not expand the irrigated area or propose to change processes and technologies. Thus, the scope of works under Project 2 does not trigger the provisions of India's EIA notification.

40. **Involuntary resettlement (category C).** The project will avoid involuntary resettlement impacts on both titled and non-title holders residing in the project area. A full due diligence of the project was undertaken in May 2018 to confirm that project works will not cause any permanent or temporary involuntary physical and/or economic displacement to people, assets, and/or their means of livelihood.¹⁸ Involuntary resettlement impacts will be avoided because the project will occur on public land, within the WRD's existing right of way. Furthermore, civil works will occur from within the canal (during the dry period) and thereby avoid any use of the embankments, which are occupied by non-titleholders in some sections. Access for machinery and contractor camps will be negotiated between the contractors and affected parties on a voluntary basis, as per the guidelines set out in the contractor bidding documents.

41. KNNL will closely monitor the implementation of the project and promptly inform ADB if it were to cause unanticipated permanent or temporary involuntary resettlement impacts. Unanticipated impacts that trigger the SPS will require KNNL to prepare and submit a resettlement plan based on the project's updated resettlement framework.¹⁹ The resettlement plan must be approved and disclosed by ADB before civil works start in the subproject area. Contractor bidding documents require ongoing monitoring, documenting, and reporting on all social impacts that would trigger the SPS. KNNL will report on safeguard compliance in all project progress reports.

42. **Indigenous peoples (category C).** The SPS indigenous peoples safeguard is not triggered even though project screening confirms the presence of different ethnic groups in the project footprint.²⁰ The project implementation focuses exclusively on pre-existing governance and water resource infrastructure. The project will not affect (i) the territories or natural or cultural resources of indigenous peoples; or (ii) the dignity, human rights, livelihood systems, or culture of indigenous peoples.

¹⁸ Social Safeguards Due Diligence Report is in Appendix 11.

¹⁹ Updated Resettlement Framework is in Appendix 10.

²⁰ Poverty and Social Assessment is in Appendix 16.

F. Summary of Risk Assessment and Risk Management Plan

43. Significant risks and mitigating measures are summarized in Table 4 and described in detail in the risk assessment and risk management plan.²¹

Table 4: Summary of Risks and Mitigating Measures

Risks	Mitigation Measures
Loss of political, community, and stakeholder support at the state level for applying IWRM to river basin planning and management.	Continue supporting ACIWRM to strengthen IWRM knowledge and practices within Karnataka, including advocacy with key stakeholders.
Trained WRD staff do not remain in office.	Ensure that WRD maintains a succession staffing plan and that young officers are engaged under the project to strengthen WRD's knowledge about the project and its themes.
State government does not operationalize policy for irrigation management transfer and WUCS collection of irrigation service fees.	Establish and strengthen WUCS to operate and maintain the tertiary irrigation systems in a financially sustainable manner, including advocacy for payment of irrigation service fees. Modernize the irrigation systems to ensure that the infrastructure is capable of quality water delivery services. Farmers are willing to pay if water delivery services meet their expectations.
Low O&M expenditure impacts long-term sustainability and performance of the irrigation system.	Operationalize an asset management system to assist with annual budget planning and workflow for O&M, and integrate it into the WUCS', WRD's and KNNL's annual budget planning.

ACIWRM = Advanced Center for Integrated Water Resources Management, IWRM = integrated water resources management, KNNL = Karnataka Neeravari Nigam Limited, O&M = operation and maintenance, WRD = Water Resources Department, WUCS = water user cooperative societies.

Source: Asian Development Bank.

V. ASSURANCES

44. The government, the state, and KNNL have assured ADB that implementation of the project shall conform to all applicable ADB policies including those concerning anticorruption measures, safeguards, gender, procurement, consulting services, and disbursement as described in detail in the updated facility administration manual and loan documents.

45. The government, the state, and KNNL have agreed with ADB on certain covenants for the project, which are set forth in the loan agreement and project agreement.

VI. THE PRESIDENT'S DECISION

46. On the basis of the approval by ADB's Board of Directors for the provision of loans under the multitranchise financing facility in an aggregate principal amount not exceeding \$150,000,000 to India for the Karnataka Integrated and Sustainable Water Resources Management Investment Program, the President has approved the tranche as described in para. 19 and such other terms and conditions as are substantially in accordance with those set forth in the draft loan and project agreements.

²¹ Updated Risk Assessment and Risk Management Plan is in Appendix 13.

DESIGN AND MONITORING FRAMEWORK FOR PROJECT 2

Impact the Project is Aligned with			
(i) Sustainable water security in selected river basins in Karnataka improved (program defined) ^a			
(ii) Water use efficiency in irrigated agriculture improved (program defined) ^a			
Results Chain	Performance Indicators with Targets and Baselines	Data Sources and Reporting	Risks
Outcome IWRM successfully implemented in selected river basins in Karnataka	By 2024: a. River basin plan for the Middle Krishna Basin (K2, K3, K4 sub-basins) endorsed by State IWRM Committee (2019 baseline: n.a.) b. 30 VNC WUCS confirmed receipt of improved water delivery service (2019 baseline: n.a.) c. VNC tertiary irrigation system operated and maintained by the VNC WUCS (2019 baseline: n.a.)	a. WRD and/or ACIWRM annual reports and scheme monitoring reports b–c. KNNL and/or CADA annual and scheme monitoring reports	Loss of political, community and stakeholder support at the state level to sustain and proceed with reforms
Outputs 1. State and basin institutions for IWRM strengthened	By 2023: 1a. River basin plan for the Middle Krishna Basin (K2, K3, K4 sub-basins) developed (2019 baseline: n.a.) 1b. At least 600 WRD staff (of which 200 are women staff) certified in IWRM (2019 baseline: 150 staff are certified under Project 1, of which 40 are women) 1c. State water resources information system expanded to include analysis and reporting modules (2019 baseline: n.a.)	1a. ACIWRM annual reports 1b. WRD, KNNL and ACIWRM progress reports 1c. ACIWRM annual reports	Trained WRD staff do not remain in office
2. VNC irrigation system infrastructure and management modernized	By 2023: 2a. 12 anicuts, intake structures, and 442 km of irrigation canals covering 11,150 ha upgraded (2019 baseline: n.a.) (RFI A) 2b. Asset management plan for the main VNC system infrastructure prepared (2019 baseline: n.a.)	2a–b. KNNL annual reports and scheme monitoring reports	State government does not operationalize policy for irrigation management transfer and WUCS collection of irrigation service fees

Results Chain	Performance Indicators with Targets and Baselines	Data Sources and Reporting	Risks
	2c. 30 WUCS established and trained on operation and maintenance of tertiary irrigation system and improved on-farm water application methods (with 30% women representation in management boards and 40% attendance in training) (2019 baseline: 0) 2d. At least 50% of all WUCS officers and members (with at least 60% men's participation) reported increased knowledge on the application of gender concepts and principles to water resource management. (2019 baseline: n.a.) 2e. At least 50% of KNNL staff (including 30% women) reported increased knowledge on the application of gender concepts and principles to water resource management. (2019 baseline: n.a.)	2c. KNNL and CADA annual reports with sex-disaggregated data. 2d-e. Pre- and post-training assessment.	
3. Program management systems operational	By 2023 3a. Project outputs delivered on time and within budget (2019 baseline: n.a.) 3b. Project MIS expanded to include VNC activities with gender indicators (2019 baseline: n.a.) 3c. Quarterly progress reports and safeguard monitoring reports submitted on time with sex-disaggregated data and analysis (2019 baseline: n.a.) 3d. TLBC periodic financing request and associated due diligence documents prepared and advance procurement actions initiated (2019 baseline: n.a.)	3a. Quarterly disbursement and contract award projections and financial reports prepared by KNNL and ACIWRM 3b-c. KNNL and ACIWRM quarterly progress reports and semi-annual safeguard monitoring reports 3d. ADB mission aide memoire	

Results Chain	Performance Indicators with Targets and Baselines	Data Sources and Reporting	Risks
Key Activities with Milestones			
1. State and basin institutions for IWRM strengthened			
1.1 Initiate preparation of profile and basin plan for Middle Krishna River basins (August 2019).			
1.2 Continue developing the Karnataka Water Resources Information System (August 2019)			
1.3 Continue providing IWRM training to WRD staff (August 2019)			
2. VNC irrigation system infrastructure and management modernized			
2.1 Initiate civil works for VNC subproject (July 2019)			
2.2 Establish VNC WUCS and initiate strengthening activities, including on-farm extension services (January 2020)			
2.3 Initiate community participation packages with VNC WUCS (January 2020)			
2.4 Complete all works and WUCS activities (July 2023)			
2.5 Conduct gender sensitization workshops for KNNL staff and 30 VNC WUCS (July 2019)			
2.6 Finalize feasibility studies, detailed project report, detailed designs, and bid documents for TLBC subproject (June 2020)			
3. Program management systems operational			
3.1 Fully staffed PIO by loan effectiveness (August 2019)			
3.2 Update MIS with gender indicators (August 2019)			
3.3 KNNL prepares PFR for Tranche 3 to ADB (July 2020)			
3.4 Project 2 outputs delivered on time and within budget (July 2023)			
Inputs			
ADB: \$91 million (regular ordinary capital resources loan)			
Government: \$39 million			
Assumptions for Partner Financing			
Not Applicable			

ACIWRM = Advanced Center for Integrated Water Resources Management, ADB = Asian Development Bank, CADA = Command Area Development Authority, GAP = gender action plan, ha = hectare, IWRM = integrated water resources management, km = kilometer, KNNL = Karnataka Neeravari Nigam Limited, MIS = management information system, n.a. = not applicable, PFR = periodic financing request, PIO = project implementation office, TLBC = Tungabhadra Left Bank Canal, VNC = Vijayanagara Channels, WRA = Water Resources Regulatory Agency, WRD = Water Resources Department, WUCS = water users cooperative society

^a This impact was formulated following the previous design and monitoring framework guidelines. However, it remains aligned with: NITI Aayog. 2017. *India Three Year Action Agenda 2017–18 to 2019–20*. New Delhi.

Contribution to ADB Results Framework

RFI A: Land improved through irrigation, drainage, and/or flood management (hectares). Target: 11,150 ha.

Source: Asian Development Bank.

LOAN NUMBER _____-IND

LOAN AGREEMENT
(Ordinary Operations)

(Karnataka Integrated and Sustainable Water Resources Management
Investment Program – Project 2)

between

INDIA

and

ASIAN DEVELOPMENT BANK

DATED _____

IND 43253

LOAN AGREEMENT (Ordinary Operations)

LOAN AGREEMENT dated _____ between INDIA, acting by its President ("Borrower"), and ASIAN DEVELOPMENT BANK ("ADB").

WHEREAS

(A) by a framework financing agreement dated 28 July 2014, as amended from time to time, between the Borrower and ADB ("FFA"), ADB has agreed to provide a multitranche financing facility to the Borrower for purposes of financing projects under the Karnataka Integrated and Sustainable Water Resources Investment Program ("Investment Program");

(B) by a periodic financing request dated 8 July 2019 ("PFR"), the Borrower has applied to ADB for a loan for the purposes of the Project described in Schedule 1 to this Loan Agreement;

(C) the Project will be carried out by the State of Karnataka ("State") acting through (i) KNNL (as defined below); and (ii) ACIWRM (as defined below) ("Project Executing Agency" or "EA") and, and for this purpose the Borrower will make available to the State the proceeds of the Loan provided for herein upon terms and conditions mutually satisfactory to ADB and the Borrower; and

(D) ADB has agreed to make a loan to the Borrower from ADB's ordinary capital resources upon the terms and conditions set forth herein and in the Project Agreement of even date herewith between ADB on the one part, and the State, KNNL and ACIWRM on the other part;

NOW THEREFORE the parties hereto agree as follows:

ARTICLE I

Loan Regulations; Definitions

Section 1.01. All the provisions of ADB's Ordinary Operations Loan Regulations, dated 1 January 2017 ("Loan Regulations"), are hereby made applicable to this Loan Agreement with the same force and effect as if they were fully set forth herein.

Section 1.02. Wherever used in this Loan Agreement, the several terms defined in the Loan Regulations have the respective meanings therein set forth unless modified herein or the context otherwise requires. Additional terms used in this Loan Agreement have the following meanings:

- (a) "ACIWRM" means the State's Advanced Centre for Integrated Water Resources Management, or any successor thereto, which shall be the

Executing Agency for Part A of the Project as described in Schedule 1 to this Loan Agreement;

- (b) “Consulting Services” means consulting services as described in the Procurement Regulations and set out in the Procurement Plan, and to be financed out of the proceeds of the Loan;
- (c) “EA” or “Project Executing Agency” for the purposes of, and within the meaning of, the Loan Regulations means as the context may require, the State acting through (i) ACIWRM, or any successors thereto which is responsible for carrying out Part A of the Project; and (ii) KNNL, or any successors thereto which is responsible for carrying out Part B and Part C of the Project;
- (d) “EARF” or “Environmental Assessment and Review Framework” means the environmental assessment and review framework for the Investment Program, including any update thereto, prepared and submitted by the Borrower through the EA and cleared by ADB and incorporated by reference in the FFA;
- (e) “EIA” or “Environmental Impact Assessment” means an environmental impact assessment for a Subproject, including any update thereto, prepared and submitted by the Borrower through the EA pursuant to the requirements set forth in the EARF and cleared by ADB;
- (f) “EMP” or “Environmental Management Plan” means an environmental management plan for a Subproject, including any update thereto, prepared and submitted by the Borrower through the EA, cleared by ADB, and incorporated in the EIA;
- (g) “Environmental Safeguards” means the principles and requirements set forth in Chapter V, Appendix 1, and Appendix 4 (as applicable) of the SPS;
- (h) “Facility” means the multitranche financing facility provided by ADB to the Borrower for the purposes of financing projects under the Investment Program;
- (i) “FAM” means the updated Facility administration manual for the Project dated August 2019 and agreed between the Borrower, EA and ADB, as updated from time to time in accordance with the respective administrative procedures of the Borrower, EA and ADB;
- (j) “FFA” means the framework financing agreement dated 28 July 2014, as amended from time to time, between ADB and the Borrower with respect to the Facility;
- (k) “Financing Arrangements” means the arrangements between the Borrower and the State as per current policy of the Borrower, and acceptable to ADB;

- (l) “Gender Action Plan” or “GAP” means a gender action plan prepared for the Project including any update thereto, and agreed between the Borrower, EA and ADB;
- (m) “Goods” means equipment and materials as set out in the Procurement Plan and to be financed out of the proceeds of the Loan, including any related Nonconsulting Services such as transportation, insurance, installation, commissioning, training, and initial maintenance;
- (n) “Investment Program” means the Karnataka Integrated and Sustainable Water Resources Management Investment Program;
- (o) “IWRM” means integrated water resources management;
- (p) “KNNL” means Karnataka Neeravari Nigam Limited, or any successor thereto which shall be the Executing Agency for Part B and Part C of the Project as described in Schedule 1 to this Loan Agreement;
- (q) “Loan Disbursement Handbook” means ADB’s Loan Disbursement Handbook (2017, as amended from time to time);
- (r) “Nonconsulting Services” means nonconsulting services as described in the Procurement Regulations and set out in the Procurement Plan, and to be financed out of the proceeds of the Loan;
- (s) “PFR” means the periodic financing request submitted or to be submitted by the Borrower, for the purposes of each loan under the Facility, and for the purpose of this Loan Agreement means the periodic financing request dated 8 July 2019;
- (t) “Procurement Plan” means the procurement plan for the Project as agreed between the Borrower, EA and ADB and included in the FAM, as updated from time to time in accordance with the Procurement Policy, the Procurement Regulations, and other arrangements agreed with ADB;
- (u) “Procurement Policy” means ADB’s Procurement Policy - Goods, Works, Nonconsulting and Consulting Services (2017, as amended from time to time);
- (v) “Procurement Regulations” means ADB’s Procurement Regulations for ADB Borrowers - Goods, Works, Nonconsulting and Consulting Services (2017, as amended from time to time);
- (w) “Project facilities” means the facilities to be improved, rehabilitated, modernized, constructed, and/or maintained, and the equipment to be installed and maintained under the Project;
- (x) “Safeguards Monitoring Report” means each report prepared and submitted by the Borrower through the EA to ADB that describes

progress with implementation of, and compliance with the EMP (as applicable), including and corrective and preventive actions;

- (y) "Services" means Consulting Services and Nonconsulting Services;
- (z) "SPS" or "Safeguard Policy Statement" means ADB's Safeguard Policy Statement (2009);
- (aa) "State" means the State of Karnataka;
- (bb) "Subproject" means a subproject under the Project that meets the subproject eligibility criteria set forth in Schedule 4 to this Loan Agreement;
- (cc) "Works" means construction or civil works as set out in the Procurement Plan and to be financed out of the proceeds of the Loan, including any related Nonconsulting Services and project related services that are provided as part of a single responsibility or turnkey contract; and
- (dd) "WUCS" means water user cooperative society(ies).

ARTICLE II

The Loan

Section 2.01. (a) ADB agrees to lend to the Borrower from ADB's ordinary capital resources an amount of ninety-one million Dollars (\$91,000,000), as such amount may be converted from time to time through a Currency Conversion in accordance with the provisions of Section 2.06 of this Loan Agreement.

(b) The Loan has a principal repayment period of 15 years, and a grace period as defined in subsection (c) hereinafter.

(c) The term "grace period" as used in subsection (b) hereinabove means the period prior to the first Principal Payment Date in accordance with the amortization schedule set forth in Schedule 2 to this Loan Agreement.

Section 2.02. The Borrower shall pay to ADB interest on the principal amount of the Loan withdrawn and outstanding from time to time at a rate for each Interest Period equal to the sum of:

- (a) LIBOR; and
- (b) 0.60% as provided by Section 3.02 of the Loan Regulations less a credit of 0.10% as provided by Section 3.03 of the Loan Regulations.

Section 2.03. The Borrower shall pay a commitment charge of 0.15% per annum. Such charge shall accrue on the full amount of the Loan (less amounts withdrawn from time to time), commencing 60 days after the date of this Loan Agreement.

Section 2.04. Interest and other charges on the Loan shall be payable semiannually on 15 March and 15 September in each year.

Section 2.05. The Borrower shall repay the principal amount of the Loan withdrawn from the Loan Account in accordance with the provisions of Schedule 2 to this Loan Agreement.

Section 2.06. (a) The Borrower may at any time request any of the following Conversions of the terms of the Loan in order to facilitate prudent debt management:

- (i) a change of the Loan Currency of all or any portion of the principal amount of the Loan, whether withdrawn and outstanding or unwithdrawn, to an Approved Currency;
- (ii) a change of the interest rate basis applicable to all or any portion of the principal amount of the Loan withdrawn and outstanding from a Floating Rate to a Fixed Rate, or vice versa; and
- (iii) the setting of limits on the Floating Rate applicable to all or any portion of the principal amount of the Loan withdrawn and outstanding by the establishment of an Interest Rate Cap or Interest Rate Collar on said Floating Rate.

(b) Any conversion requested pursuant to subsection (a) hereinabove that is accepted by ADB shall be considered a "Conversion", as defined in Section 2.01(f) of the Loan Regulations, and shall be effected in accordance with the provisions of Article V of the Loan Regulations and the Conversion Guidelines.

ARTICLE III

Use of Proceeds of the Loan

Section 3.01. The Borrower shall make the proceeds of the Loan available to the EA upon terms and conditions mutually agreeable to ADB and the Borrower, and shall cause the EA to apply such proceeds to the financing of expenditures on the Project in accordance with the provisions of this Loan Agreement and the Project Agreement.

Section 3.02. The proceeds of the Loan shall be allocated and withdrawn in accordance with the provisions of Schedule 3 to this Loan Agreement, as such Schedule may be amended from time to time by agreement between the Borrower and ADB.

Section 3.03. Except as ADB may otherwise agree, the Borrower shall ensure, or cause the EA to ensure, that the items of expenditure to be financed out of the

proceeds of the Loan are procured in accordance with the provisions of Schedule 4 to this Loan Agreement.

Section 3.04. The Loan Closing Date for the purposes of Section 9.02 of the Loan Regulations shall be 31 March 2024 or such other date as may from time to time be agreed between the Borrower and ADB.

ARTICLE IV

Particular Covenants

Section 4.01. In the carrying out of the Project and operation of the Project facilities, the Borrower shall perform, or cause to be performed by the State and the EA, all obligations set forth in this Loan Agreement and the Project Agreement.

Section 4.02. The Borrower shall enable ADB's representatives to inspect the Project, the Goods, Works and Services, and any relevant records and documents.

Section 4.03. ADB shall disclose the annual audited financial statements for the Project and the opinion of the auditors on the financial statements within 14 days of the date of ADB's confirmation of their acceptability by posting them on ADB's website.

Section 4.04. In so far as it relates to the Project, the Borrower shall take all actions or cause the State to take all actions, which shall be necessary on its part to enable the EA to perform its obligations under the Project Agreement, and shall not take or permit any action which would interfere with the performance of such obligations.

Section 4.05. (a) In so far as it relates to the Project, the Borrower shall exercise its rights under the Financing Arrangements in such a manner as to protect the interests of the Borrower and ADB and to accomplish the purposes of the Loan.

(b) In so far as it relates to the Project, no rights or obligations under the Financing Arrangements shall be assigned, amended, abrogated or waived without the prior notice to ADB.

ARTICLE V

Effectiveness

Section 5.01. A date 90 days after the date of this Loan Agreement is specified for the effectiveness of this Loan Agreement for the purposes of Section 10.04 of the Loan Regulations.

ARTICLE VI**Miscellaneous**

Section 6.01. The Secretary, Additional Secretary, Joint Secretary, Director, or Deputy Secretary, in the Department of Economic Affairs of the Ministry of Finance of the Borrower is designated as representative of the Borrower for the purposes of Section 12.02 of the Loan Regulations.

Section 6.02. The following addresses are specified for the purposes of Section 12.01 of the Loan Regulations:

For the Borrower

The Secretary to the Government of India
Department of Economic Affairs
Ministry of Finance
North Block
New Delhi – 110001
India

Facsimile Number:

(91-11) 2309-4075
(91-11) 2309-2511.

For ADB

Asian Development Bank
6 ADB Avenue
Mandaluyong City
1550 Metro Manila
Philippines

Facsimile Numbers:

(632) 636-2444
(632) 636-2391.

IN WITNESS WHEREOF the parties hereto, acting through their representatives thereunto duly authorized, have caused this Loan Agreement to be signed in their respective names as of the day and year first above written and to be delivered at the principal office of ADB.

INDIA

By _____
[Name and designation]

ASIAN DEVELOPMENT BANK

By _____
[Name and designation]

SCHEDULE 1

Description of the Project

1. The objective of the Project is to implement IWRM in selected river basins in the State.
2. The Project shall comprise following parts as more fully described in the PFR:
 - (a) Part A: Provision of support and capacity development to strengthen State and basin institutions for IWRM, including
 - (i) preparing the middle Krishna Basin Plan (for sub-basins K2, K3 and K4);
 - (ii) expanding the Karnataka Water Resources Information System; and
 - (iii) further strengthening Water Resources Department staff with IWRM training;
 - (b) Part B: Modernization of irrigation system infrastructure and management, including
 - (i) modernizing the Vijayanagara Channels including repairing anicuts and intake structures and about 442 kilometers of main, distributary and minor canals;
 - (ii) establishing and strengthening WUCS; and
 - (iii) providing farmers with extension services to improve on-farm water management and crop production, and developing an asset database and management plan; and
 - (c) Part C: Operationalization of program management systems to provide support for procurement, coordination, safeguards, monitoring and reporting, and preparation of the subsequent project under the Investment Program.
4. The Project is expected to be completed by 30 September 2023.

SCHEDULE 2

Amortization Schedule

1. The following table sets forth the Principal Payment Dates of the Loan and the percentage of the total principal amount of the Loan payable on each Principal Payment Date (Installment Share). If the proceeds of the Loan shall have been fully withdrawn as of the first Principal Payment Date, the principal amount of the Loan repayable by the Borrower on each Principal Payment Date shall be determined by ADB by multiplying (a) the total principal amount of the Loan withdrawn and outstanding as of the first Principal Payment Date; by (b) the Installment Share for each Principal Payment Date, such repayment amount to be adjusted, as necessary, to deduct any amounts referred to in paragraph 4 of this Schedule, to which a Currency Conversion applies.

Date Payment Due	Installment Share (Expressed as a %)
15 March 2025	3.333333
15 September 2025	3.333333
15 March 2026	3.333333
15 September 2026	3.333333
15 March 2027	3.333333
15 September 2027	3.333333
15 March 2028	3.333333
15 September 2028	3.333333
15 March 2029	3.333333
15 September 2029	3.333333
15 March 2030	3.333333
15 September 2030	3.333333
15 March 2031	3.333333
15 September 2031	3.333333
15 March 2032	3.333333
15 September 2032	3.333333
15 March 2033	3.333333
15 September 2033	3.333333
15 March 2034	3.333333
15 September 2034	3.333333
15 March 2035	3.333333
15 September 2035	3.333333
15 March 2036	3.333333
15 September 2036	3.333333
15 March 2037	3.333333
15 September 2037	3.333333
15 March 2038	3.333333
15 September 2038	3.333333
15 March 2039	3.333333
15 September 2039	3.333343
Total	100.000000

2. If the proceeds of the Loan shall not have been fully withdrawn as of the first Principal Payment Date, the principal amount of the Loan repayable by the Borrower on each Principal Payment Date shall be determined as follows:

- (a) to the extent that any proceeds of the Loan shall have been withdrawn as of the first Principal Payment Date, the Borrower shall repay the amount withdrawn and outstanding as of such date in accordance with paragraph 1 of this Schedule; and
- (b) any withdrawal made after the first Principal Payment Date shall be repaid on each Principal Payment Date falling after the date of such withdrawal in amounts determined by ADB by multiplying the amount of each such withdrawal by a fraction, the numerator of which shall be the original Installment Share specified in the table in paragraph 1 of this Schedule for said Principal Payment Date (the Original Installment Share) and the denominator of which shall be the sum of all remaining Original Installment Shares for Principal Payment Dates falling on or after such date, such repayment amounts to be adjusted, as necessary, to deduct any amounts referred to in paragraph 4 of this Schedule, to which a Currency Conversion applies.

3. Withdrawals made within 2 calendar months prior to any Principal Payment Date shall, for the purposes solely of calculating the principal amounts payable on any Principal Payment Date, be treated as withdrawn and outstanding on the second Principal Payment Date following the date of withdrawal and shall be repayable on each Principal Payment Date commencing with the second Principal Payment Date following the date of withdrawal.

4. Notwithstanding the provisions of paragraphs 1 and 2 of this Schedule, upon a Currency Conversion of all or any portion of the withdrawn principal amount of the Loan to an Approved Currency, the amount so converted in said Approved Currency that shall be repayable on any Principal Payment Date occurring during the Conversion Period, shall be determined by ADB by multiplying such amount in its currency of denomination immediately prior to said Conversion by either (i) the exchange rate that reflects the amounts of principal in said Approved Currency payable by ADB under the Currency Hedge Transaction relating to said Conversion; or (ii) if ADB so determines in accordance with the Conversion Guidelines, the exchange rate component of the Screen Rate.

5. If the principal amount of the Loan withdrawn and outstanding from time to time shall be denominated in more than one Loan Currency, the provisions of this Schedule shall apply separately to the amount denominated in each Loan Currency, so as to produce a separate amortization schedule for each such amount.

SCHEDULE 3

Allocation and Withdrawal of Loan Proceeds

General

1. The table attached to this Schedule sets forth the Categories of items of expenditure to be financed out of the proceeds of the Loan and the allocation of the Loan proceeds to each such Category ("Table"). (Reference to "Category" in this Schedule is to a Category of the Table).

Basis for Withdrawal from the Loan Account

2. Except as ADB may otherwise agree, the proceeds of the Loan shall be allocated to items of expenditure, and disbursed on the basis of the withdrawal percentage for each item of expenditure set forth in the Table.

Reallocation

3. Notwithstanding the allocation of Loan proceeds and the withdrawal percentages set forth in the Table,

- (a) if the amount of the Loan allocated to any Category appears to be insufficient to finance all agreed expenditures in that Category, ADB may, in consultation with and by notice to the Borrower and the EA, (i) reallocate to such Category, to the extent required to meet the estimated shortfall, amounts of the Loan which have been allocated to another Category but, in the opinion of ADB, are not needed to meet other expenditures; and (ii) if such reallocation cannot fully meet the estimated shortfall, reduce the withdrawal percentage applicable to such expenditures in order that further withdrawals under such Category may continue until all expenditures thereunder shall have been made; and
- (b) if the amount of the Loan allocated to any Category appears to exceed all agreed expenditures in that Category, ADB may, in consultation with and by notice to the Borrower and the EA, reallocate such excess amount to any other Category.

Disbursement Procedures

4. Except as ADB may otherwise agree, the Loan proceeds shall be disbursed in accordance with the Loan Disbursement Handbook, and detailed arrangements agreed upon between ADB and the Borrower.

Retroactive Financing

5. Withdrawals from the Loan Account may be made to finance eligible expenditures incurred under the Project before the Effective Date, but not earlier than 12 months before the date of this Loan Agreement in connection with items to be retroactively financed, subject to a maximum amount equivalent to 20% of the Loan amount.

TABLE

ALLOCATION AND WITHDRAWAL OF LOAN PROCEEDS			
Number	Item	Total Amount Allocated for ADB Financing \$	Basis for Withdrawal from the Loan Account
		Category	
1	Works	56,480,000	73% of total expenditure claimed
2	Mechanical and equipment; Consulting Services (other than as provided under Item 3 below); on-farm extension Services; ACIWRM office operations; training and workshops; studies	14,530,000	68% of total expenditure claimed
3	ACIWRM support consultants (international)	1,310,000	100% of total expenditure claimed*
4	Unallocated	18,680,000	
	TOTAL	91,000,000	

* Exclusive of taxes and duties imposed within the territory of the Borrower

SCHEDULE 4

Execution of Project; Financial Matters

Implementation Arrangements

1. The Borrower, the State, and the EA shall ensure that the Project is implemented in accordance with the detailed arrangements set forth in the FAM. Any subsequent change to the FAM shall become effective only after approval of such change by the Borrower, the State, relevant EA and ADB. In the event of any discrepancy between the FAM and this Loan Agreement, the provisions of this Loan Agreement shall prevail.

2. (a) The Borrower shall ensure or cause the EA to ensure towards smooth implementation of the Project, that grievances if any from stakeholders, relating to Project or any Subproject implementation or use of funds are addressed effectively and efficiently.

(b) The Borrower shall ensure or cause the State, KNNL and ACIWRM to ensure compliance with all the requirements and obligations in their part as included in this Loan Agreement and the Project Agreement to meet the objectives of the Project in a timely and efficient manner.

Counterpart Support; Funds

3. The Borrower shall cause the State to make available the Loan proceeds to the EA under appropriate arrangements acceptable to ADB; and ensure:

(a) sufficient counterpart funds from its budget for each fiscal year, in a timely manner, for the efficient implementation of the Project; and

(b) adequate funds towards operations and maintenance of Project facilities, through budgetary allocations or other means, to be provided to KNNL and ACIWRM, during and after completion of a Subproject and Project completion.

4. The Borrower shall also cause the State to ensure that (i) the WUCS make a 10% upfront contribution in regards to command area development activities undertaken in their respective areas; (ii) any financing gap created by a failure on the part of the WUCS or any other implementing agency to provide relevant counterpart funds for the Project is filled by funds from its own resources; (iii) any shortfall between cost and revenues for the operation and maintenance of the facilities created or rehabilitated under the Project are met; (iv) any unforeseen environmental and social impacts are mitigated; and (v) additional costs arising from design changes, price escalation in construction costs and/or unforeseen circumstances are met.

Procurement

5. The Borrower shall ensure, or cause the EA to ensure, that:
- (a) the procurement of Goods, Works and Services is carried out in accordance with the Procurement Policy and the Procurement Regulations;
 - (b) Goods, Works and Services shall be procured based on the detailed arrangements set forth in the Procurement Plan, including the procurement and selection methods, the type of bidding documents, and ADB's review requirements. The Borrower through the EA may modify the detailed arrangements set forth in the Procurement Plan only with the prior agreement of ADB, and such modifications must be set out in updates to the Procurement Plan; and
 - (c) (i) all Goods and Works procured and Services obtained (including all computer hardware, software and systems, whether separately procured or incorporated within other goods and services procured) do not violate or infringe any industrial property or intellectual property right or claim of any third party; and (ii) all contracts for the procurement of Goods, Works and Services contain appropriate representations, warranties and, if appropriate, indemnities from the contractor, supplier, consultant or service provider with respect to the matters referred to in this subparagraph.
6. The Borrower shall ensure, or shall cause the EA to, announce the Project and business opportunities (i.e., invitation of potential bids) associated with the Project on its website. The website shall disclose the following information in relation to Works and Services procured for the Project: (a) the list of participating bidders; (b) the name of the selected bidder; (c) the amount of the contracts awarded; and (d) the goods and services procured.
7. The Borrower shall ensure, or cause the EA to ensure, that:
- (a) no Works contract is awarded for a Subproject which involves environmental impacts until the EA has incorporated the relevant provisions from the EMP into the Works contract; and
 - (b) no commencement of Works is allowed under any Works contract which involves environmental impacts and requires environmental clearances, until the EA has obtained the final approval of (i) the EIA from ADB, and (ii) environmental clearance, including approval of the EIA and any other environmental assessment report, from the appropriate environmental authority(ies) of the Borrower and/or State.

Coordination among the Borrower's and the State's Agencies

8. The Borrower shall ensure or cause, the State, KNNL and ACIWRM as applicable to ensure that all ministries, agencies, and divisions involved in the implementation of the Project give their full cooperation to ensure smooth implementation of the Project. Specifically, the Borrower shall ensure or cause the EA to give full, timely and efficient cooperation, including (a) providing relevant counterpart funds or other support for the Project; (b) carrying out their respective roles and responsibilities, and fulfilling their obligations, in accordance with the FAM; and (c) issuing any licenses, permits or approvals required in connection with any Works within their respective area.

Subproject Selection and Implementation

9. The Borrower shall ensure or cause the EA to ensure that each Subproject is selected and approved in accordance with the provisions of Schedule 4 to the FFA.

10. The Borrower shall ensure or cause the EA to ensure that all documents forming the basis for screening, selection, processing, and implementation of each Subproject are made available to ADB upon request and are kept for such purposes for a minimum of 5 years from the date of the projection completion report of the Project.

SafeguardsEnvironment

11. The Borrower shall ensure, or cause the EA to ensure, that the preparation, design, construction, implementation, operation and decommissioning of the Project, and all Project facilities comply with (a) all applicable laws and regulations of the Borrower and the State relating to environment, health, and safety; (b) the Environmental Safeguards; (c) the EARF; and (d) all measures and requirements set forth in the respective EIA and EMP, and any corrective or preventative actions set forth in a Safeguards Monitoring Report.

Land Acquisition and Involuntary Resettlement

12. The Borrower shall ensure or cause the EA to ensure that the Project does not have any involuntary resettlement impacts, all within the meaning of the SPS. In the unforeseen event of any involuntary resettlement impact in the Project, the Borrower shall ensure, or cause the EA to take all steps required to ensure that compliance with the applicable laws and regulations of the Borrower and the State and the SPS.

Indigenous Peoples

13. The Borrower shall ensure or cause the EA to ensure that the Project does not have any indigenous peoples impacts, all within the meaning of the SPS. In the unforeseen event of any indigenous peoples impacts in the Project, the Borrower shall ensure, or shall cause the EA to take all steps required to ensure compliance with the applicable laws and regulations of the Borrower and the State and the SPS.

Human and Financial Resources to Implement Safeguards Requirements

14. The Borrower shall ensure, or cause the EA to ensure, that all necessary budgetary and human resources to fully implement the EMP, as required, are made available on a timely basis.

Safeguards – Related Provisions in Bidding Documents and Works Contracts

15. The Borrower shall ensure, or cause the EA to ensure, that all bidding documents and contracts for Works contain provisions that require contractors to:

- (a) comply with the measures and requirements relevant to the contractor set forth in the EIA and the EMP (to the extent they concern impacts on affected people during construction), and any corrective or preventative actions set out in a Safeguards Monitoring Report;
- (b) make available a budget for all such environmental and social measures;
- (c) provide the EA with a written notice of any unanticipated environmental, resettlement or indigenous peoples risks or impacts that arise during construction, implementation or operation of the Project that were not considered in the EIA or the EMP;
- (d) adequately record the condition of roads, agricultural land and other infrastructure prior to starting to transport materials and construction; and
- (e) fully reinstate pathways, other local infrastructure, and agricultural land to at least their pre-project condition upon the completion of construction.

Safeguards Monitoring and Reporting

16. The Borrower shall ensure, or cause the EA to ensure the following:

- (a) submit semi-annual Safeguards Monitoring Reports to ADB and disclose relevant information from such reports to affected persons promptly upon submission;
- (b) if any unanticipated environmental and/or social risks and impacts arise during construction, implementation or operation of the Project that were not considered in the EIA or the EMP, promptly inform ADB of the occurrence of such risks or impacts, with detailed description of the event and proposed corrective action plan;
- (c) no later than the date of award of Works contract, engage qualified and experienced external expert or qualified NGO under a selection process and terms of reference acceptable to ADB, to verify information

produced through the Project monitoring process, and facilitate the carrying out of any verification activities by such external expert; and

- (d) report any breach of compliance with the measures and requirements set forth in the EMP promptly after becoming aware of the breach.

Prohibited List of Investments

17. The Borrower shall ensure, or cause the EA to ensure, that no proceeds of the Loan under the Project are used to finance any activity included in the list of prohibited investment activities provided in Appendix 5 of the SPS.

Labor Standards, Health and Safety

18. The Borrower shall ensure, or cause the EA to ensure, that Works contracts under the Project follow all applicable labor laws of the Borrower and the State and that these further include provisions to the effect that contractors; (a) carry out HIV/AIDS awareness programs for labor and disseminate information at worksites on risks of sexually transmitted diseases and HIV/AIDS as part of health and safety measures for those employed during construction; and (b) follow and implement all statutory provisions on labor (including not employing or using children as labor, equal pay for equal work), health, safety, welfare, sanitation, and working conditions. Such contracts shall also include clauses for termination in case of any breach of the stated provisions by the contractors.

Communications and Participation

19. The Borrower shall ensure, or cause the EA to ensure, that the Project is undertaken in conformity with the communication strategy as agreed between ADB, the Borrower, the State, and the EA and referred in the FAM.

Gender and Development

20. The Borrower shall ensure, or cause the EA to ensure, that (a) the GAP is implemented in accordance with its terms; (b) the bidding documents and contracts include relevant provisions for contractors to comply with the measures set forth in the GAP; (c) adequate resources are allocated for implementation of the GAP; and (d) progress on implementation of the GAP, including progress toward achieving key gender outcome and output targets, are regularly monitored and reported to ADB.

Governance and Anticorruption

21. The Borrower shall ensure and cause the State, KNNL and ACIWRM to comply with ADB's Anticorruption Policy (1998, as amended to date) and (a) shall ensure that the anticorruption provisions acceptable to ADB, the Borrower, the State and EA are included in all bidding documents and contracts financed by ADB in connection with the Project, including provisions specifying the right of ADB to review and examine the records and accounts of the State and EA and all contractors, suppliers, consultants, and other service providers as they relate to the Subprojects and the Project, and as included in the FAM; (b) shall allow and assist ADB's representatives to carry out random spot checks on the work in progress and utilization of funds for the Project; (c) acknowledge that ADB reserves the right to investigate directly or

through its agents any alleged corrupt, fraudulent, collusive or coercive practice relating to the Project; and (d) cooperate with any such investigation and extend all necessary assistance for satisfactory completion of such investigation, and as included in the FAM.

Operation and Maintenance

22. (a) The Borrower shall ensure, or cause the EA to ensure, that the Project facilities are adequately maintained and that proper technical supervision and adequate funds for this purpose are provided. The funds required for operation and maintenance shall be allocated by the EA annually and released on a timely basis.

(b) Without limiting the generality of the foregoing, by no later than 31 December 2022, the Borrower shall ensure or cause the State to ensure that KNNL has (i) prepared, a geographic information system (GIS)-based asset management system and operation and maintenance plan for the Project facilities, which shall be operationalized and applied following Project implementation; and (ii) shall ensure that all equipment and spare parts financed under the Project shall exclusively be used for the operation and maintenance of the Project.

LOAN NUMBER ____-IND

PROJECT AGREEMENT

(Karnataka Integrated and Sustainable Water Resources Management
Investment Program – Project 2)

between

ASIAN DEVELOPMENT BANK

and

STATE OF KARNATAKA

KARNATAKA NEERAVARI NIGAM LIMITED

ADVANCED CENTRE FOR INTEGRATED WATER RESOURCES MANAGEMENT

DATED _____

IND 43253

PROJECT AGREEMENT

PROJECT AGREEMENT dated _____ between ASIAN DEVELOPMENT BANK ("ADB") on the one part, and the STATE OF KARNATAKA, acting by its Governor ("State"), the KARNATAKA NEERAVARI NIGAM LIMITED ("KNNL") and the ADVANCED CENTRE FOR INTEGRATED WATER RESOURCES MANAGEMENT ("ACIWRM") on the other part.

WHEREAS

(A) by a Loan Agreement of even date herewith between India ("Borrower") and ADB, ADB has agreed to make to the Borrower a loan of ninety-one million Dollars (\$91,000,000) on the terms and conditions set forth in the Loan Agreement, but only on the condition that the proceeds of the loan be made available to the State, and through the State to KNNL and ACIWRM, and that the State acting through KNNL and ACIWRM agrees to undertake certain obligations towards ADB set forth herein; and

(B) the State, KNNL, and ACIWRM in consideration of ADB entering into the Loan Agreement with the Borrower, have agreed to undertake the obligations set forth herein;

NOW THEREFORE the parties hereto agree as follows:

ARTICLE I

Definitions

Section 1.01. Wherever used in this Project Agreement, unless the context otherwise requires, the several terms defined in the Loan Agreement and in the Loan Regulations (as so defined) have the respective meanings therein set forth, except that for purposes of this Project Agreement, the term "Project" means (a) Part A of the Project for ACIWRM, and (b) Part B and Part C of the Project for KNNL, as described in Schedule 1 to the Loan Agreement.

ARTICLE II

Particular Covenants

Section 2.01. (a) The State, KNNL, and ACIWRM shall carry out the Project with due diligence and efficiency, and in conformity with sound applicable technical, financial, business, and development practices.

(b) In the carrying out of the Project and operation of the Project facilities, the State, KNNL, and ACIWRM shall perform all obligations set forth in the Loan Agreement to the extent that they are applicable to the State, KNNL and ACIWRM, respectively.

Section 2.02. The State, KNNL and ACIWRM shall make available, promptly as needed, and on terms and conditions mutually acceptable to ADB and the Borrower, the funds, facilities, services, land and other resources as required, in addition to the proceeds of the Loan, for the carrying out of the Project.

Section 2.03. (a) In the carrying out of the Project, the State, KNNL, and ACIWRM, respectively, shall engage competent and qualified consultants and contractors, consistent with the Procurement Policy and the Procurement Regulations, to an extent and upon terms and conditions mutually satisfactory to ADB and the Borrower.

(b) Except as ADB and the Borrower may otherwise agree, the State, KNNL and ACIWRM shall procure all items of expenditures to be financed out of the proceeds of the Loan in accordance with the provisions of Schedule 4 to the Loan Agreement. ADB may refuse to finance a contract where any such item has not been procured under procedures substantially in accordance with those agreed between the Borrower and ADB or where the terms and conditions of the contract are not satisfactory to ADB.

Section 2.04. The State, KNNL and ACIWRM shall carry out the Project in accordance with plans, design standards, specifications, work schedules and construction methods mutually acceptable to ADB and the Borrower. The State, KNNL and ACIWRM shall furnish, or cause to be furnished, to ADB, promptly after their preparation, such plans, design standards, specifications and work schedules, and any material modifications subsequently made therein, in such detail as ADB shall reasonably request.

Section 2.05. (a) The State, KNNL and ACIWRM, as appropriate, shall take out and maintain with responsible insurers, or make other arrangements satisfactory to ADB for, insurance of Project facilities to such extent and against such risks and in such amounts as shall be consistent with sound practice.

(b) Without limiting the generality of the foregoing, the State, KNNL and ACIWRM, as appropriate, shall insure, or cause to be insured, the Goods to be procured, including imported for the Project against hazards incident to the acquisition, transportation and delivery thereof to the place of use or installation, and for such insurance any indemnity shall be payable in a currency freely usable to replace or repair such Goods.

Section 2.06. The State, KNNL and ACIWRM shall maintain, or cause to be maintained, records and accounts adequate to identify the items of expenditure financed out of the proceeds of the Loan, to disclose the use thereof in the Project, to record the progress of the Project (including the cost thereof) and to reflect, in accordance with consistently maintained sound accounting principles, its operations and financial condition.

Section 2.07. (a) ADB, the State, KNNL and ACIWRM shall cooperate fully to ensure that the purposes of the Loan will be accomplished.

(b) The State, KNNL and ACIWRM shall promptly inform ADB through the Borrower of any condition which interferes with, or threatens to interfere with, the progress of

the Project, the performance of its obligations under this Project Agreement, or the Financing Arrangements, or the accomplishment of the purposes of the Loan.

(c) ADB, the State, KNNL and ACIWRM shall each from time to time, at the request of either party, exchange views through their representatives with regard to any matters relating to the Project, the State, KNNL, ACIWRM and the Loan.

Section 2.08. (a) In so far as it relates to the Project, the State, KNNL and ACIWRM shall furnish to ADB all such reports and information as ADB shall reasonably request concerning (i) the Loan and the expenditure of the proceeds thereof; (ii) the items of expenditure financed out of such proceeds; (iii) the Project; (iv) the administration, operations, financial status of the State, and financial condition of KNNL and ACIWRM; and (v) any other matters relating to the purposes of the Loan.

(b) Without limiting the generality of the foregoing, the State and/acting through KNNL and ACIWRM as applicable shall furnish to ADB periodic reports on the execution of the Project and on the operation and management of the Project facilities. Such reports shall be submitted in such form and in such detail and within such a period as ADB shall reasonably request, and shall indicate, among other things, progress made and problems encountered during the period under review, steps taken or proposed to be taken to remedy these problems, and proposed program of activities and expected progress during the following period.

(c) Promptly after physical completion of the Project, but in any event not later than 3 months thereafter or such later date as ADB may agree for this purpose, the State, KNNL and ACIWRM shall prepare and furnish to ADB a report, in such form and in such detail as ADB shall reasonably request, on the execution and initial operation of the Project, including its cost, the performance by the State, KNNL and ACIWRM of their respective obligations under this Project Agreement and the accomplishment of the purposes of the Loan.

Section 2.09. (a) The State as applicable, and KNNL and ACIWRM shall (i) maintain separate accounts and records for the Project; (ii) prepare annual financial statements for the Project in accordance with financial reporting standards prevalent in the country; (iii) have such financial statements audited annually by independent auditors whose qualifications, experience and terms of reference are acceptable to ADB, in accordance with auditing standards acceptable to ADB; (iv) as part of each such audit, have the auditors prepare a report, which includes the auditors' opinion(s) on the financial statements and the use of the Loan proceeds, and a management letter (which sets out the deficiencies in the internal control of the Project that were identified in the course of the audit, if any); and (v) furnish to ADB, no later than 6 months after the close of the fiscal year to which they relate, copies of such audited financial statements, audit report and management letter, all in the English language, and such other information concerning these documents and the audit thereof as ADB shall from time to time reasonably request.

(b) ADB shall disclose the annual audited financial statements for the Project and the opinion of the auditors on the financial statements within 14 days of the date of ADB's confirmation of their acceptability by posting them on ADB's website.

(c) In addition to annual audited financial statements referred to in subsection (a) hereinabove, each of KNNL and ACIWRM shall (i) provide its annual financial statements prepared in accordance with financing reporting standards acceptable to ADB;

(ii) have its financial statements audited annually by independent auditors whose qualifications, experience and terms of reference are acceptable to ADB, in accordance with auditing standards acceptable to ADB; (iii) as part of each such audit, have the auditors prepare the auditors' opinion(s) on the financial statements and compliance with the financial covenants of the Loan Agreement; and (iv) furnish to ADB, no later than 1 month after approval by the relevant authority, copies of such audited financial statements and auditors' opinion(s), all in the English language, and such other information concerning these documents and the audit thereof as ADB shall from time to time reasonably request.

(d) The State, KNNL, and ACIWRM shall enable ADB, upon ADB's request, to discuss the financial statements for the Project, KNNL, and ACIWRM and their respective financial affairs where they relate to the Project with the auditors appointed by the State, KNNL and ACIWRM respectively pursuant to subsections (a)(iii) and (c)(ii) hereinabove, and shall authorize and require any representative of such auditors to participate in any such discussions requested by ADB. This is provided that such discussions shall be conducted only in the presence of an authorized officer of the State, KNNL, and ACIWRM, as applicable, unless the State, KNNL, and ACIWRM, as applicable, shall otherwise agree.

Section 2.10. The State, KNNL, and ACIWRM shall enable ADB's representatives to inspect the Project, the Goods and Works and any relevant records and documents.

Section 2.11. (a) The State, KNNL and ACIWRM respectively, shall, promptly as required, take all action within its powers to maintain its constitutional (in case of State), and corporate existence (in case of KNNL and ACIWRM), to carry on its operations, and to acquire, maintain and renew all rights, properties, powers, privileges and franchises which are necessary in the carrying out of the Project or in the conduct of its operations.

(b) In relation to the Project, the State, KNNL and ACIWRM respectively shall at all times conduct its operations in accordance with sound applicable technical, financial, business, development and operational practices, and under the supervision of competent and experienced management and personnel.

(c) In relation to the Project, the State, KNNL and ACIWRM respectively shall at all times operate and maintain its plants, equipment and other property, and from time to time, promptly as needed, make all necessary repairs and renewals thereof, all in accordance with sound applicable technical, financial, business, development, operational and maintenance practices.

Section 2.12. Except as ADB may otherwise agree, the State, KNNL and ACIWRM respectively shall not sell, lease or otherwise dispose of any of its assets which shall be required for the efficient carrying on of its operations or the disposal of which may prejudice its ability to perform satisfactorily any of its obligations under this Project Agreement.

Section 2.13. Except as ADB may otherwise agree, in relation to the Project, the State, KNNL and ACIWRM shall apply the proceeds of the Loan to the financing of expenditures on the Project in accordance with the provisions of the Loan Agreement and this Project Agreement, and shall ensure that all items of expenditures financed out of such proceeds are used exclusively in the carrying out of the Project.

Section 2.14. Except as ADB may otherwise agree, the State shall duly perform all its obligations under the Financing Arrangements, and shall not take, or concur in, any action which would have the effect of assigning, amending, abrogating or waiving any rights or obligations of the parties under the Financing Arrangements.

Section 2.15. The State, KNNL and ACIWRM shall promptly notify ADB of any proposal to amend, suspend or repeal any provision of their respective constitutional documents, which, if implemented, could adversely affect the carrying out of the Project or the operation of the Project facilities. The State, KNNL and ACIWRM shall afford ADB an adequate opportunity to comment on such proposal in taking any affirmative action thereon.

ARTICLE III

Effective Date; Termination

Section 3.01. This Project Agreement shall come into force and effect on the date on which the Loan Agreement comes into force and effect. ADB shall promptly notify the State, KNNL, and ACIWRM of such date.

Section 3.02. This Project Agreement and all obligations of the parties hereunder shall terminate on the date on which the Loan Agreement shall terminate in accordance with its terms.

Section 3.03. All the provisions of this Project Agreement shall continue in full force and effect notwithstanding any cancellation or suspension under the Loan Agreement.

ARTICLE IV

Miscellaneous

Section 4.01. Any notice or request required or permitted to be given or made under this Project Agreement and any agreement between the parties contemplated by this Project Agreement shall be in writing. Such notice or request shall be deemed to have been duly given or made when it shall be delivered by hand, mail or facsimile to the party to which it is required or permitted to be given or made at its address hereinafter specified, or at such other address as such party shall have designated by notice to the party giving such notice or making such request. The addresses so specified are:

For ADB

Asian Development Bank
6 ADB Avenue
Mandaluyong City
1550 Metro Manila
Philippines

Facsimile Numbers:

(632) 636-2444
(632) 636-2391.

For the State of Karnataka

Principal Secretary
Water Resources Department
Government of Karnataka
Vikas Soudha
Bengaluru - 560001
Karnataka
India

Facsimile Number:

+91 80 2225-7064.

For KNNL

Managing Director
Karnataka Neeravari Nigam Limited
No. 1, 4th Floor
Coffee Board Building
Dr. Ambedkar Veedhi
Bengaluru - 560001
Karnataka
India

Facsimile Number:

+91 80 2238-6015.

For ACIWRM

Director General
ACIWRM
Government of Karnataka
Vikas Soudha
Bengaluru - 560001
Karnataka
India

Facsimile Number:

+91 80 2225-7064.

Section 4.02. (a) Any action required or permitted to be taken, and any documents required or permitted to be executed, under this Project Agreement by or on behalf of (i) the State may be taken or executed by its Principal Secretary, Water Resources Department, (ii) for KNNL by its Managing Director, and (iii) for ACIWRM by its Registrar & Chief Engineer, or by such other person or persons as he or she shall so designate in writing notified to ADB.

(b) The State, KNNL, and ACIWRM, respectively, shall furnish to ADB sufficient evidence of the authority of each person who will act under subsection (a) hereinabove, together with the authenticated specimen signature of each such person.

Section 4.03. No delay in exercising, or omission to exercise, any right, power or remedy accruing to either party under this Project Agreement upon any default shall impair any such right, power or remedy or be construed to be a waiver thereof or an acquiescence in such default; nor shall the action of such party in respect of any default, or any acquiescence in any default, affect or impair any right, power or remedy of such party in respect of any other or subsequent default.

IN WITNESS WHEREOF the parties hereto, acting through their representatives thereunto duly authorized, have caused this Project Agreement to be signed in their respective names as of the day and year first above written, and to be delivered at the principal office of ADB.

ASIAN DEVELOPMENT BANK

By _____
[Name and Designation]

STATE OF KARNATAKA

By _____
[Name and Designation]

KARNATAKA NEERAVARI NIGAM
LIMITED

By _____
[Name and Designation]

ADVANCED CENTRE FOR
INTEGRATED WATER RESOURCES
MANAGEMENT

By _____
[Name and Designation]

Facility Administration Manual

Project Number: 43253
Loan Numbers: Project 1 – 3172
Project 2 –

September 2019

India: Karnataka Integrated and Sustainable Water
Resources Management Investment Program

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Facility Administration Manual Purpose and Process

The facility administration manual (FAM) describes the essential administrative and management requirements to implement the Multitranche Financing Facility (MFF) on time, within budget, and in accordance with Government of India and Asian Development Bank (ADB) policies and procedures. This FAM will also serve as project administration manual for Project 1. The FAM should include references to all available templates and instructions either through linkages to relevant URLs or directly incorporated in the FAM.

The executing agency will be the State acting through Karnataka Neeravari Nigam Limited (KNNL) and the Advanced Centre for Integrated Water Resources Management (ACIWRM), and together with the implementing agencies—Command Area Development Authorities (CADA), and water user cooperative societies (WUCS). The executing agency will be wholly responsible for the implementation of ADB financed projects, as agreed jointly between the borrower and ADB, and in accordance with Government and ADB's policies and procedures. ADB staff is responsible to support implementation including compliance by KNNL, ACIWRM, CADA, and WUCS of their obligations and responsibilities for project implementation in accordance with ADB's policies and procedures.

At Loan Negotiations the borrower and ADB have agreed to the FAM and ensure consistency with the Loan Agreement. Such agreement is reflected in the minutes of the Loan Negotiations. In the event of any discrepancy or contradiction between the FAM and the Loan Agreement, the provisions of the Loan Agreement shall prevail.

After ADB Board approval of the project's report and recommendation of the President (RRP), changes in implementation arrangements are subject to agreement and approval pursuant to relevant Government and ADB administrative procedures (including the Project Administration Instructions) and upon such approval they will be subsequently incorporated in the FAM.

ABBREVIATIONS

ADB	=	Asian Development Bank
ACIWRM	=	Advanced Centre for Integrated Water Resources Management
CADA	=	command area development authorities
CNNL	=	Cauvery Neeravari Nigam Limited
CSO	=	civil society organization
EARF	=	environmental assessment and review framework
FAM	=	facility administration manual
FFA	=	framework financing agreement
ha	=	hectare
IEE	=	initial environmental examination
IWRM	=	integrated water resources management
KISWRMIP	=	Karnataka Integrated and Sustainable Water Resources Management Investment Program
KNNL	=	Karnataka Neeravari Nigam Limited
LIBOR	=	London interbank offered rate
MFF	=	Multitranchise Financing Facility
MIS	=	management information system
PD	=	Program Director
PIO	=	project implementation office
PIU	=	project implementation unit
PMU	=	program management unit
PSC	=	project support consultants
SOE	=	statement of expenditure
SPS	=	Safeguard Policy Statement
SSS	=	single source selection
TLBC	=	Tungabhadra Left Bank Canal
VNC	=	Vijayanagara Channels
WRA	=	Water Resources Authority
WRD	=	Water Resources Department
WUCS	=	water user cooperative societies

I. INVESTMENT PROGRAM DESCRIPTION

1. The State of Karnataka, in the south of India, has a population of 61.1 million and covers an area of 191,976 square kilometers (5.83% of the area of India).¹ Although it is the sixth most urbanized state of India and has a rapidly growing economy, about 55% of the total population remains engaged in irrigated agriculture with a rural poverty rate of 26.1%. The State is water stressed with increasing inter-sector water demands. The Karnataka Integrated and Sustainable Water Resources Management Investment Program (KISWRMIP, the investment program) will improve water availability to meet competing water demands in selected river basins by implementing integrated water resources management (IWRM) in the state. It will improve water use efficiency in irrigated agriculture to provide economic opportunities to improve rural incomes.

2. **Impact and Outcome.** The impact of the investment program is improved sustainable water security in selected river basins in Karnataka. The expected outcome will be IWRM successfully implemented in selected river basins in Karnataka.²

3. **Outputs.** The investment program will support three outputs: (i) state and basin institutions strengthened for IWRM; (ii) irrigation system infrastructure and management modernized; and (iii) program management systems operational.

4. **Output 1: State and basin institutions strengthened for IWRM.** The output will strengthen institutional capacities of the Water Resources Department (WRD) and the Advanced Centre for IWRM (ACIWRM)³ to implement IWRM in selected river basins in Karnataka. This will include: (i) development and implementation of river basin plans to better manage water resources; (ii) capacity building and certification⁴ of the WRD (including its associated agencies such as Karnataka Neeravari Nigam Limited [KNNL]) in IWRM with at least 30% of trainees being women; and (iii) improved water resources information systems. The activities to be undertaken are detailed in the IWRM road map in Attachment 1 and the design and monitoring framework (DMF) for the program in Attachment 2.

5. The indicators for this output are, by 2024: (i) at least three river basin plans are developed and implemented; (ii) at least 600 WRD staff (of which 200 are women staff) are certified in IWRM; and (iii) state water resources information systems established. The main activities to be undertaken will include: (i) appointment of staff for ACIWRM; (ii) Institutional/capacity assessment (WRD/KNNL/ACIWRM) completed; (iii) staff are appointed for ACIWRM; (iv) IWRM certification program in implementation; (v) river basin resources inventories prepared; (vi) state IWRM strategy endorsed by IWRM steering committee; (vii) water resources system established and under use for river basin planning; (viii) pilot projects completed for land and water management, remote sensing of water consumption, water use efficiency and water productivity; and river basin plans prepared and under implementation in three sub-basins.

6. The output is being implemented by ACIWRM. Specialist support in IWRM is being provided by international and national consulting services who are recruited as individual consultants to provide specific expertise in topics like river basin planning, basin modeling, communications etc.

¹ Government of India, Ministry of Home Affairs, Office of the Registrar General and Census Commissioner. Census 2011. <http://www.indiaonlinepages.com/population/karnataka-population.html> (accessed 9 June 2012).

² The design and monitoring framework for the investment program is in Attachment 2.

³ The State Government of Karnataka established the ACIWRM through cabinet endorsement on 4 January 2012 and registration under the Societies Act on 10 Feb 2012 with its rules and regulations.

⁴ The ACIWRM will be accredited by UNESCO-IHE as an institute for professional training in IWRM.

7. **Output 2: Irrigation system infrastructure and management modernized.** This output will include: (i) modernization of irrigation infrastructure through three irrigation subprojects within the selected Krishna-8 (K-8) sub-basin (Gondi subproject under Project 1, and Vijayanagara Channels (VNC) subproject under Project -2, and Tungabhadra Left Bank Canal (TBLC) subproject under Project 3); (ii) strengthening asset management and main system operation and maintenance (O&M); and (iii) capacity building of inclusive water user cooperative societies (WUCS), with women representation, for improved operation, maintenance and water management within their respective command area. Project 2 also includes strengthening agricultural extension (demonstrations and training on improved/modern on-farm water management).

8. The indicators for the output are, by 2024: (i) 300 km of irrigation canal upgraded; (ii) irrigation services improved over 50,000 ha of land; (iii) asset management plans implemented for main system infrastructure on 3 selected systems; (iv) more than 82 of 110 WUCS operate and maintain transferred infrastructure, including fee collection; and (v) at least 30% of WUCS management boards are women to engage in the decision-making process on water management. Activities to be undertaken will include: (i) the identification of locations and installation of telemetry equipment for flow measurements within the sub-basin; (ii) preparation of feasibility studies and detailed project reports for the identified irrigation subprojects; (iii) implementation of civil works at Gondi, VNC and TBLC irrigation systems; (iv) preparation of main system asset management and operation and maintenance plan; (v) capacity building of KNNL and WUCS in improved system management.

9. The output is being implemented by KNNL and project implementation offices (PIO) which have been established in the corresponding irrigation zone for each subproject. The command area development authority (CADA) is also a part of the PIO providing specific technical support for WUCS strengthening. Specialist inputs are being provided by international and national program support consultancy (PSC) which has been recruited to provide specific expertise in irrigation systems, environmental management, social and gender development, etc.

10. **Output 3: Program management systems operational.** This output includes: (i) the establishment of a program management unit (PMU) within KNNL to supervise program implementation; (ii) provision of technical expertise from project support consultants (PSC); and (iii) the establishment of project monitoring and evaluation systems for project implementation activities. The indicator for this output is that the project outputs will be delivered on time and within budget. The main activities to be undertaken include: (i) a fully staffed PMU and project implementation offices with an appointed program director by loan effectiveness; (ii) procurement of required equipment; (iii) PSC recruited within 6 months of loan effectiveness and monitored thereafter; (iv) program management information systems database is operational by July 2017; (v) annual disbursement projections are met; and (vi) midterm and program completion effects monitoring is undertaken.

11. Project management support (including office equipment and operational costs) is being provided to the PMU and ACIWRM, and PIO is located according to the individual irrigation subprojects to be modernized. The PMU is located within KNNL and Managing Director, KNNL is the program director (PD). The PMU staff will be based in the PD office in Bangalore. The PMU comprises the following units: (i) monitoring and communication; and (ii) budget and administration; and (iii) irrigation management. The PMU is being supported by PSC services. Staffing requirements for PMU are presented below:

Table 1. Program Management Staffing Requirements

Staff Position	Input Person-Months (p-m)
Program Director	120.0
Monitoring and Communication Cell	
Program Officer (management and geographic information systems)	120.0
Communication Expert	120.0
Budget and Administration Cell	
Program Officer Procurement	120.0
Program Officer Budget and Accounts	120.0
Irrigation Management Cell	
Executive Engineer, Irrigation	120.0
Assistant Executive Engineer, Irrigation	120.0
Assistant Engineer, Irrigation	120.0
Total	960.0

A. Special Features

12. The investment program finances key activities in the state's IWRM road map (Attachment 1). It will introduce an IWRM approach to effective irrigation service delivery to enhance efficiency, productivity, and sustainability in water use by implementing key policy principles, furthering reforms for improved sector governance, and by incorporating the following lessons:

Table 2: Value Addition to Program Outputs

IWRM	The investment program will implement the key principles of the state's "Vision for Development—2020" (State Planning Board, 2009) emphasizes: (i) greater visibility of agriculture and allied activities to increase rural incomes; and (ii) achieving a sustainable and orderly process of industrialization and urbanization for poverty reduction and sustained human development. Also, the State's Water Policy (2002), the National Water Mission, 2011 (Ministry of Water Resources), National Water Policy, 2012 (Ministry of Water Resources) and the 12 th Five Year Plan 2012–2017 (Planning Commission). This will be in the context of making river basin management fully operational to ensure integrated water resource management helping to conserve water, minimize wastage and ensure more equitable distribution both across and within States, and integrated water policies to cope with variability in rainfall and river flow at the basin level with associated policy enhancements owned by the state.
	Institutional strengthening: The State has established the Advanced Centre for IWRM (ACIWRM) - a think tank to provide best practice, knowledge and guidance on water resources management at the state level. The Program will continue to support ACIWRM. A sub basin level institution will be established for long term exemplary river basin management.
	IHE Delft Institute for Water Education (IHE) will contribute training peer-review, advisors, and possibly suggestions for improvement at the end of the first cycle.
	International best practice: The investment program builds on regional and global best practices in IWRM. Specifically focusing on short and medium terms goals in the context of IWRM road map implementation specifically, monitoring and evaluation, sub- basin planning, performance benchmarking and policy analysis for improving resource management.

	Sub-basin planning: The investment program will undertake a sub-basin planning approach considering modeling of the overall water resources system rather than piecemeal and irrigation centric approach. Other water users, upstream reservoir operation and existing availability will be incorporated in the planning and design process. This is a more advanced approach to conventional irrigation system design where the sub-basin interconnectivity of water uses, and water users are not incorporated into the design process.
	Decision support systems: comprehensive measurement (using state of the art technology and automated systems) and system benchmarking will be established to determine baseline situation and for real time decision making on system operation.
Irrigation Service Delivery Improvement	Components are designed consistently with step-by-step implementation procedures to establish participatory irrigation management (PIM)-based productive agriculture, incorporating best practices collected through national and international studies. The program has a specific focus on extending field channel networks and promoting conjunctive use of groundwater, as essential elements for improving water-use efficiency, crop intensification, and diversification. It will strengthen WUCS to progressively take on O&M and implementation roles, with their capacity developed to facilitate links in input delivery, extension, and product marketing to achieve productive agriculture.
	Piped distribution network: proposed to provide buried pipe network at the tertiary system to avoid losses as observed in open channels and prevent tampering with irrigation allocations to tertiary units. This is being considered for inclusion under TLBC modernization plan.
	Micro-irrigation systems: providing high efficiency irrigation and improving water use efficiency. This will be introduced on all irrigation subprojects to increase productivity through less water. Where possible and if sufficient hydraulic head is available, they can be gravity-fed systems.
	Online and offline storages: where feasible, these will be provided along the main canal and other suitable locations within the irrigation system. The aim will be to improve water storage during surplus periods and provide during low-flow period.

B. Implementation Duration

13. The investment program will be implemented over a 10-year period, with up to three projects, subject to the government's submission of related periodic financing requests, execution of related loan and project agreements for each project, and fulfillment of terms and conditions and undertakings set forth in the framework financing agreement (FFA). Project 1 is being implemented between 2015 and 2019. The Project 1 loan became effective on 13 July 2015. The activities under Project 1 loan closing is scheduled for 31 December 2019. It is expected that Project 2 loan will be approved and made effective in Q4 2019. The activities under Project 2 are expected to be completed by 30 September 2023, and loan closing is scheduled for 31 March 2024. Project 3 is currently being prepared under Project 1 and is likely expected to commence in 2020 and be completed in 2024. Alternatively, Project 3 may also be processed as a stand-alone project; this will be decided during implementation of Project 2 and prior to processing Project 3.

C. Project 1

14. The Government of India submitted the PFR for Project 1 on 28 July 2014. Project 1 of the investment program will support the initial progress in introducing IWRM by (i) commencing

the strengthening of relevant institutions, specifically the ACIWRM, for advancing policy review, river basin planning, water resources information systems establishment; and (ii) modernization of irrigation system infrastructure and management for the Gondi subproject, including WUCS strengthening. Project 1 will have outputs as described below:

- (i) **Output 1:** State and basin institutions strengthened for IWRM – the output will initiate IWRM activities including operationalizing the ACIWRM (ensuring it is fully staffed); commencing the IWRM certification process which will be undertaken by an accredited agency like IHE, and preparation of the state IWRM strategy which will be approved by the state IWRM steering committee.
- (ii) **Output 2:** Irrigation system infrastructure and management modernized – the output will commence irrigation modernization activities at Gondi irrigation subproject, specifically the implementation of civil works for canal lining, distribution network and command area development. Also, there will be capacity building of 11 WUCS and increasing women representation in the WUCS management boards.
- (iii) **Output 3:** Project and management systems operational – the output will establish operational requirements for project implementation and management. This will specifically include the establishment of PMU and PIO for Gondi irrigation subproject, preparation of annual work plans and activity schedules, and commencement of contract awards and disbursements.

15. The design and monitoring framework for Project 1 is in Attachment 3.

D. Project 2

16. Project 2 of the investment program will support the initial progress in introducing IWRM by (i) continuing IWRM training program, preparing basin plans for Middle Krishna sub-basin (sub-basins K2, K3 and K4) and continuing development of piloting concepts and building of water productivity, water use efficiency, crop and agricultural productivity; and (ii) modernization of irrigation system infrastructure and management for the VNC subproject, including strengthening of WUCS and irrigation extension in the project area. Project 2 will have outputs as described below:

- (i) **Output 1:** State and basin institutions for IWRM strengthened. Project 2 will: (a) prepare the middle Krishna Basin Plan (for sub-basins K2, K3 and K4); (b) expand the state water resources information system; and (c) continue strengthening WRD staff with IWRM training with women comprising at least 30% of participants.
- (ii) **Output 2:** VNC irrigation system infrastructure and management modernized. Project 2 will modernize the VNC subproject that includes repairing 12 anicuts and intake structures and about 442 kilometers of main, distributary and minor canals. The project will establish and strengthen 30 WUCS located within the project area that will then be responsible for improving their farmer irrigation canals. Each WUCS will be governed by a board comprising at least 30% women. Project 2 will also provide KNNL staff and WUCS with gender sensitization workshops, and farmers with extension services to improve on-farm water management and crop production; and will operationalize an asset database and management plan.
- (iii) **Output 3:** Program management systems operational. This output ensures that investment program outputs are delivered on time and within budget. Project 2 will include (a) establishing a fully staffed project implementation office (PIO) at KNNL's Munirabad's office with an appointed project implementation officer;

(b) procuring works and equipment financed under Project 2; (c) coordinating with ACIWRM and managing the project support consultant; (d) coordinating with other agencies responsible for command area development and on-farm improvement activities; (e) expanding the program management information system database that was established under Project 1 and including gender indicators; (f) meeting annual contract award and disbursement projections; (g) conducting periodic safeguard and performance monitoring reviews; (h) preparing the technical, economic and safeguard due diligence for proposed Project 3 and preparing the associated draft periodic financing request; and (i) undertaking advanced procurement actions for proposed Project 3.

17. The design and monitoring framework for Project 2 is in Attachment 4.

II. IMPLEMENTATION PLANS

A. Project Readiness Activities for Project 1

Table 3: Project Readiness Activities for Project 1

Indicative Activities	2013				2014				Responsible Agency
	May	Jun	Nov	Dec	Feb	Jul	Sep	Oct	
Preparation of Detailed Design									EA
Submission of detailed design for CWC approval									EA
Preparation of ICB bidding documents									EA
Preparation of RFP for PSC services									
Establish project implementation arrangements (complete)									EA
Commence bidding process for civil works and request for proposal									EA
Bids received for NCB works									
Loan negotiations									GOI, EA and ADB
ADB Board approval									ADB
Loan signing									GOI and ADB
Government legal opinion provided									GOI
Government budget inclusion (complete)									SGOK
Loan effectiveness									

ADB = Asian Development Bank, CWC = Central Water Commission, EA = executing agency, GOI = Government of India, ICB = international competitive bidding, NCB = national competitive bidding, PSC = program support consultants, RFP = request for proposals, SGOK = State Government of Karnataka.

B. Project Readiness Activities for Project 2

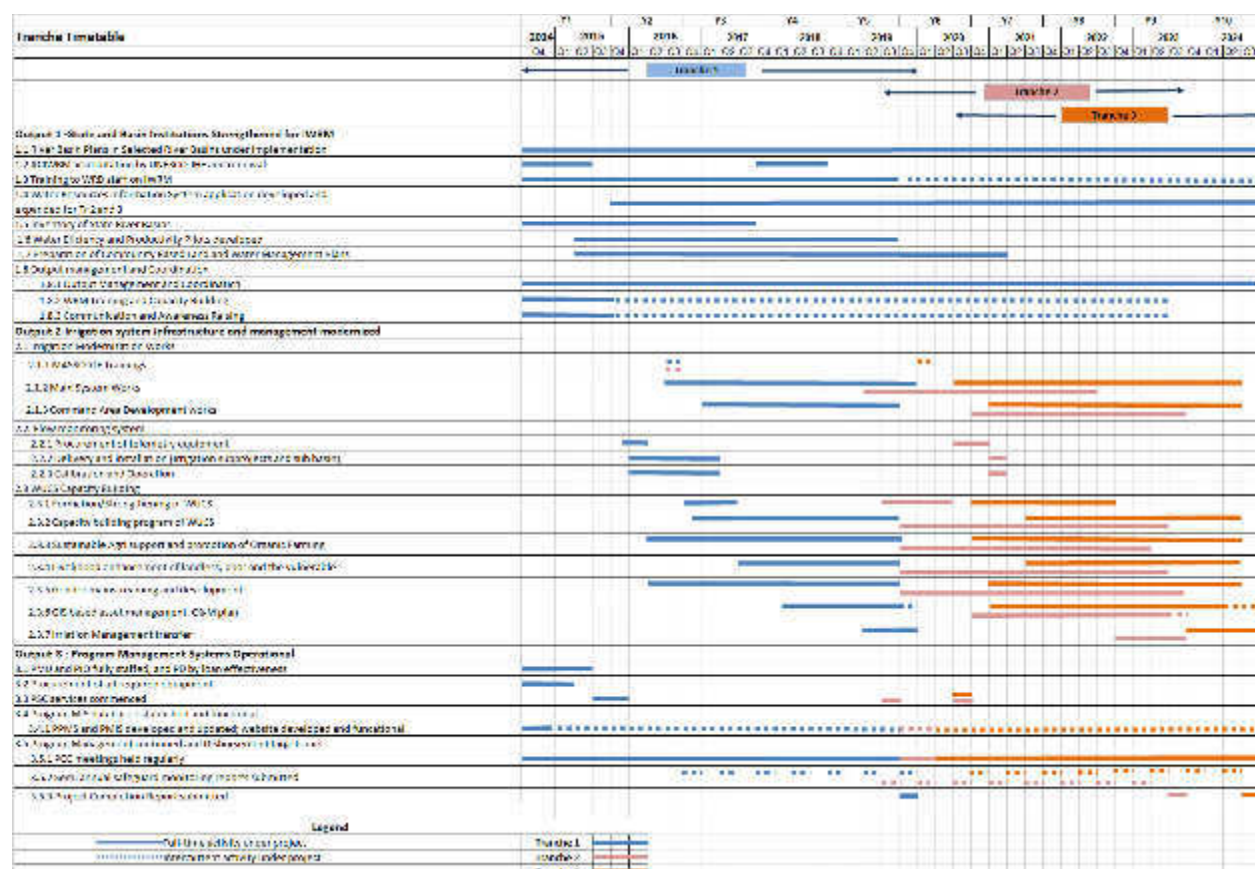
Table 4: Project Readiness Activities for Project 2

Activity	2018	2019									Responsible
	Jun	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	
Establish Project Implementation Office	x										EA
Fact Finding Mission	x										EA, ADB
Award VNC Package 1		x									EA
Mobilize T2 PSC						x					EA
Loan Negotiations							x				DEA, EA, ADB
ADB Management Approval									x		ADB
Loan Signing									x		DEA, EA, ADB
Loan Effectiveness										x	ADB
Government Loan Inclusion					x						SGOK

ADB = Asian Development Bank, DEA = Department of Economic Affairs, EA = executing agency, PSC = program support consultants, SGOK = State Government of Karnataka, VNC = Vijayanagara Channels.

C. Overall Investment Program Implementation Plan

Table 5: Investment Program Implementation Plan



ACIWRM = Advanced Center for Integrated Water Resources Management, IWRM = integrated water resources management, MASSCOTE = Mapping System and Services for Canal Operation Techniques, MIS = management information system, O&M = operations and maintenance, PCC = program coordination committee, PIO = project implementation office, PMIS = program management information system, PMU = program management unit, PPMS = program performance management system, PSC = project support consultant, UNESCO = United Nations Educational, Scientific and Cultural Organization, WRD = Water Resources Department, WRM = water resources management, WUCS = water user cooperative societies

D. Implementation Plan for Project 1 (2014–2018)

Table 6: Implementation Plan for Project 1 (2014–2019)

	Y1				Y2				Y3				Y4				Y5				Y6	
Tranche Timetable	2014	2015				2016				2017				2018				2019				
	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	
										Tranche 1												
Output 1 -State and Basin Institutions Strengthened for IWRM																						
1.1 River Basin Plans in Selected River Basins under implementation																						
1.2 ACIWRM accreditation by UNESCO-IHE and renewal																						
1.3 Training to WRD staff on IWRM																						
1.4 Water Resources Information System application developed and expanded for Tr 2 and 3																						
1.5 Inventory of State River Basins																						
1.6 Water Efficiency and Productivity Pilots developed																						
1.7 Preparation of Community Based Land and Water Management Plans																						
1.8 Output management and Coordination																						
1.8.1 Output Management and Coordination																						
1.8.2 WRM Training and Capacity Building																						
1.8.3 Communication and Awareness Raising																						
Output 2-Irrigation system infrastructure and management modernized																						
2.1 Irrigation Modernization Works																						
2.1.1 MASSCOTE Trainings																						
2.1.2 Main System Works																						
2.1.3 Command Area Development works																						
2.2. Flow monitoring system																						
2.2.1 Procurement of telemetry equipment																						
2.2.2 Delivery and installation (irrigation subprojects and sub basin)																						
2.2.3 Calibration and Operation																						
2.3 WUCS Capacity Building																						
2.3.1 Formation/Strengthening of WUCS																						
2.3.2 Capacity building program of WUCS																						
2.3.3 Sustainable Agri support and promotion of Organic Farming																						
2.3.4 Livelihood enhancement of landless, poor and the vulnerable																						
2.3.5 Gender mainstreaming and development																						
2.3.6 GIS based asset management, O&M plan																						
2.3.7 Irriation Management transfer																						
Output-3 : Program Management Systems Operational																						
3.1 PMU and PIO fully staffed, and PD by loan effectiveness																						
3.2 Procurement of all required equipment																						
3.3 PSC services commenced																						
3.4 Program MIS database established and functional																						
3.4.1 PPMS and PMIS developed and updated; website developed and functional																						
3.5 Program Management continued and Disbursement targets met																						
3.5.1 PCC meetings held regularly																						
3.5.2 Semi annual safeguard monitoring reports submitted																						
3.5.3 Project Completion Report submitted																						
Legend																						
	Full-time activity under project																					
	Intermittent activity under project																					

ACIWRM = Advanced Center for Integrated Water Resources Management, IWRM = integrated water resources management, MASSCOTE = Mapping System and Services for Canal Operation Techniques, MIS = management information system, O&M = operations and maintenance, PCC = program coordination committee, PIO = project implementation office, PMIS = program management information system, PMU = program management unit, PPMS = program performance management system, PSC = project support consultant, UNESCO = United Nations Educational, Scientific and Cultural Organization, WRD = Water Resources Department, WRM = water resources management, WUCS = water user cooperative societies

E. Implementation Plan for Project 2 (2019–2023)

Table 7: Implementation Plan for Project 2 (2019–2023)

Indicative Activities	2019			2020				2021				2022				2023	
	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2
Output 1: State and basin institutions strengthened for integrated water resources management																	
- preparation of K2, K3, K4 basin profiles and plans																	
- IWRM training and capacity building																	
- Expanding Karnataka Water Resources Information System																	
- Preparing TLBC modernization																	
- Expanding inventory of Karnataka's river basins																	
- Water quality data base and assessments																	
- Water productivity and water accounting assessments																	
Output 2: irrigation system infrastructure and management modernized																	
- VNC Package 1 implementation																	
- VNC Package 2 implementation																	
- Establishment strengthening of VNC WUCS																	
- WUCS CPP packages implementation																	
- initiate establishment and strengthening of TLBC WUCS																	
- implement on-farm extension activities in Gondi and VNC command areas																	
- implement gender sensitization workshops																	
- preparation of TLBC feasibility studies, detailed project report, detailed designs and bid documents																	
- complete all works and installation of flow measurement structures, gates and WUCS office buildings																	
Output 3: program management systems operational																	
- PCC meeting held every quarter and minutes included within QPR (PCC: Quarterly)																	
- Quarterly coordination meetings between PMU, PIO, ACIWRM, CADA, PSC, WALMI, WUCS																	
- QPR to be submitted to ADB one month after reporting period																	
- Semi-annual safeguard monitoring reports submitted to ADB																	
- Project completion report submitted to ADB																	
Periodic financing request no.3 and associated due diligence documents submitted to ADB																	
- PPMS and PMIS updated																	
- Quarterly submissions of withdrawal applications to CAAA, one month prior to quarter end																	
- APFS submitted to ADB																	

ACIWRM = Advanced Center for Integrated Water Resources Management, ADB = Asian Development Bank, APFS = audited project financial statements, CADA = Command Area Development Authority, QPR = quarterly progress reports, PCC = program coordination committee, PIO = project implementing office, PMU = program management unit, PSC = program support consultants, TLBC = Tungabhadra Left Bank Canal, VNC = Vijayanagara Channels, WALMI = Water and Land Management Institute, WUCS = water user cooperative society.

Source: Asian Development Bank.

III. PROJECT MANAGEMENT ARRANGEMENTS

A. Project Implementation Organizations – Roles and Responsibilities

Table 8: Project Implementation Organizations – Roles and Responsibilities

Project Implementation Organizations	Management Roles and Responsibilities
State IWRM Steering Committee	State IWRM Steering Committee (IWRM SC) was constituted vide Government Order dated 19 July 2010 to coordinate the activities of key stakeholders associated with IWRM. The IWRM SC serves to: (i) coordinate activities and provide guidance and resolve issues for the investment program; (ii) provide policy direction on matters relating to IWRM in Karnataka; (iii) provide strategic guidance and oversee the coordination IWRM programs in the State; and (iv) ensure coordination on water resource management among State agencies. The IWRM SC comprises: (i) Chief Secretary of the State (chair), (ii) Principal Secretary, Water Resources Department (member secretary), (iii) Principal Secretary, Urban Development Department, (iv) Principal Secretary, Rural Development Department, (v) Principal Secretary, Agriculture Department, (vi) Principal Secretary, Commerce and Industries Department, (vii) Principal Secretary, Forest, Ecology and Environment Department, (viii) Secretary, Minor Irrigation, (ix) Managing Director, Karnataka Urban Infrastructure Development and Finance Corporation, (x) Managing Director, Karnataka Urban Water Supply and Drainage Board, and (xi) Chairman, Bangalore Water Supply and Sanitation Board. The IWRM SC meets as required however, at least twice in a year.
Program Coordination Committee (PCC)	The PCC was established on 27 November 2015 under the chairmanship of Additional Chief Secretary/Principal Secretary, WRD to provide: (i) program oversight, (ii) policy guidance, and (iii) facilitate inter-agency coordination among all relevant departments and agencies related to KISWRMIP activities, support institutionalize effective mechanisms to plan, improve and manage the program initiatives. The members of the PCC include: (i) Secretaries WRD and Minor Irrigation Department, (ii) Managing Directors of Nigams (KNNL, KBJNL, CNNL), (iii) Director ACIWRM, (iv) Director of Water and Land Management Institute, (v) Director Karnataka Engineering Research Stations,

Project Implementation Organizations	Management Roles and Responsibilities
	(vi) Chief Engineers of Water Resources Development Organization, (vii) Principal of the Engineering Staff College, (viii) zonal Chief Engineers where the projects under the program are under implementation; Chief Engineer Command Areas Development Authority (CADA) Directorate, and (ix) relevant CADAs in-charge officers and representative of Agriculture Department. The Member Secretary of the PCC is the Managing Director of KNNL. The PCC meet as and when required, but at least quarterly to (i) review program progress; (ii) approve sub-project selection; and (iii) act on any issues in need of attention at the state level.
Water Resources Department (WRD)	WRD is a major department in the State Government of Karnataka and is headed by the Minister for Major and Medium Irrigation. Irrigation projects are classified under two categories, as indicated below depending upon the extent of culturable command area (CCA). Major Irrigation Projects are those having CCA of 10,000 ha or more and medium subprojects, are of 2,000 to 10, 000 ha. WRD consists of four major corporations: KBJNL (established in 1994), KNNL (established in 1998), CNNL (established in 2003) and Visvesvaraya Jala Nigam Ltd (established in 2017). These corporations are unique establishments of the State of Karnataka as these are envisaged as Special Purpose Vehicles for speedy implementation of irrigation projects and to enable the Government to borrow funds from external sources. In addition to these three corporations, there are some projects which are being directly implemented by WRD. The main three divisions of WRD are the Water Resources Development Organization (which is responsible for investigation and planning of all major and medium irrigation projects in Karnataka and provides information to the State on interstate water disputes), the Command Area Development Authorities (CADAs) and the Nigams. These are described in more detail below.
Executing Agency: Karnataka Neeravari Nigam Limited (KNNL)	The KNNL is responsible for major and medium irrigation in the Upper Tungabhadra Sub-Basin. It was incorporated in 1998 under the Companies Act 1956, as a wholly owned company of the Government of Karnataka to expedite the completion of the on-going irrigation projects in lower Krishna River Basin. The State government has transferred 10 projects to the KNNL for implementation, out of 28 major and medium irrigation projects in the Krishna River Basin. The KNNL provides support for: (i) Collection of irrigation flow data, compilation, processing and hydrological designs; (ii) Planning and investigation of major and medium Irrigation projects; (iii) Construction and maintenance of new and existing major and medium irrigation projects; (iv) Monitoring and evaluation of the irrigation projects; (v) Resolving interstate water dispute problems;

Project Implementation Organizations	Management Roles and Responsibilities
	(vi) Command area development activities; (vii) Research, development and training activities; and (viii) Maintenance and monitoring of machinery and equipment. KNNL's focus is on the main and secondary canals and CADA is responsible for the tertiary irrigation systems. KNNL will be responsible for all aspects of program execution including reviewing policy and overall coordination of all activities. KNNL, as the executing agency, will be responsible (but not limited to) for execution of the entire program, coordination of all activities particularly those requiring other government ministerial actions such as the Planning Commission, Department of Economic Affairs of the Ministry of Finance, Ministry of Water Resources etc., disbursement support such as reimbursement of expenditures etc. It will also be responsible for top-level coordination with ADB and other relevant development partners.
Program Management Unit (PMU)	PMU is located within KNNL and Managing Director, KNNL is the Program Director, or may delegate that responsibility to another senior officer having the rank of Superintending Engineer with ADB's prior agreement (with the Managing Director becoming the Nodal Officer). The PMU staff is based in the PMU office in Bangalore. The PMU will operate under the overall guidance and with the support of the Principal Secretary, WRD, and Secretary, WRD. The PMU comprises of the following units: (i) technical wing; and (ii) financial and establishment wing. The PMU is being supported by Program Support Consulting (PSC) services. WRD should post PMU staff on a full-time basis to the PMU. The PMU is overall responsible for implementing the investment program. The PMU has full authority to execute the investment program and to liaise with ADB. Specifically, the PMU will (i) coordinate with other agencies relevant to the investment program; (ii) prepare an overall implementation plan and annual investment program budget; (iii) review and approve project feasibility studies; (iv) monitor the activities of investment program implementation offices (PIO); (iv) maintain program financial records and accounts and ensure timely submission of required documents; (v) prepare periodic reports on implementation progress; (vi) establish and maintain a program management information system (MIS) database; and (vii) monitor overall investment program progress and assess social and environmental impact and benefits. The PMU, working with the PSC, will overall monitor the execution of the investment program with respect to agreed procedures and compliances. The investment program related services that will be provided by the PMU include: (i) Ensuring that investment program activities are adequately coordinated, and budgets are placed in an appropriate and timely manner; (ii) Ensuring headquarter-level coordination with the activities of KNNL, Command Area Development Authority Directorate, ACIWRM and any complementary government schemes that might be available from departments such as agriculture, horticulture, etc.;

Project Implementation Organizations	Management Roles and Responsibilities
	(iii) Assisting in identifying requisite staff and placing them in the PIO to ensure that a full complement of staff is maintained; (iv) Providing guidance and direction to the PIOs; (v) Conducting regular meetings with PIOs to monitor, coordinate, and guide their activities and to assist with any specific problems that may arise; (vi) Monitoring, through a purpose designed MIS database, the activities of the program. This will involve establishing the data input and retrieval systems and producing outputs for servicing the needs of the State, India, ADB (and other relevant partners), program offices and other concerned stakeholders; (vii) Preparing media information (like leaflets, newsletters etc.) for distribution among the WUCS and other interested people highlighting program related issues, experiences, learning and successes; (viii) Provide State-wide training for WUCS; and (ix) Supporting any other program management related needs of the State, India and ADB. The Program Director will undertake the following responsibilities: (i) oversee the overall implementation of the investment program and each of its Projects; (ii) liaise and correspond with ADB on all issues relating to the investment program and each project; (iii) coordinate with (a) national and state agencies to resolve any interdepartmental issues; and (b) other development agencies for implementation of non-physical activities; (iv) Procure consultancy services (including for ACIWRM in the initial stages of investment program implementation); (v) Oversee and manage the procurement of services, works and goods by the respective implementing agencies; (vi) manage program support consultants, and contractors and suppliers with the support of zonal Chief Engineers and their field staff working under PIOs; (vii) monitor the adequate implementation of • compliance with ADB Safeguard Policy Statement (2009); • resettlement plans (RPs), as may be required; • environment management plans (EMPs); • corrective action plans consistent with safeguard frameworks; • including, the timely review and submission of annual and semiannual safeguards monitoring reports to ADB; • ensure that RPs, Initial Environmental Examinations (IEEs), Environmental Impact Assessments (EIAs), and EMPs are updated by consultants during detailed design; • confirm that EMPs are included in the bidding and contract documents of all civil works packages; and, • the timely disclosure of safeguards documents to affected persons, in an accessible place and in a form and language understandable to them. (viii) maintain investment program accounts and submit consolidated quarterly reports;

Project Implementation Organizations	Management Roles and Responsibilities
	(ix) establish and maintain an investment program performance monitoring system (PPMS) at each subproject level, as well as a project and investment program levels; (x) monitor physical and non-physical investment activities under each project; (xi) obtain necessary data for establishing (baseline), maintaining and updating the PPMS; (xii) prepare and submit: • reports to the Program Steering Committee for consideration and approval; • periodic progress reports on each investment program activity; • Periodic Financing Requests (through DEA) for ADB's consideration; • audit reports; and • reports as mandated under the loan and project agreements. (xiii) consolidate and submit withdrawal applications to ADB in a timely manner; (xiv) secure annual budgetary allocations for the investment program; and (xv) disseminate information related to the investment program and each project to the public and media.
Executing Agency: Advanced Centre for Integrated and Water Resources Management (ACIWRM)	A PIO for Output 1 is established within ACIWRM. The Director of the ACIWRM is the program implementation officer of the PIO. Administrative support is provided by the ACIWRM's administration unit comprising Registrar, Joint Registrar, Assistant Registrar, Section Officers and support staff. Required administrative staff are sourced from WRD and KNNL (or other Nigams as applicable). Consulting services provide technical support to ACIWRM, specifically the Chief Technical Advisor (CTA) who is the IWRM policy specialist. The Director of ACIWRM, will report to the ACIWRM Director General and coordinate with PMU. It will be supported by a core team of consultants who will provide support throughout the activity. The ACIWRM PIO: (i) coordinate preparation of annual work and procurement plans, budgeting, accounting, procurement, annual reporting, monitoring and evaluation of ACIWRM activities in coordination with the PMU; (ii) provide on-the job-training in implementing the project IWRM activities; (iii) organize and facilitate international and national study tours, training courses, work placements, academic fellowships and, a tri-annual international conferences; (iv) provide expert advice on IWRM; (v) undertake pilot activities in IWRM including water use efficiency, policy reviews, conjunctive use of surface, ground and rainwater etc. in accordance with the tasks identified in the roadmap; (vi) coordinate closely with PMU to ensure IWRM concepts are incorporated in design and implementation of field activities; (vii) be responsible for all activities relating to decision support systems (including hydrological measurements, establishment of database, use of data for planning purposes etc.);

Project Implementation Organizations	Management Roles and Responsibilities
	(viii) liaise with relevant state, national, regional and international agencies to share advice/knowledge /technical skills and expertise etc. in IWRM; (ix) be responsible for implementation of the IWRM accreditation program in conjunction with IHE Delft Institute for Water Education; (x) receive and coordinate capacity building training with various agencies including state and national research agencies, international agencies such as eWater, Network of Asian River Basin Organization, etc.; and, (xi) provide expert advice and support to irrigation modernization plans for sub-projects under the investment program and elsewhere.
Project Implementation Offices	For field implementation (Output 2), each irrigation subproject to be modernized will have a full-time dedicated PIO based within the existing irrigation division. The project implementation officer will be of the rank of the Chief Engineer of that irrigation zone reporting to the PD. The project implementation officer will be supported by the zonal Superintending and Executive Engineers. Each field PIO will have a full complement of engineering staffing which will be dedicated full-time for implementation of the investment program's activities. These will include: (i) Assistant Executive Engineers; (ii) Assistant Engineers; and (iii) Junior Engineers and other technical staff. The number of dedicated sub-divisions under the PIO will be determined by the extent of command area and quantum of works to be undertaken. The CADA staff deputed to the PIO will be an Assistant Engineer, Assistant Agriculture Officer and the Senior Inspector of Cooperative Society. The PIO will be provided with WUCS Support Service Teams (SST) to maximize field outreach and supporting guidance from consulting services.^a Under the support and guidance of PMU, provided through regular PMU-PIO meetings, the PIO will: (i) coordinate with the field staff of the concerned line departments, (ii) prepare an annual work plan for approval by PMU, (iii) implement the work plan, and (iv) establish reporting systems to provide information on physical and institutional progress and impacts. The PIO will work closely with WUCS and establish participatory decision-making system through regular meetings at minor canal and on farm levels. The specific tasks of the PIO will include: (i) provide inputs to project planning and design process; (ii) undertake WUCS strengthening and micro-planning for CAD and agriculture development; (iii) implement safeguards actions following the relevant plans; (iv) execute civil works; (v) coordinate for and/or implement support services for agriculture and livelihoods;

Project Implementation Organizations	Management Roles and Responsibilities
	(vi) support project operation, and maintenance in collaboration with WUCSs while ensuring the capacities and resources for the latter; and (vii) arrange training programs for the staff including service providers, and WUCS. The PIO will modernize irrigation systems to maximize irrigation reliability, efficiency and equity in service delivery. The PIO will strengthen WUCS into independent self-sustaining entities capable of fulfilling their responsibilities including irrigation management, equitable distribution of water to farmers, operation and maintenance of minor canal system, collection of irrigation water charges and capable of interacting with service agencies including KNNL/CADA, Agriculture Department, Horticulture Department and other Departments to ensure that they receive necessary services. The goal is a significant improvement in irrigation service delivery through water use efficiency and increased agricultural productivity.
Command Area Development Authority (CADA)	Command Area Development Authorities (CADAs) are a division under WRD. They were constituted in 1980 for providing tertiary irrigation management support for the major irrigation schemes in Karnataka. Within the K-8 sub-basin there are two CADAs operating these for the Bhadra Reservoir Project and the Tungabhadra Project CADA. The CADAs will provide support for: (i) All round development of areas (within the irrigation schemes) for agriculture; (ii) Construction of field irrigation channels; (iii) On-farm development; (iv) Adoption of <i>warabandi</i> (rostering) system for distribution of water; (v) Adoption and enforcement of suitable cropping pattern; (vi) Conjunctive use of surface and ground water; (vii) Reclamation of water-logged areas; (viii) Extension training and demonstration programs; and (ix) Promotion of participatory irrigation management (PIM). The CADAs will provide tertiary and field system support for implementation of the program as part of the PIOs.
Water User Cooperative Societies (WUCS)	WUCS are community-based organizations which are already in existence or to be formed and registered under the Karnataka Cooperative Societies Act, 1959. The WUCS has voluntary membership. The WUCS will be strengthened by the program to undertake their mandated role which is to: (i) assist KNNL and CADA in implementing irrigation and drainage works; (ii) implement or execute on-farm development works; (iii) procure water in bulk on volumetric basis from the Water Resources Department or Krishna Jala Bhagya Nigam or Karnataka Neeravari Nigam and distribute it to the land holders in accordance with the principles laid down by the General Body for equitable distribution of water; (iv) prepare water budget and financial budget for each irrigation season; (v) operate and maintain canals situated within its jurisdiction; (vi) collect water charges and service charges from the land holders; (vii) resolve disputes that may arise among landholders; and

Project Implementation Organizations	Management Roles and Responsibilities
	(viii) educate and train landholders in the efficient and economical use of water and adoption of new technology as well as to implement necessary programs
Consulting Services	Consulting services will provide technical support for program implementation specifically in the context of IWRM which remains an innovative approach in the south Asia context. There will be three main consulting services components: (i) PSC; (ii) IWRM; and (iii) monitoring and evaluation. The services will be recruited by PMU for PSC and monitoring and evaluation consultants and by PIO for IWRM for IWRM services. Each of the consulting services will work in coordination with the PD and/or program implementation officer responsible for the activity.
ADB	ADB will be responsible for periodic review and supervision of investment program implementation through regular loan review missions, midterm and project completion missions, monitoring financial disbursement, and ensuring compliance with loan assurances including institutional arrangements as agreed and stated in the loan agreement.

ACIWRM = Advanced Centre for Integrated Water Resources Management, ADB = Asian Development Bank, CADA = Command Area Development Authority, CNNL = Cauvery Neeravari Nigam Limited, IWRM = integrated water resources management, KBJNL = Krishna Bhagya Jala Nigam Limited, KNNL = Karnataka Neeravari Nigam Limited, PCC = program coordination committee, PIO = project implementing office, PMU = program management unit, PSC = project support consultants, TLBC = Tungabhadra Left Bank Canal, VNC = Vijayanagara Channels, WALMI = Water and Land Management Institute, WUCS = water user cooperative society.

^a The organization arrangement of PMU and PIO may be adjusted in later tranches based on implementation experiences.

B. Key Persons Involved in Implementation

Executing Agency

Karnataka Neeravari Nigam
Limited

Mallikarjuna B. Gunge
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ADB

South Asia Environment, Natural
Resources and Agriculture
Division (SAER)

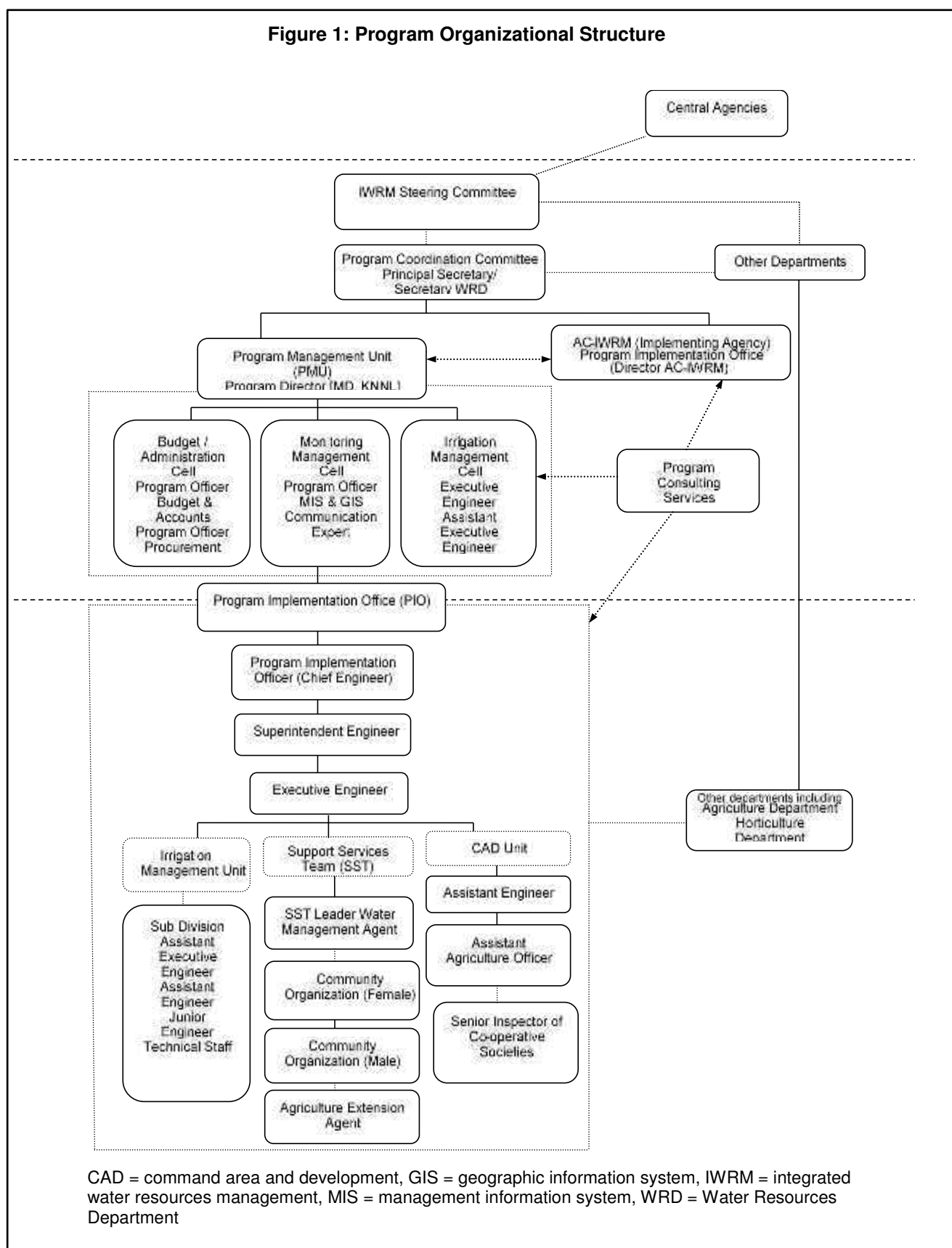
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C. Program Organizational Structure

Figure 1: Program Organizational Structure



IV. COSTS AND FINANCING

18. The investment program at approval is estimated to cost \$225 million inclusive of taxes duties, interest and other charges on the loan during construction. The indicative cost and financing plan for the investment program is in Tables 9 to 12.

Table 9: Investment Program
(\$ million)

Item	Amount ^a
A. Base Cost^b	
Output 1: State and basin institutions strengthened for IWRM	21
Output 2: Irrigation system infrastructure and management modernized	163
Output 3: Program management systems operational	16
Subtotal (A)	200
B. Contingencies^c	20
C. Financing Charges During Implementation^d	5
Total (A+B+C)	225

^a Includes taxes and duties of \$9 million to be financed from government resources.

^b In mid-2013 prices.

^c Physical contingencies computed at 5% for civil works, 10% for consulting services and equipment and, zero for other items. Price contingencies computed at 1.8 to 2.2% on foreign exchange costs and 8.0% to 9.6% on local currency costs using ADB Domestic Cost Escalation Factors (2012-2016) (November 2013); includes provision for potential exchange rate fluctuation under the assumption of a purchasing power parity exchange rate.

^d Includes interest and commitment charges. Interest during construction for ADB loan(s) has been computed at the 6-month LIBOR Floating Rate (ADB, 6 January 2014) 0.3452% plus a spread of 0.5%, plus maturity premium 0.1%, less rebate 0.19%, or 0.7550% overall. Commitment charges for an ADB loan are 0.15% per year to be charged on the undisbursed loan amount. Both paid from government resources.

Source: Asian Development Bank estimates.

19. As originally approved, the investment program will be implemented over a 7-year period, and comprised tentatively of two tranches, subject to the government's submission of related periodic financing requests, execution of the related loan and project agreements for each tranche, and fulfillment of terms and conditions and undertakings set forth in the FFA.⁵ During preparation of Project 2 the government agreed with ADB for the investment program to comprise three tranches by separating activities originally planned under Project 2 into Projects 2 and 3, and to extend the MFF period to 10 years. The government will make the relevant portion of the proceeds of each tranche available to the state, promptly and on terms and conditions acceptable to ADB, for the implementation of the investment program.

⁵ Framework Financing Agreement is in [Appendix 1 of the Report and Recommendation of the President \(MFF 0085-IND\)](#).

A. Detailed Cost Estimates by Expenditure Category for the Investment Program

Table 10: Detailed Cost Estimates by Expenditure Category for the Investment Program

Item	Amount (\$ '000)			%	% Total
	Foreign	Local	Total	Foreign Exchange	Base Costs
A. Investment Costs					
1. Civil Works	-	149,276	149,276	-	75
2. Equipment and Supplies	778	3,033	3,811	11	2
3. Consultancy Services & Training					
a. Training	-	2,662	2,662	-	1
b. Support Services Team for WUCs	-	7,850	7,850	-	4
c. Surveys and Studies					
Effects Monitoring	-	919	919	-	-
Survey, Design and Studies	4,465	5,438	9,902	65	5
Subtotal c. Surveys and Studies	4,465	6,357	10,821	65	5
d. Support Consultants					
International Consultants	897		897	13	-
National Consultants		593	593	-	-
Consultancy Support	85	701	786	1	-
Sub Consultant During Project-1	1,307	3,203	4,510		2
Subtotal d. Support Consultants	2,289	4,497	6,786	34	3
Subtotal 3. Consultancy Services & Training	6,753	21,366	28,120	99	14
4. Staff Costs	-	10,539	10,539	-	5
5. Duties & Taxes	-	7,905	7,905	-	4
Total BASELINE COSTS	6,831	192,819	199,650	100	100
Physical Contingencies	111	8,170	8,281	1	4
Price Contingencies	469	11,681	12,151	4	6
Total PROJECT COSTS	7,411	212,670	220,082	3	110
Interest During Implementation	-	4,374	4,374	-	2
Commitment Charges	-	544	544	-	-
Total Costs to be Financed	7,411	217,589	225,000	3	113

WUCS = water user cooperative society

Source: Asian Development Bank estimates.

B. Detailed Cost Estimates by Financier for the Investment Program

Table 11: Detailed Cost Estimates by Financier for the Investment Program
(\$ '000)

Item	Govt.	%	ADB	%	Beneficiaries	%	Total	%
A. Investment Costs								
1. Civil Works	47,288	27.3	120,992	70.1	4,657	2.6	172,937	76.9
2. Equipment and Supplies	700	20	3,236	80	0	-	3,936	1.7
3. Consultancy Services & Training								
a. Training	317	10	2,849	90	0	-	3,166	1.4
b. Support Services Team for WUCS	449	5	8,532	95	0	-	8,981	4.0
c. Surveys and Studies								
Effects Monitoring	110	10	986	90	0	-	1,096	0.5
Survey, Design and Studies	4,082	38.3	6,586	61.7	0	-	10,668	4.7
Subtotal c. Surveys and Studies	4,192	35.6	7,572	64.4	0	-	11,764	5.2
d. Support Consultants								
International Consultants		-	1,154	100	0	-	1,154	0.5
National Consultants	248	12.5	1,731	87.5	0	-	1,979	0.9
Consultancy Support	71	20.5	335	79.5	0	-	406	0.2
Subtotal d. Support Consultants	677	15	3833	85			4,510	2.0
Subtotal d. Support Consultants	996	48	7053	352	0		8,049	
Subtotal 3. Consultancy Services & Training	5,676	17.8	26,284	82.2	0	-	31,960	14.2
4. Staff Costs	11,249	100	0	-	0	-	11,249	5.0
Total PROJECT COSTS	65,191	29.6	150,235	68.3	4,657	2.1	220,082	97.8
Interest During Implementation			-	-	-	-	4,374	1.9
Commitment Charges			-	-	-	-	544	0.2
Total Disbursement	70,109	31.2	150,235	66.8	4,657	2.1	225,000	100

WUCS = water user cooperative societies.

Source: Asian Development Bank estimates.

C. Detailed Cost Estimates by Outputs/Components for the Investment Program

Table 12: Detailed Cost Estimates by Outputs/Components for the Investment Program (\$ '000)

Item	State & Basin Institutions Strengthened for IWRM	Irrigation System Infrastructure & Management Modernized	Project Management	Total	Physical Contingencies %	Amount
A. Investment Costs						
1. Civil Works	-	155,641	-	155,641	5	7,782
2. Equipment and Supplies	136	3,656	96	3,888	10	388
3. Consultancy Services & Training						
a. Training	2,957	-	-	2,957	-	-
b. Support Services Team for WUCs	-	1,882	6,419	8,301	-	-
c. Surveys and Studies						
Effects Monitoring	-	-	1,021	1,021	-	-
Survey, Design and Studies	9,245	765	-	10,010	-	-
Subtotal c. Surveys and Studies	9,245	765	1,021	11,031	-	-
d. Support Consultants						
International Consultants	186	-	710	896	4.7	103
National Consultants	419	-	1,117	1,536	5.2	182
Consultancy Support	129	-	221	350	-	-
Subtotal d. Consultant During Project-1	1,361	412	2,737	4,510		
Subtotal Support Consultants	2,095	412	4,785	7,292		
Subtotal Consultancy Services & Training	14,297	3,356	11,928	29,581	1	285
4. Staff Costs	6,259	-	4,281	10,539	-	-
Total BASELINE COSTS	20,692	162,654	16,304	199,650	4.2	8,281
Physical Contingencies	14	7,999	295	8,308	-	-
Price Contingencies						
Inflation						
Local	5,291	46,004	5,326	56,620	-	-
Foreign	353	1	115	469	-	-
Subtotal Inflation	5,644	46,005	5,441	57,089	-	-
Devaluation	-4,234	-36,449	-4,256	-44,939	-	-
Subtotal Price Contingencies	1,410	9,586	1,185	12,151	3.8	467
Total PROJECT COSTS	22,115	180,182	17,784	220,082	4	8,748
Taxes	537	7,311	884	8,732	4.2	368
Foreign Exchange	5,678	36	1,698	7,411	1.6	119

WUCS = water user cooperative societies.

Source: Asian Development Bank estimates.

1. Investment Cost and Financing Plan for Project 1

20. The Government of India submitted the first PFR in line with the draft FFA. The cost of Project 1 subprojects is estimated at \$48.0 million equivalent, including taxes and duties of \$1.7 million equivalent and interest and commitment fee of \$1.1 million to be financed by the government.

21. For Project 1, the government requested a loan of \$31.0 million from ADB ordinary capital resources,⁶ with \$16.6 million to be financed by the government and \$0.4 million by beneficiaries which also forms part of the state share of financing.⁷ The loan will have a 25-year term, including a grace period of 5 years, an annual interest rate determined in accordance with ADB's London interbank offered rate (LIBOR)-based lending facility,⁸ a commitment charge of 0.15% per year, and such other terms and conditions set forth in the draft loan and project agreements. The first tranche will be implemented between 2015 and 2019. The financing plan is in Table 13.

Table 13: Investment Plan for Investment Program and Project 1

Source	Investment Program		Project-1	
	Amount (\$ million)	Share of Total (%)	Amount (\$ million)	Share of Total (%)
Asian Development Bank	150	67	31.0	64
Government	70	31	16.6	35
Beneficiaries	5	2	0.4	1
Total	225	100	48.0	100

Source: Asian Development Bank estimates.

23. The cost estimates for Project 1 is in Table 14.

Table 14: Financing Plan for Project 1
(\$ million)

Project/ Component	Counterpart	ADB	Beneficiaries	Total
1. State and basin institutions strengthened for IWRM.	6.4	6.5	0.0	12.9
2. Irrigation system infrastructure and management modernized	7.0	21.2	0.4	28.6
3. Program and management systems operational	2.0	3.3	0.0	5.3
Interest and Commitment Fee	1.2			1.2
Total	16.6	31.0	0.4	48.0

IWRM = integrated water resources management

Source: Asian Development Bank estimates.

⁶ \$5 million was partially cancelled on 6 May 2019 at the request of the government.

⁷ Beneficiaries are WUCS, who will provide upfront contributions for implementations of CAD works. The contributions will be the equivalent of 10% the cost of the CAD works. However, failing such contributions, the government will be responsible for the beneficiaries' portion of financing.

⁸ The interest includes a maturity premium of 10 basis points. This is based on the above loan terms and the government's choice of repayment option and dates.

a. Detailed Cost Estimates by Expenditure Category for Project 1

22. Detailed cost estimates for Project 1 are presented in Tables 15 to 20. These reflect changes to the financing arrangements that were agreed with the government with a loan amendment signed in June 2016.

Table 15: Detailed Cost Estimates by Expenditure Category for Project 1 (US\$ '000)

Item	Foreign	Local	Total	% Foreign	% Base
A. Investment Costs					
1. Civil Works	-	21,628	21,628	-	49
2. Equipment and Supplies	741	2,965	3,706	16	8
3. Consultancy Services & Training					
a. Training	-	1,703	1,703	-	4
b. Support Services Team for WUCs	-	-	0	-	-
c. Surveys and Studies					
i. Effects Monitoring	-	634	634	-	1
ii. Survey, Design and Studies	2,682	3,610	6,292	57	14
Subtotal c. Surveys and Studies	2,682	4,245	6,927	57	16
4. Support Consultants	1,307	3,203	4,510	28	10
B. Staff Costs	-	5,240	5,240	-	12
Total BASELINE COSTS	4,730	38,983	43,713	100	100
C. C. Contingencies					
1. Physical Contingencies	51	1,374	1,425	4	3
2. Price Contingencies	164	1,468	1,632	10	4
Total PROJECT COSTS	4,295	42,476	46,771	9	107
D. Financial Charges During Implementation					
1. Interest During Implementation	-	1,170	1,170	-	2
2. Commitment Charges	-	59	59	-	-
Total Costs to be Financed	4,295	43,708	48,000	9	109

WUCS = water user cooperative societies.

Source: Asian Development Bank estimates.

b. Allocation and Withdrawal of Loan Proceeds for Project 1

Table 16: Allocation and Withdrawal of Loan Proceeds for Project 1

Number	Item	Total Amount Allocated for ADB Financing (\$)		Percentage and Basis for Withdrawal from the Loan Account
		Category	Subcategory	
1	Civil Works	17,828,212		74% total expenditure claimed
2	Equipment & Supplies	2,469,370		82% total expenditure claimed
3	Training	1,741,171		90% total expenditure claimed
5	Surveys and Studies	438,974		
5A	Monitoring and Evaluation		0	90% total expenditure claimed
5B	Survey, Design and Studies		438,974	64% total expenditure claimed
6	Support Consultants	3,522,273		85% total expenditure claimed
7	Unallocated	0		
	Total	26,000,000		

c. Detailed Cost Estimates by Financier for Project 1

Table 17: Detailed Cost Estimates by Financier for Project 1

Item	ADB Amount	%	Govt. Amount	%	Beneficiaries Amount	%	Total Amount	%
A. Investment Costs								
1. Civil Works ^a	15,870	74.6	5,416	25.4	-	-	21,286	44
2. Command area development works (by beneficiaries)					342	100	342	1
3. Equipment and Supplies	3,052	82.4	654	17.6	-	-	3,706	8
4. Consultancy Services & Training	-		-		-		-	
a. Training	1,533	90	170	10	-	-	1,703	4
b. Support Services Team for WUCs								-
c. Surveys and Studies	-		-		-		-	
i. Effects Monitoring	571	90	63	10	-	-	634	1
ii. Survey, Design and Studies	3,997	63.5	2,295	36.5	-	-	6,292	13
Subtotal c. Surveys and Studies	4,569	66	2,358	34	-	-	6,927	14
d. Support Consultants	3,833	85	677	15	-		4,510	9
Subtotal Consultancy Services & Training			2,909	22.1	-	-	13,140	27
B. Staff Costs	-	-	5,240	100	-	-	5,240	11
Total BASELINE COSTS	28,856	66	14,456	33.2	342	0.8	43,654	91
C. Contingencies	2,144	70.2	942	28.9	30	0.9	3,116	6
Total PROJECT COSTS	31,000	66.3	15,398	32.9	372	0.8	46,771	
D. Financial Charges During Implementation								
1. Interest During Implementation			1,170	100	-	-	1,170	2
2. Commitment Charges	-	-	59	100	-	-	59	-
Total Disbursement	31,000	64.6	16,627	34.6	372	0.8	48,000	100

^a Includes irrigation works for main and distributary canal structures and earthworks and command area development works.

WUCS = water user cooperative societies.

Source: Asian Development Bank estimates.

d. Detailed Cost Estimates by Outputs/Components for Project 1

Table 18: Detailed Cost Estimates by Outputs/Components for Project 1

Item	State & Basin Institutions Strengthened for IWRM	Irrigation System Infrastructure & Management Modernized	Project Management	Total	Physical Contingencies %	Amount
A. Investment Costs						
1. Civil Works	-	21,628	-	21,628	5	1,081
2. Equipment and Supplies	136	3,517	53	3,706	10	370
3. Consultancy Services & Training						
a. Training	1,703	-	-	1,703	-	-
b. Support Services Team for WUCS	-				-	-
i. Surveys and Studies						
ii. Effects Monitoring	-	-	634	634	-	-
c. Survey, Design and Studies	5,527	765	-	6,292	-	-
Subtotal c. Surveys and Studies	5,527	765	634	6,927	-	-
d. Support Consultants	1361	412	2737	4510		
Subtotal Consultancy Services & Training	8,592	1,177	3,371	13,140	1.1	148
B. Staff Costs	3,632	-				
Total BASELINE COSTS	12,359	26,322	1,608	5,240	-	-
C. Contingencies			5,033	43,714	3.3	1,425
Price Contingencies						
Physical Contingencies						
Inflation			153	1,425	-	-
Local	1,718	4,418				
Foreign	136	0	857	6,993	-	-
Subtotal Inflation		4,418	28	164	-	-
Devaluation		-3,491	885	7,157	-	-
Subtotal Price Contingencies	497	928	-677	-5,525	-	-
Total PROJECT COSTS	12,869	28,508	208	1,632	3	48
Taxes	313	1,195	5,393	46,771	3.2	1,473
Foreign Exchange	3,478	4	317	1,824	3.8	69
			813	4,295	1.2	53

WUCS = water user cooperative societies.

Source: Asian Development Bank estimates.

e. Detailed Cost Estimates by Year for Project 1

Table 19: Detailed Cost Estimates by Year for Project 1

Item	2016	2017	2018	2019	Total
A. Investment Costs					
1. Civil Works	7,529	9,387	4,528	184	21,628
2. Equipment and Supplies	1,820	900	-	986	3,706
3. Consultancy Services & Training					
a. Training	326	513	473	391	1,703
b. Support Services Team for WUCs					0
c. Surveys and Studies					
i. Effects Monitoring	248	44	299	44	634
ii. Survey, Design and Studies	1,459	1,617	2,003	1,214	6,292
Subtotal c. Surveys and Studies	1,706	1,661	2,047	1,513	6,927
d. Support Consultants	650	1460	1450	950	4510
Subtotal Consultancy Services & Training	2,683	3,634	4,225	2,599	13,140
B. Staff Costs	1,409	1,293	1,269	1,269	5,240
Total BASELINE COSTS	12,318	12,862	12,245	6,049	43,714
Total BASELINE COSTS	434	447	402	133	1,425
C. Contingencies					
Physical Contingencies	121	402	649	437	1,632
Price Contingencies	12,874	13,710	13,296	6,619	46,771
Taxes	531	546	492	243	1,824
Foreign Exchange	1,103	970	1,266	957	4,295

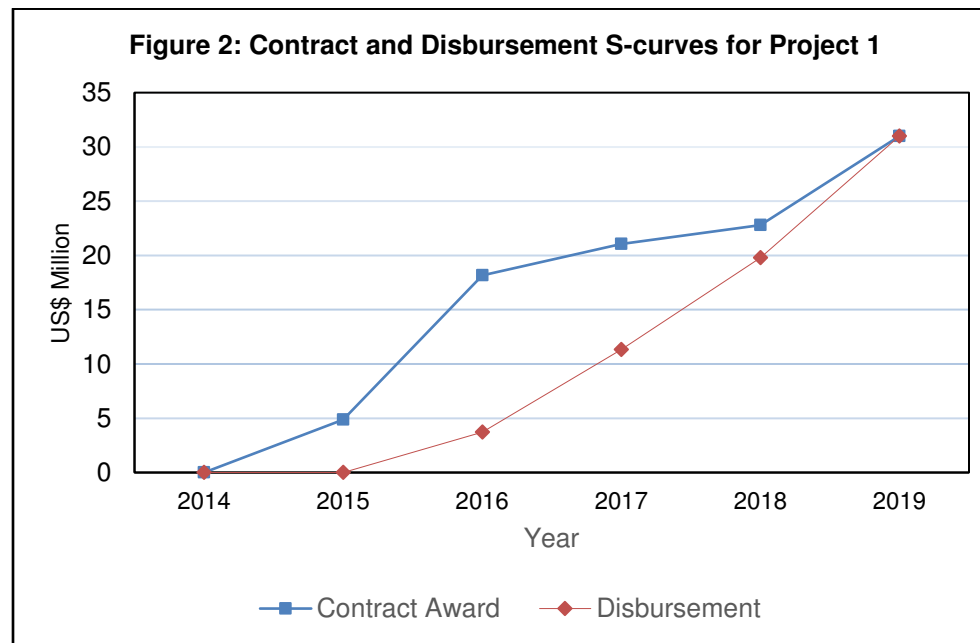
WUCS = water user cooperative societies.

Source: Asian Development Bank estimates.

f. **Contract and Disbursement S-curve for Project 1**

Table 20: Contract and Disbursement S-curve for Project 1

Year	Contract Awards					Disbursements				
	Q1	Q2	Q3	Q4	Total	Q1	Q2	Q3	Q4	Total
2015	0.00	0.00	0.00	4.87	4.87	0.00	0.00	0.00	0.00	0.00
2016	12.75	0.14	0.39	0.01	13.29	0.00	0.17	2.51	1.03	3.71
2017	0.88	0.68	1.19	0.15	2.91	1.40	2.49	2.04	1.69	7.63
2018	0.41	0.27	0.67	0.37	1.72	0.28	2.70	0.96	4.51	8.45
2019	8.22	0.00	0.00	0.00	8.22	11.21	0.00	0.00	0.00	11.21
	Total Contract Awards				31.00	Total Disbursements				31.00



2. Investment Cost and Financing Plan for Project 2

24. The State submitted the second PFR in line with the draft FFA. The cost of Project 2 is estimated at \$130 million equivalent, including taxes and duties of \$11.67 million equivalent and interest and commitment fees of \$6.60 million, both to be financed by the government.

25. The following key assumptions underpin Project 2 cost estimates and financing plan:

- (i) Exchange rate: ₹69.12 = \$1.00 (as of 1 May 2019)
- (ii) Price contingencies based on expected cumulative inflation over the implementation period are shown in Table 21.

Table 21: Escalation Rates for Price Contingency Calculation

Item	2018	2019	2020	2021	2022	2023 and after
Foreign rate of price inflation	1.5%	1.5%	1.5%	1.6%	1.6%	1.6%
Domestic rate of price inflation	3.4%	4.3%	4.6%	5.0%	5.0%	5.0%

Sources: Asian Development Bank estimates based on MUV Index (foreign rate of price inflation) and ADB staff estimates (domestic rate of price inflation).

26. The government has requested a regular loan of \$91 million from ADB's ordinary capital resources to help finance the project. The loan will have a 20-year term, including a grace period of 5 years; an annual interest rate determined in accordance with ADB's London interbank offered rate (LIBOR)-based lending facility; a commitment charge of 0.15% per year (the interest and other charges during construction to be capitalized in the loan); and such other terms and conditions set forth in the draft loan and project agreements. Based on the straight-line method, the average maturity is 12.75 years, and the maturity premium payable to ADB is 0.00% per year.

The financing plan is in Table 22 and detailed cost estimates are shown in Tables 23 to 28.

Table 22: Financing Plan for Project 2

Source	Amount (\$ million)	Share (%)
Asian Development Bank		
Ordinary capital resources (regular loan)	91.00	70
Government	39.00	30
Total	130.00	100

Source: Asian Development Bank estimates.

a. Detailed Cost Estimates by Expenditure Category for Project 2

Table 23: Detailed Cost Estimates by Expenditure Category for Project 2

Item	Amount (\$ million)		Total Cost	% of Total Base Cost
	Foreign Exchange	Local Currency		
A. Investment Costs				
1. Civil works ^a				
a. VNC Package-1	-	49.93	49.93	49.71
b. VNC Package-2	-	11.64	11.64	11.59
c. WUCS Packages	-	16.14	16.14	16.07
2. Mechanical and Equipment ^b	-	4.90	4.90	4.88
3. Consulting services				
a. Program Support Consultants	0.93	6.39	7.32	7.29
b. ACIWRM Support Consultants, Foreign	1.31	-	1.31	1.31
c. ACIWRM Support Consultants, National	-	0.96	0.96	0.96
d. On-farm Extension Services	-	5.91	5.91	5.89
Subtotal (A)	2.25	95.86	98.11	97.68
B. Recurrent Costs				
1. ACIWRM Office Operations ^c	-	0.31	0.31	0.31
2. Training and Workshops	0.22	0.97	1.19	1.18
3. Studies	-	0.83	0.83	0.82
Subtotal (B)	0.22	2.10	2.33	2.32
Total Base Cost (A + B)	2.47	97.97	100.43	100.00
C. Contingencies				
1. Physical	0.25	8.54	8.79	8.75
2. Price	0.13	14.03	14.16	14.10
Subtotal (C)	0.38	22.58	22.95	22.85
D. Interest and Commitment Charges				
1. Interest during implementation	6.29	-	6.29	6.27
2. Commitment charges	0.31	-	0.31	0.30
Subtotal (D)	6.60	-	6.60	6.57
Total Project Cost (A+B+C+D) ^d	9.44	120.54	130.00	129.42

ACIWRM = Advanced Centre for Integrated Water Resources Management, VNC = Vijayanagara Channels, WUCS = water user cooperative societies.

^a Includes EMP implementation costs.

^b Includes but is not limited to hydrometeorological and water quality monitoring equipment, office equipment and vehicles, software, data, and printing.

^c ACIWRM Operation Costs include office rent, vehicle hire, communications, and office and vehicle operation and maintenance costs.

^d Any incidental expenditures relating to bank charges, local transport, freight and insurance are eligible for ADB financing.

Source: Asian Development Bank estimates.

b. Allocation and Withdrawal of Loan Proceeds for Project 2

Table 24: Allocation and Withdrawal of Loan Proceeds for Project 2

Allocation and Withdrawal of ADB Regular Loan Proceeds				
Number	Item	Amount Allocated (\$)		Percentage and Basis for Withdrawal from the Loan Account
		Category	%	
1	Civil works.	56,480,000	73	percent of total expenditure claimed.
2	Mechanical and equipment; consulting services, excluding ACIWRM support consultants, foreign; on-farm extension services; ACIWRM office operations; training and workshops; and studies.	14,530,000	68	
3	ACIWRM support consultants, foreign.	1,310,000	100*	
	Unallocated.	18,680,000		
	Total	91,000,000		

* net of taxes and duties.

c. Detailed Cost Estimates by Financier for Project 2 (\$ million)

Table 25: Detailed Cost Estimates by Financier for Project 2 (\$ million)

Item	Total	ADB Regular Loan		Government ^a			
	Amount	Amount	%	Costs	Taxes & duties ^b	Total	%
A. Investment Costs							
1. Civil works							
1a. VNC Package 1	49.93	36.30	73	8.24	5.38	13.62	27
1b. VNC Package 2	11.64	8.42	73	1.92	1.29	3.21	27
1c. WUCS Packages	16.14	11.75	73	2.66	1.73	4.39	27
2. Mechanical and Equipment ^c	4.90	3.32	68	0.83	0.75	1.58	32
3. Consulting services							
3a. Program Support Consultant	7.33	4.97	68	1.24	1.12	2.36	32
3b. ACIWRM Support Consultants, Foreign	1.31	1.31	100	-	-	-	-
3c. ACIWRM Support Consultants, National	0.96	0.65	68	0.16	0.15	0.31	32
3d. On-farm Extension Services	5.91	4.01	68	1.00	0.90	1.90	32
Subtotal (A)	98.12	70.74	72	16.06	11.32	27.37	28
B. Recurrent Costs							
1. ACIWRM Office Operations ^d	0.31	0.21	68	0.05	0.05	0.10	32
2. Training and Workshops	1.19	0.81	68	0.20	0.18	0.38	32
3. Studies	0.83	0.56	68	0.14	0.13	0.27	32
Subtotal (B)	2.33	1.58	68	0.39	0.35	0.75	32
Total Base Cost (A + B)	100.44	72.32	72	16.45	11.67	28.12	28
C. Contingencies							
1. Physical	8.79	7.16	81	1.63	-	1.63	19
2. Price	14.16	11.52	81	2.65	-	2.65	19
Subtotal (C)	22.95	18.68	81	4.27	-	4.27	19
D. Interest and Commitment Charges							
1. Interest during implementation	6.29	-	-	6.29	-	6.29	100
2. Commitment charges	0.31	-	-	0.31	-	0.31	100
Subtotal (D)	6.60	-	-	6.60	-	6.60	100
Total Project Cost (A+B+C+D)	130.00	91.00	70	27.32	11.67	39.00	30

ACIWRM = Advanced Centre for Integrated Water Resources Management, VNC = Vijayanagara Channels, WUCS = water user cooperative societies.

^a Audit cost is included as part of the government cost; ^b Cash contributions; ^c Includes but is not limited to hydrometeorological and water quality monitoring equipment, office equipment and vehicles, software, data, and printing; ^d ACIWRM Operation Costs includes office rent, vehicle hire, communications, and office and vehicle operation and maintenance costs.

Source: Asian Development Bank estimates.

d. Detailed Cost Estimates by Component for Project 2

Table 26: Detailed Cost Estimates by Component for Project 2

Item	Cost \$ million	1. State and basin institutions for IWRM strengthened		2. VNC irrigation system infrastructure and management modernized		3. Program management systems operational	
		\$ million	%	\$ million	%	\$ million	%
A. Investment Costs							
1. Civil works							
1a. VNC Package-1	49.93	-	-	49.93	100	-	-
1b. VNC Package-2	11.64	-	-	11.64	100	-	-
1c. WUCS Packages	16.14	-	-	16.14	100	-	-
2. Mechanical and Equipment	4.90	3.96	81	0.94	19	-	-
3. Consulting services							
3a. Program Support Consultant	7.32	-	-	0.75	10	6.57	90
3b. ACIWRM Support Consultants, Foreign	1.31	1.31	100	-	-	-	-
3c. ACIWRM Support Consultants, National	0.96	0.96	100	-	-	-	-
3d. On-farm Extension Services	5.91	-	-	5.91	100	-	-
Subtotal (A)	98.11	6.23	6	85.30	87	6.57	7
B. Recurrent Costs							
1. ACIWRM Office Operations ^a	0.31	0.31	100	-	-	-	-
2. Training and Workshops	1.19	1.19	100	-	-	-	-
3. Studies	0.83	0.83	100	-	-	-	-
Subtotal (B)	2.33	2.33	100	-	-	-	-
Total Base Cost (A + B)	100.43	8.56	9	85.30	85	6.57	7
C. Contingencies							
1. Physical	8.79	0.73	8	7.50	85	0.56	6
2. Price	14.16	0.89	6	12.48	88	0.79	6
Subtotal (C)	22.95	1.63	7	19.98	87	1.35	6
D. Interest and Commitment Charges							
1. Interest during implementation	6.29	0.54	9	5.34	85	0.41	7
2. Commitment charges	0.31	0.03	9	0.26	85	0.02	7
Subtotal (D)	6.60	0.56	9	5.60	85	0.43	7
Total Project Cost (A+B+C+D)	130.00	10.75	8	110.89	85	8.35	6

ACIWRM = Advanced Centre for Integrated Water Resources Management, VNC = Vijayanagara Channels, WUCS = water user cooperative societies.

Source: Asian Development Bank estimates.

e. Detailed Cost Estimates by Year for Project 2

Table 27: Detailed Cost Estimates by Year for Project 2

Item	(\$ million)					
	Total	2019	2020	2021	2022	2023
A. Investment Costs						
1. Civil works						
1a. VNC Package-1	49.93	7.98	10.34	10.02	14.54	7.04
1b. VNC Package-2	11.64	-	3.05	2.96	2.86	2.77
1c. WUCS Packages	16.14	-	3.39	4.92	4.76	3.07
2. Mechanical and Equipment	4.90	1.28	1.24	1.21	1.17	-
3. Consulting services						
3a. Program Support Consultant	7.32	1.13	1.52	1.47	2.13	1.07
3b. ACIWRM Support Consultants, Foreign	1.31	0.21	0.27	0.26	0.38	0.19
3c. ACIWRM Support Consultants, National	0.96	0.15	0.20	0.19	0.28	0.14
3d. On-farm Extension Services	5.91	0.95	1.23	1.19	1.72	0.83
Subtotal (A)	98.11	11.70	21.24	22.22	27.85	15.10
B. Recurrent Costs						
1. ACIWRM Office Operation Costs ^a	0.31	0.05	0.06	0.06	0.09	0.04
2. Training and Workshops	1.19	0.19	0.25	0.24	0.35	0.17
3. Studies	0.83	0.13	0.17	0.17	0.24	0.12
Subtotal (B)	2.33	0.37	0.48	0.47	0.68	0.33
Total Base Cost	100.43	12.07	21.72	22.68	28.53	15.43
C. Contingencies						
1. Physical	8.79	1.05	1.90	1.99	2.50	1.35
2. Price	14.16	0.44	1.76	2.97	5.26	3.74
Subtotal (C)	22.95	1.49	3.65	4.95	7.76	5.09
D. Interest and Commitment Charges						
1. Interest during implementation	6.29	0.08	0.59	1.18	1.90	2.55
2. Commitment charges	0.31	0.06	0.11	0.08	0.04	0.01
Subtotal (D)	6.60	0.14	0.69	1.26	1.95	2.56
Total Project Cost (A+B+C+D)	130.00	13.71	26.07	28.90	38.24	23.07
%Total Project Cost	100	10.55	20.05	22.23	29.42	17.75

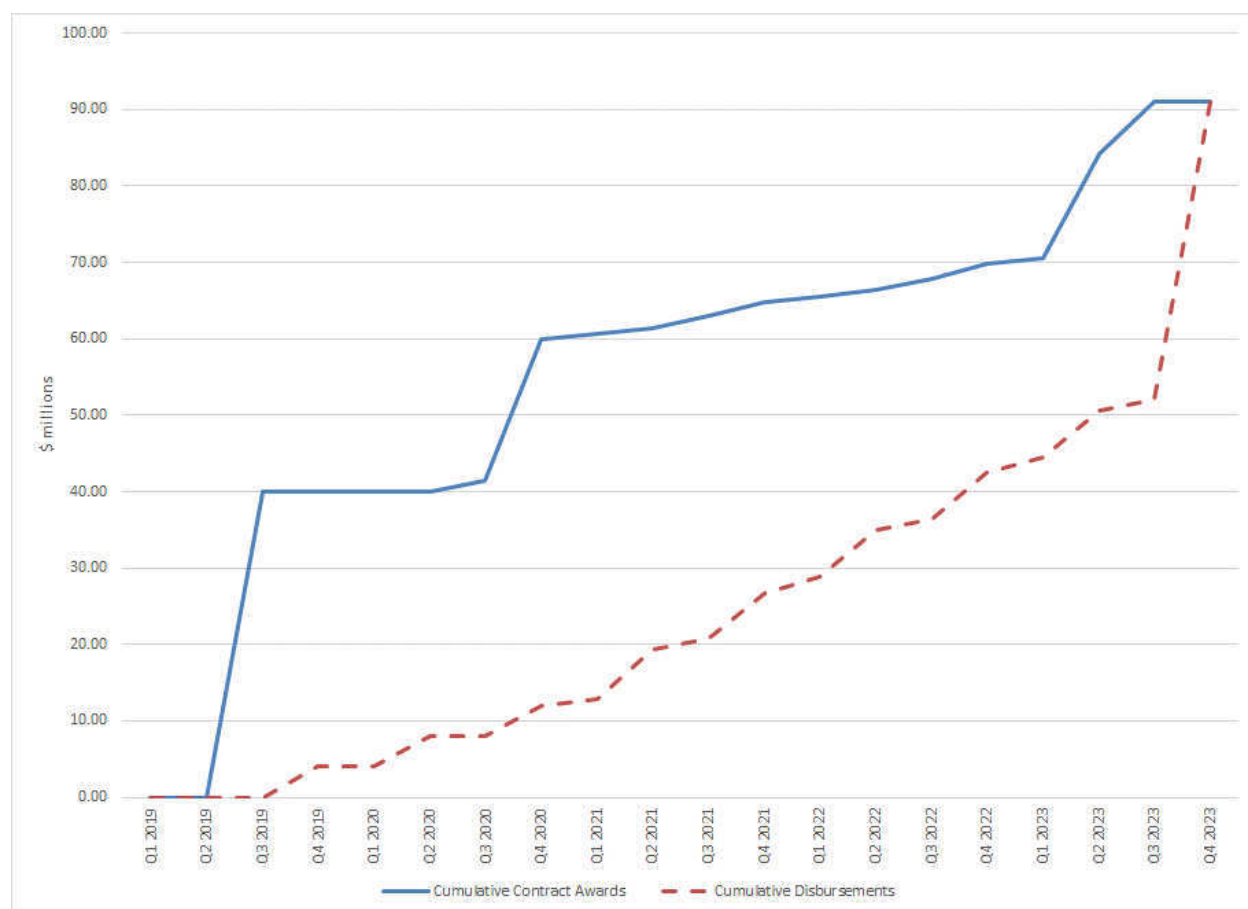
ACIWRM = Advanced Centre for Integrated Water Resources Management, VNC = Vijayanagara Channels, WUCS = water user cooperative societies.
Source: Asian Development Bank estimates.

f. **Contract and Disbursement S-curve for Project 2**

Table 28: Projected Contract and Disbursement for Project 2

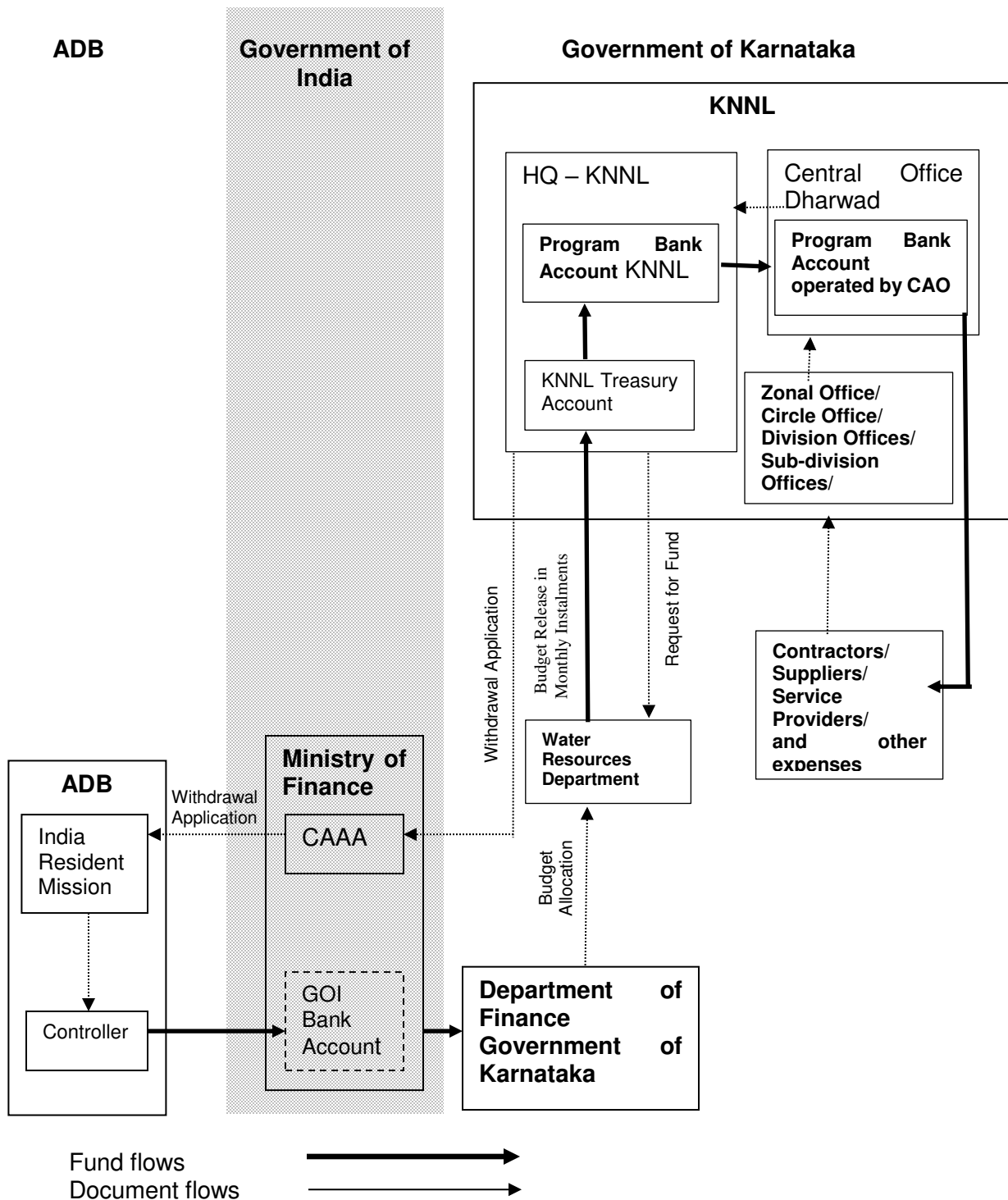
Year	Contract Awards					Disbursements				
	Q1	Q2	Q3	Q4	Total	Q1	Q2	Q3	Q4	Total
2019	-	-	40.00	-	40.00	-	-	-	4.00	4.00
2020	-	-	1.48	18.48	19.96	-	4.00	-	4.00	8.00
2021	0.73	0.73	1.48	1.96	4.91	0.85	6.44	1.39	6.09	14.78
2022	0.73	0.73	1.48	1.96	4.91	2.09	6.09	1.39	6.09	15.67
2023	0.73	13.67	6.82	-	21.22	2.09	6.09	1.39	38.97	48.55
Total	2.20	15.13	51.27	22.40	91.00	5.03	22.63	4.18	59.16	91.00

Figure 3: Contract and Disbursement S-curve for Project 2



2. Investment Program Fund Flow Diagram for KNNL

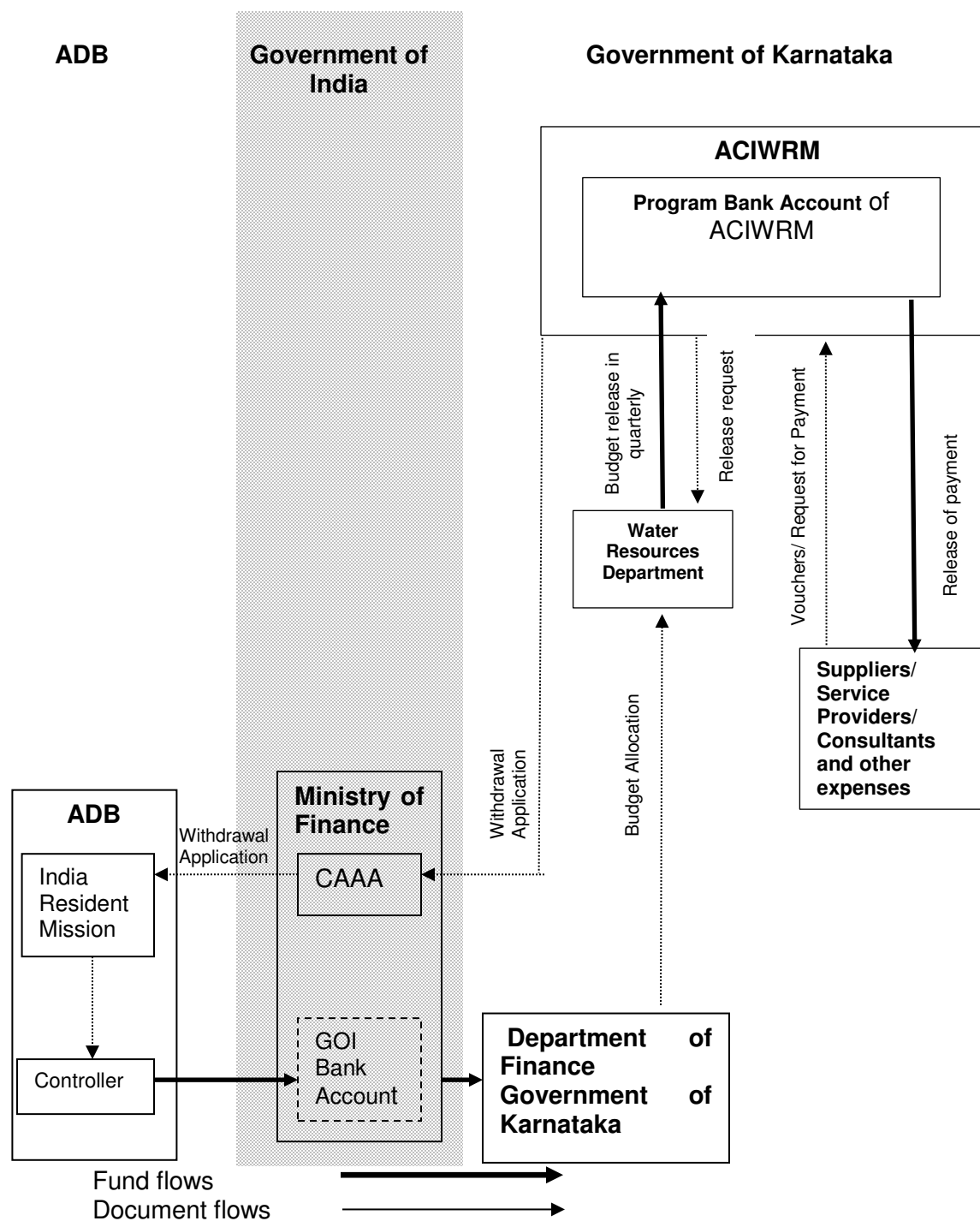
Figure 4: Investment Program Fund Flow Diagram for KNNL



ADB = Asian Development Bank, CAAA = Controller of Aid Accounts and Audit, CAO = Chief Accounts Officer, GOI = Government of India, KNNL = Karnataka Neeravari Nigam Limited

3. Investment Program Fund Flow Diagram for ACIWRM

Figure 5: Investment Program Fund Flow Diagram for ACIWRM



ADB = Asian Development Bank, ACIWRM = Advance Centre for Integrated Water Resources Management, CAAA = Controller of Aid Accounts and Audit, GOI = Government of India

V. FINANCIAL MANAGEMENT

A. Financial Management Assessment

27. A financial management assessment of KNNL and ACIWRM was initially undertaken in 2014 in accordance with ADB's Guidelines for the Financial Management and Analysis of Projects (2005) and Financial Due Diligence: A Methodology Note (2009). The assessment was primarily focused on the financial management capacity of these agencies pertaining to planning, budgeting, fund flow arrangements, accounting and financial reporting, internal controls, audit and asset management. The objective is to assess the overall capacity to utilize funds provided for the intended purposes, and to maintain proper documentation. The findings from this assessment were considered while drawing up the financial management arrangements and oversight for the project.

28. The financial management assessment was updated in June 2018. It was undertaken to ensure that KNNL and ACIWRM are maintaining and developing their institutional and financial management capacities to implement the investment program successfully and effectively operate the facilities built by the investment program. It was also conducted to review fund-flow arrangements, staffing, accounting and financial reporting systems, financial information systems and internal controls, as well as internal and external auditing arrangements.

29. KNNL is a government company of the Government of Karnataka, registered under the Indian Companies Act 1956 on 9 December 1998. It is the investment program's executing agency and is one of two implementing agencies. KNNL has its headquarter office and Chief Administrative/Accounts Office - Central Office at Dharwad. The Project 2 PIO is located at Munirabad with Zonal Office, Circle Office, Division Office and Sub-division Office. ACIWRM was established under WRD by the Government of Karnataka. It was registered under the Society Registration Act in 2012.

30. Both KNNL and ACIWRM have improved in financial management capacity comparing with the FMA done during Project 1 of the investment program. There is improvement in the financial management capacity of both organization with regular funds flow to the project as well as trained staff. Both agencies also have satisfactory financial management capability to (i) record the required financial transactions; (ii) provide regular and reliable financial statements; (iii) provide reliable monitoring reports; and (iv) safeguard the financial assets. Given these capabilities, the implementing agencies have the capacity to handle the proposed project. The minimum required policies and procedures are in place. The statutory audit function of KNNL and Project 1 is currently carried out by chartered accountants empaneled by the Comptroller and Auditor General of India. KNNL and ACIWRM meet ADB's minimum financial management requirements for acting as the executing and implementing agency with the recommended mitigation measures: (i) closely monitor the finance and accounting staff of the Project Implementing Office (PIO); (ii) appoint the vacant position of Deputy Manager Finance in KNNL and post one additional finance staff in ACIWRM; and (iii) address the audit findings on the entity level audit report by KNNL.

31. The overall financial management risk is rated as moderate.⁹ There are no high-risk areas and, overall, KNNL's and ACIWRM's financial management arrangements are considered adequate. Financial management risks will be considered and updated throughout the life of the

⁹ Updated Financial Management Assessment, including an updated financial risk management assessment, is annexed to the Periodic Financing Request Report for Tranche 2.

investment program along with appropriate risk mitigation measures. The financial management and internal control risk assessment is shown in Table 29.

Table 29: Financial Management and Internal Control Risk Assessment

Risk	Risk Assessment	Risk Description	Mitigation Measures or Action Plans
Inherent Risk			
1. Country-specific risks	L	Country specific risk	No country specific risk to this project.
2. Entity-specific Risks	L	Experience of the implementing agency for implementing the project	The implementing agencies are KNNL and ACIWRM, both of which were the implementing agencies responsible for implementing Tranche 1. Most Finance and Accounts Staff in KNNL HQ are permanent staff of KNNL while staff in PIO office Munirabad, CAO central office at Darwad and District Office are on deputation from WRD and some are new. Since base document for the statement of expenditure (SOE) is prepared at the KNNL field level office, it is recommended to give training to all concerned staff once a year to have uniformity and quality in the preparation of SOE and withdrawal applications (WAs). In case of ACIWRM, it is functioning with limited staff in Finance and Accounts.
3. Project-specific Risks	L	There are no project specific risks in this project.	Not applicable.
Overall Inherent Risk	L		
Control Risk			
1. Implementing Entity	L	KNNL and ACIWRM will manage and implement the proposed project	KNNL is a Government Company registered under the Indian Companies Act 1956 in 1998. ACIWRM is a society registered under the Society Registration Act in 2012. Both organizations are under the Water Resources Department (WRD) of the state. Output 1 is being implemented by ACIWRM; Outputs 2 will be implemented by KNNL through its Project Implementation Office (PIO) situated at Munirabad through its Sub-division, and Output 3 is being implemented by KNNL through its Program Management Unit (PMU) located at its Central Office in Dharwad, Bangalore. KNNL is managed by the Managing Director (MD) with support staff and ACIWRM is managed by Registrar with support staff. Output 2 includes works that will be implemented by Water User Cooperative Societies (WUCS). The WUCS will be managed by the Command Area Development Authority (CADA).

Risk	Risk Assessment	Risk Description	Mitigation Measures or Action Plans
2. Fund Flow	L	Timely release of counterpart fund to the project	ADB disbursement will follow reimbursement procedure. Timely availability of counterpart funds will be ensured by the KNNL and ACIWRM implementing agencies through WRD as per the proposed fund flow arrangements. No systematic problems related to funds have been reported in the current projects handled by KNNL and ACIWRM.
3. Staffing	M	Dedicated finance and accounting staff for KNNL and ACIWRM	Full-time finance and accounting staff are now at KNNL HQ and are responsible for ADB related work. Staffing issues, like frequent transfers in WRD, need to be addressed. Staff who are posted for ADB related work should be retained till project end, so that the training imparted will not go to waste. The vacant position of Deputy Manager Finance in KNNL needs to be filled immediately. In the case of ACIWRM, there may be a shortage of Finance and Accounts staff while implementing the proposed project. Currently, there is only one finance and accounts staff. It is recommended to have one more staff to take care of the increase in workload.
4. Accounting Policies and Procedures	L	The accounting policies and procedures and its updating	KNNL follows accrual basis double entry accounting system. Since it is a government-owned company registered under the Indian Companies Act, it has to follow all procedures laid down in the Act. Day-to-day activities are governed by governmental rules and regulations including the Public Works Department Code, Public Works Account Code, Finance Rule and Treasury Code. ACIWRM follows Governmental Accounting System in India on cash basis. The Public Works Department Code, Public Works Account Code, Finance Rule and Treasury Code are followed for the preparation of its Accounts by using "Tally ERP (Enterprise Resource Planning)" software.
5. Internal Audit	M	Arrangements for Internal Audit function	KNNL outsources the internal audit function to a Chartered Accountant who conducts the internal audit on quarterly basis. An audit committee also review the recommendations in the audit report and ensure recommendations are implemented. ACIWRM: There is no internal audit arrangement in ACIWRM. The statutory auditor does the audit quarterly. The payment vouchers are pre-audited by the Finance Staff before ACIWRM makes payments. The ACIWRM is planning to post a separate Internal Auditor to audit the project activities on a quarterly basis. The management committee reviews the audit recommendations for compliance.

Risk	Risk Assessment	Risk Description	Mitigation Measures or Action Plans
6. External Audit	M	The audit of the project financial statements.	The statutory audit function is carried out by a Chartered Accountant selected from the panel approved by the CAG. The statutory auditors prepare separate certification work for the ADB-funded project expenditures. The review of the entity audit reports of the last five years of KNNL are all qualified mainly for non-compliance with some of the Indian Accounting Standards. KNNL need separate attention to rectify the weaknesses pointed out by the auditors.
7. Reporting and Monitoring	M	Financial and disbursement reports should form part of quarterly progress report.	There is room for improvement in the existing reporting systems. Quarterly and annual progress reports should be made mandatory for the proposed project including Physical and financial Reports (with variance analysis) and disbursement reports.
8. Information Systems	L	Optimal use of technology in the finance and accounts wing for the preparation of accounts and reporting requirements.	KNNL is in the process of implementing the Tally ERP accounting software. The finance and accounts wings of KNNL and ACIWRM are presently using the Tally software for accounting and recording in addition to manual recording as per Government Accounting procedures. Reports other than those prepared from Tally are in spread sheets.
Overall Control M Risk			

H = High, S = Substantial, M = Moderate, L = Low

ADB = Asian Development Bank, ACIWRM = Advanced Centre for Integrated Water Resources Management, CADA = Command Area Development Authority, CAG = Comptroller and Auditor General, CAO = Chief Administrative/Accounts Office - Central Office, IA = Implementing Agency, KNNL = Karnataka Neeravari Nigam Limited, PIO = Project Implementing Office, SOE = state-owned enterprise, WRD = Water Resources Department, WUCS = Water User Cooperative Society.

B. Fund Flow Arrangements

32. **Bank Account.** KNNL and ACIWRM have opened separate project bank accounts to receive and disburse funds allocated for the program, to avoid co-mingling with overall funds received by KNNL and to facilitate withdrawals/onward transfers to implementing agencies, chief accounts officer (CAO) and PIO thereof for effecting payments to contractors, service providers, etc. The implementing agencies and CAO-PIO shall open individual program bank accounts to receive funds from KNNL to finance approved sub-projects/ activities only under the Program. The investment program funds will not be transferred to any bank accounts maintained for receipt of other funds and/or be used for temporary bridging of resource gaps for other activities outside the scope of this investment program.

33. **Fund Flow Arrangements.** The fund flow arrangements for the project will be on a reimbursement basis, wherein eligible expenditures (inclusive of ADB's share and counterpart contributions) will be pre-financed by the State from its consolidated account to the entity deposit accounts of KNNL and ACIWRM, respectively. Fund flow from the treasury account to KNNL's and ACIWRM's entity deposit accounts will be either on a quarterly or monthly basis or more frequently based on the expenditures incurred by the program. KNNL and ACIWRM will transfer funds to separate program bank accounts in the respective CAO based on the indent raised for

disbursement against verified and approved bills/invoices, and or other expenditure duly approved by the CAO.

34. On receipt of confirmation of transfer of funds by KNNL and ACIWRM, the CAO shall authorize payments through bank transfers from their respective program bank accounts to the respective contractors/ service providers. The PIO shall submit the utilization certificates for the expenditure duly authorized by CAO to KNNL after completion of the transaction against each bill/invoice certified for payment by the CAO. Similarly, the registrar in ACIWRM shall submit utilization certificates for the expenditure duly authorized by the accounts superintendent/ audit officer to PMU for the latter to consolidate the expenditure claims for submission to ADB for reimbursement through Controller of Aid Accounts and Audit (CAAA), as applicable.

35. Upon receipt of documentation of completion of the financial transaction, KNNL and ACIWRM will submit to CAAA withdrawal application with the statement of expenditures (SOE) eligible for ADB financing under various categories and corresponding percentage of financing as detailed in the Allocation and Withdrawal of Loan Proceeds.

36. The KNNL, as a registered company, and ACIWRM, as a registered society, can make financial transactions with each other. For CADA and other agencies assisting with implementing arrangements, it is required that their staff be deputed under the PIO and payments made through this arrangement.

C. Disbursement Procedures

37. Disbursement of loan proceeds under the investment program will be in accordance with ADB's *Loan Disbursement Handbook* (2017, as amended from time to time)¹⁰ and the detailed arrangements agreed upon between the government and ADB.

38. The State has opted to pre-finance the expenditure under the investment program and seek reimbursement of eligible expenditures from ADB. This arrangement was communicated by the State to the DEA. The disbursement arrangements for the investment program will be on a reimbursement basis.

39. **Disbursement Arrangements.** Under the investment program and tranches, reimbursement procedure (with or without full supporting documents) will be followed where ADB pays from the loan account to the borrower's account; or in some cases, to the project account for eligible expenditures, which have been incurred and paid for by the project out of its budget allocation or its own resources.

40. Required actions prior to submission of withdrawal application include, but are not limited, to the following:¹¹

- (i) Authorized signatory letter from the State authorizing signatories and authentication of their signatures addressed to DEA, CAAA and ADB;
- (ii) Submission of Treasury Account details to DEA and CAAA for transfer of reimbursements from ADB to the State against eligible expenditures under the project.

¹⁰ Available at: <http://www.adb.org/sites/default/files/loan-disbursement-handbook.pdf>

¹¹ Any additional requirements by India/CAAA enforced from time to time need to be complied with by the State and executing agency.

41. **Statement of Expenditure Procedure.** The SOE procedure will be used to reimburse/liquidate eligible expenditures using ADB's Client Portal Disbursement (CPD) at <https://cpd.adb.org/>. All claims beginning 2 February 2018 may use SOE without any limit. Previously, SOE was used only for claims not exceeding \$100,000 equivalent (as stated in ADB Loan Disbursement Handbook, 2017); and for all individual payments exceeding \$100,000 equivalent, documented claim procedure was followed.¹² KNNL and ACIWRM have been provided with CPD user accounts and are encouraged to use the CPD in all withdrawal applications. Manual submission is permitted when there are technical issues or for cases requiring detailed summary sheets and remarks, such as when any refund or recovery on disbursements is necessary. Withdrawal applications for payments made to contractors/consultants in currencies other than Indian Rupees, if any, will be submitted for reimbursement through CAAA to ADB following the documented claim procedure and will be in the currency paid to contractors/consultants. SOE sheets must carry the corresponding Procurement Contract Summary Sheet (PCSS) number provided for each contract under the investment program. SOE records should be maintained and made readily available for review by ADB's disbursement and review missions, or upon ADB's request for submission of supporting documents on a sampling basis, and for independent audit.

42. Before the submission of the first withdrawal application, the state, KNNL or ACIWRM should submit to ADB sufficient evidence of the authority of the person(s) who will sign the withdrawal applications on behalf of the borrower, together with the authenticated specimen signatures of each authorized person.

43. **Minimum value per Withdrawal Application.** The minimum value per withdrawal application is US\$100,000, unless otherwise approved by ADB. KNNL will consolidate expenditure claims of PMU, PIOs, and ACIWRM (as applicable) to meet this limit for reimbursement claims. ACIWRM may directly submit withdrawal applications through CAAA to ADB upon authorization of the state for Registrar and Chief Engineer, ACIWRM to be a signatory to withdrawal applications in addition to Program Director, KNNL. Withdrawal applications, along with the SOEs and supporting documents, will demonstrate, among other things, that the goods and/or services were produced in, or from ADB member countries, and are eligible for ADB financing. Similar eligibility condition will be detailed in the SOE should civil work contracts, if any, are awarded to a foreign contractor/company or joint venture.

44. All disbursements under government financing will be carried out in accordance with regulations of Government of India relevant to co-financing of the projects financed by the multilateral financial organizations.

45. **Asset Inventory Management:** The internal auditor appointed for the investment program will be assigned with the responsibility of providing the assurance and details of the end use of funds for creation of assets, by conducting a physical verification of the assets created before final payments for the same are released to contractors. The internal auditor shall draw the assistance and reports of project supervision consultants (PSC) and PIO engineers on the technical quality, effective use and durability of the assets, and incorporate the findings in its consolidated report. The internal auditor will regularly review the records and documentation maintained by ACIWRM and KNNL and will seek to link these with the physical assets created. The report of the internal auditor will feed into the asset inventory management registry of the management information system (MIS) maintained and operated by KNNL.

¹² All approved supporting invoices, bills, receipts of payment.

D. Finance Accounting and Auditing Requirements

46. KNNL and ACIWRM, through the PMU and PIOs, will:
- (i) ensure that separate books and records by funding source for all expenditure incurred on the project following international accounting standards¹³ and Government of India financial regulations;
 - (ii) separately prepare project financial statements in accordance with the government's accounting law and regulations which are consistent with international accounting principles and practices. Separate project financial statements will be prepared for each tranche;
 - (iii) cause the detailed project financial statements to be audited in accordance with the International Standards of Supreme Audit Institutions (INTOSAI), as appropriate and the guidelines and procedures set out by the CAG of India, by independent auditor acceptable ADB. The audited project financial statements together with the auditor's opinion will be presented in English language to ADB within 6 months of the end of the fiscal year by KNNL and ACIWRM. The terms of Reference (TOR) for Audit of ADB assisted Projects by the Comptroller & Auditor General (C&AG) of India reg. shall be followed;
 - (iv) Ensure that annual contract awards and disbursement projections are prepared and submitted to ADB at least one month before the start of each calendar year; and
 - (v) Ensure that all funds received from the state and ADB are recorded in a transparent manner.
47. The annual audit report for the project financial statements will include a management letter and auditor's opinions, which cover (i) whether the project financial statements present an accurate and fair view or are presented fairly, in all material respects, in accordance with the applicable financial reporting standards; (ii) whether loan proceeds were used only for the purposes of the project; and (iii) whether the recipient or executing agencies were in compliance with the financial covenants contained in the legal agreements for the project. In the absence of a management letter, a letter from the auditor will be submitted certifying that no management letter was provided as there were no issues identified.
48. To ensure the timely submission of audited project financial statements (APFS), KNNL and ACIWRM shall formally request the CAG to include KNNL and ACIWRM project audits in their yearly work plan through the DEA, at the time of loan negotiations. In addition, unaudited project financial statements should be submitted for audit within 3 months of the end of the fiscal year.
49. In addition, audited entity financial statements (AEFS), together with the auditor's report and management letter of KNNL and ACIWRM, will be submitted in English language to ADB within one month of their approval by their respective governing body. Compliance with financial reporting and auditing requirements will be monitored by review missions and during normal program supervision, and followed up regularly with all concerned, including the external auditors.
50. The government, KNNL and ACIWRM have been made aware of ADB's policy on delayed submission, and the requirements for satisfactory and acceptable quality of the audited financial

¹³ Cash Basis International Public Sector Accounting Standards

statements.¹⁴ ADB reserves the right to require a change in the auditor (in a manner consistent with the constitution of the recipient), or for additional support to be provided to the auditor, if the audits required are not conducted in a manner satisfactory to ADB, or if the audits are substantially delayed. ADB reserves the right to verify the project's financial accounts to confirm that the share of ADB's financing is used in accordance with ADB's policies and procedures.

51. Public disclosure of the project financial statements, including the audit report on the project financial statements, will be guided by ADB's Access to Information Policy.¹⁵ After review, ADB will disclose the project financial statements for the project and the opinion of the auditors on the financial statements no later than 14 days of ADB's confirmation of their acceptability by posting them on ADB's website. The management letter, additional auditor's opinions, and audited entity financial statements will not be disclosed.

¹⁴ ADB Policy on delayed submission of audited project financial statements:

- When audited project financial statements are not received by the due date, ADB will write to the executing agency advising that (i) the audit documents are overdue; and (ii) if they are not received within the next six months, requests for new contract awards and disbursement such as new replenishment of Imprest accounts, processing of new reimbursement, and issuance of new commitment letters will not be processed.
- When audited project financial statements have not been received within 6 months after the due date, ADB will withhold processing of requests for new contract awards and disbursement such as new replenishment of Imprest accounts, processing of new reimbursement, and issuance of new commitment letters. ADB will (i) inform the executing agency of ADB's actions; and (ii) advise that the loan may be suspended if the audit documents are not received within the next six months.
- When audited project financial statements have not been received within 12 months after the due date, ADB may suspend the loan.

¹⁵ Available from <http://www.adb.org/documents/pcp-2011?ref=site/disclosure/publications>

VI. PROCUREMENT AND CONSULTING SERVICES

A. Advance Contracting and Retroactive Financing

52. All advance contracting and retroactive financing will be undertaken in conformity with ADB's *Procurement Regulations for ADB Borrowers* (2017, as amended from time to time).¹⁶ The issuance of invitations to bid under advance contracting and retroactive financing will be subject to ADB approval. The government, KNNL and ACIWRM have been advised that approval of advance contracting and retroactive financing does not commit ADB to finance the projects.

53. Advance contracting for consultant and NGO engagement and procurement of goods and civil works, and retroactive financing will be applied for the investment program. These arrangements must be approved in principle for all projects during the processing of individual PFRs. Total eligible expenditure under retroactive financing will not exceed an amount equivalent to 20% of the proposed amount for the respective loans under the Facility, i.e., loan amount for each project, and must have been incurred before effectiveness of the relevant loan, but not earlier than 12 months before the signing of the relevant legal agreements.

54. Under Project 2, advance contracting includes tendering of the VNC Package 1 works package, processing an addendum to the PSC consultant services package that was awarded under Project 1 to continue providing services under Project 2, and shopping for goods. Table 30 shows the status of advance contracting under Project 2.

Table 30: Project 2 Advance Actions

Activity	Responsible	Due date	Comment
VNC Package-1			
Advertise Bid Notice	PMU	5 Feb 2018	Actual
Receipt of bids	PMU	16 Mar 2018	Actual
Technical bid evaluation report submission to ADB	PMU	19 Nov 2018	Actual
Price bid evaluation report submission to ADB	PMU	19 Feb 2019	Actual
Issue of contract acceptance letter / award	PMU	8 Mar 2019	Actual
Commencement date	PMU	June 2019	Actual

ADB = Asian Development Bank, PMU = program management unity, VNC = Vijayanagara Channels.

B. Procurement of Goods, Works and Consulting Services

55. Procurement under Project 1 has been undertaken under the ADB *Procurement Guidelines* (2015). For the subsequent projects, all procurement of goods and works will be undertaken in accordance with ADB's *Procurement Regulations for ADB Borrowers* (2017, as amended from time to time). ADB and the government will review the public procurement laws of the central and provincial governments to ensure consistency with ADB's procurement guidelines as and when necessary.

56. An 18-month procurement plan indicating threshold and review procedures, goods, works, and consulting service contract packages and national competitive bidding guidelines is in Section C.

¹⁶ Procurement under Project 1 follows the ADB Procurement Guidelines (2015), available at <https://www.adb.org/sites/default/files/procurement-guidelines-april-2015.pdf>. Procurement under Project 2 follows the ADB Procurement Regulations for ADB Borrowers (2017), available at: <https://www.adb.org/documents/procurement-regulations-adb-borrowers>

57. Project 2 will also finance additional inputs to the PSC contract that was awarded under Project 1. The additional inputs are to program management, construction quality assurance and WUCS strengthening for Project 2. This will be addressed through a contract amendment to the PSC's existing contract. The updated terms of reference are in Attachment 7.

C. Procurement Plans

58. The procurement plan for Project 1 is in Attachment 8 and for Project 2 is in Attachment 9.

1. Civil Works

59. ADB's prior review and concurrence is required for the first set of bidding documents. However, all bid evaluation reports will require prior review. Once the system is established to attain the conformity with ADB's *Procurement Regulations for ADB Borrowers* (2017, as amended from time to time) and technical standards to be confirmed by ADB, bidding documents are forwarded to ADB for information.

60. For simple civil works of low value costing less than \$100,000, request for quotation (RFQ) procedure may be followed. Small community works (undertaken by the water user cooperative societies or WUCS) associated with the minor canal and CAD works costing \$30,000 or less in value may be directly contracted with the concerned WUCS that include the relevant facilities in its command area and are in charge of their O&M. Procurement through the concerned WUCS through community participation modality (WUCS to execute works through transferred fund from the PIO) will also be followed subject to the acceptable terms and conditions of the state and ADB.

61. For minor works, ADB's prior review and concurrence is required for the initial bid documents and bid evaluation reports during the initial period until ADB confirms that the bid evaluation report (BER) and award of contract will be reviewed on a post-facto basis.

62. Force account will be applicable only for emergent nature of work with the concurrence of KNNL to be met out of the funds of the state and to be accounted for separately under the project.

2. Goods and Related Services

63. In procuring goods and related services for Project 2 onwards, open competitive bidding (OCB) through international advertisement will be used if the estimated cost is at least \$1 million and OCB through national advertisement if the cost is between \$100,000 and \$1 million. Request for quotation (RFQ) procedure may be followed if the estimated contract amount is less than \$100,000 for procuring low-value, readily available off-the-shelf goods.

64. For OCB, ADB's prior review and concurrence is required for the first set of bidding documents. However, all bid evaluation reports will require prior review. Once the system is established to attain the conformity with ADB's *Procurement Regulations for ADB Borrowers* (2017, as amended from time to time) to be confirmed by ADB, bidding documents are to be forwarded to ADB for information. For other types of procurement, the first draft English language version of the procurement documents will be submitted for ADB review and approval. ADB-approved procurement documents should be used as a model for all procurement financed by ADB for the project and need not be subjected to further review unless specified in the procurement plan or required under special arrangements. Likewise, the initial bid evaluation

reports and/or draft negotiated contracts will be submitted to ADB for concurrence. They will be reviewed on a post-facto basis once ADB confirms that prior submission is no longer required.

3. Consulting Services

65. For Project 2 onwards, consulting services will be engaged following the *Procurement Regulations for ADB Borrowers* (2017, as amended from time to time). Under the investment program, they will be selected using quality-based and cost-based selection (QCBS) method, using 90:10 technical to financial weightage.

66. In addition to the above, (international and national) individual specialists will be engaged through individual consultant selection procedure to provide technical expertise for the ACIWRM. For individual engagements, agreements acceptable to the state and ADB will be signed and kept on record as supporting documents of the reimbursement claims.

67. Single source selection can be adopted in case of specialist organizations for action research and training (such as piloting of IWRM activities and demonstration programs). However, the value of such contracts shall generally not exceed US\$100,000. Adequate and acceptable justification will be required for such SSS.

a. E-procurement

68. An eProcurement assessment consistent with requirements in Project Administration Instructions 3.13 was done and adoption of eProcurement under the Project had been confirmed based on system assessment by ADB. KNNL's eProcurement systems are appropriate and consistent with the procurement guidelines and with the principles of economy, efficiency, transparency, and award of contract to the lowest evaluated priced, substantially responsive bid. It may be applied to the OCB, Limited Competitive Bidding and RFQ modes of procurement. The use of eProcurement does not apply to recruitment of consultants.

69. The use of electronic systems does not excuse deviation from any current accepted practices, except those specific to the electronic handling and management of bidding documents and bids. Opportunities may be advertised electronically, bidding documents may be issued electronically, correspondence may be electronic, and bids may be received electronically. All other requirements, such as use of standard bidding documents are retained.¹⁷

D. Contract Management

70. The PMU shall forward to ADB's, South Asia Environment, Natural Resources and Agriculture Division copies of each signed contract with a value of more than US\$100,000. For contracts US\$100,000 or less, the PMU shall forward copies of the signed contract to ADB only if requested.

71. The PIO will obtain necessary clearance for contract variations and report the same to PMU. The PMU shall also obtain ADB's approval for contract variations that are subject to ADB's prior review and with an aggregate increase in the original value of the contract by more than 15% of the original amount. In case of any variation, approval / concurrence of the zonal Chief Engineer will have to be obtained before execution, by the relevant executive engineer of the subproject

¹⁷ More details are in the eGP System Assessment Report (supplementary documents—available upon request).

and inform the PMU on these developments. If variation is more than 15%, PMU shall obtain ADB prior.

72. Monitoring of performance under the various contracts shall be done as per the Management Information System for the investment program, and will include the following:

- (i) Monthly Program Status Report
- (ii) Expenditure Statement
- (iii) Physical and Financial Progress Reports

73. A Register of Contracts shall be maintained at PIO level with each contract identified by a unique number. The Register should record pertinent details including: (i) title & start and end dates; (ii) validity period; (iii) name of contractor / consultant / supplier; (iv) amount; (v) variations if any; (vi) time extension if any. The PMU will monitor quarterly and annually through procurement plan.

74. Contracts will be managed by the concerned executive engineer of the PIO as per provision in the agreement. The relevant state guidelines as amended from time to time will be followed.

75. Payments to contractors shall be made as per the conditions of the contract. However, every month at least one bill may be paid to the agency for civil works. For consulting services, the bills will be raised in the first week of the month for the preceding months.

VII. SAFEGUARDS

A. Key Safeguard Documents and Responsibilities

76. The following safeguard documents were prepared for the investment program as per ADB's *Safeguards Policy Statement, 2009* (SPS) and the government rules:¹⁸

- (i) Environmental assessment and review framework (EARF) was initially prepared for Project 1 in June 2013 and updated for Project 2 in June 2018.
- (ii) Initial environmental examination (IEE) for Project 1 subproject (Gondi) dated June 2013;
- (iii) Environmental Impact Assessment (EIA) for Project 2 VNC subproject dated June 2018;
- (iv) Environmental management plan (EMP) for Gondi subproject dated June 2013;
- (v) EMP for VNC subproject dated June 2018;
- (vi) Resettlement framework (RF) was initially prepared for Project 1 in June 2013 and updated for Project 2 in June 2018; and
- (vii) Social safeguards due diligence report was prepared for Project 2 in June 2018 to confirm the Category C status for involuntary resettlement and indigenous people safeguards.

77. The EARF and the RF serve to guide safeguard assessments and the preparation of relevant safeguard documents in all Projects of the MFF.

78. The safeguard frameworks cover investment program specific information and requirements in accordance with ADB's SPS: (i) the general anticipated impacts of the projects likely to be financed under the Facility on the environment, involuntary resettlement, and indigenous peoples; (ii) the safeguard criteria that are to be used in selecting projects; (iii) the requirements and procedure that will be followed for screening and categorization, impact assessments, development of management plans, public consultation and information disclosure (including the 120-day disclosure rule, if required), and monitoring and reporting; (iv) the institutional arrangements (including budget and capacity requirements) and Republic of India and ADB's responsibilities and authorities for the preparation, review and clearance of safeguard documents.

79. The state acting through KNNL is responsible for the overall compliance of all the requirements prescribed in Schedule 5 of the FFA and the safeguard frameworks that have been prepared with respect to the Facility (and are deemed incorporated herein by reference), during the processing and implementation of the investment program.

80. Pursuant to ADB's SPS, ADB will not reimburse expenditures incurred on activities where the agreed Safeguard policy is not applied and or those activities described in the ADB's Prohibited Investment Activities List set forth in *Appendix 5* of the SPS. All institutions under the investment program will ensure that their investments are in compliance with applicable national laws and regulations and will apply the prohibited investment activities list to subprojects financed by ADB.

¹⁸ Available at: <http://www.adb.org/Documents/Policies/Safeguards/Safeguard-Policy-Statement-June-2009.pdf>

B. Environmental Safeguards

1. Safeguard concerns

81. Output 1 of the investment program focuses on institutional strengthening for IWRM and therefore has no environmental impacts. Under Project 1 of the investment program and as part of Output 2, canal modernization is to be undertaken for the Gondi irrigation subproject and flow measurement devices are to be placed in identified locations in the Bhadra canal system and throughout the K-8 (Tungabhadra) sub basin. Modernization works will comprise channel desilting and reshaping, lining and provision of canal structures. These will take place within the existing canal sections and there is no widening or extension expected.

82. Also, identified are activities to improve water use efficiency and agriculture productivity for Gondi (Project 1), VNC (Project 2) and TLBC (Project 3). This will be through on-farm improvements like command area development, high efficiency (drip) irrigation systems and small on-line and off-line storages (adjacent to irrigation canals). Projects 2 and 3 will comprise of canal modernization works for VNC and TLBC, respectively.

83. An IEE has been prepared for the Gondi irrigation subproject which is categorized as an ADB environment Category B project.¹⁹ A short note has been prepared to guide the activities of Bhadra and remaining sub-basins for installation of telemetry equipment. This is categorized as an ADB environment Category C project, not requiring an IEE or EIA. An EARF has also been developed for the investment program which will guide the program for environment actions for TLBC and VNC and can also guide for any additional proposed activities within the sub-basin which may be included in subsequent Projects of the investment program.

84. According to the existing assessment there are unlikely to be any major adverse environment impacts of the Project 1 subprojects and all identified impacts are likely to be managed through the EMP. Overall the program will have a positive environmental impact due to improved water resources management at the sub-basin level, reduction in habitat of disease vectors, reduced damage to houses and community infrastructure from waterlogging and increased agricultural productivity. These benefits are expected throughout the whole irrigation system. There are also other expected positive impacts and are discussed in greater detail in the project's EARF.

85. For Project 2, the scope of works under VNC trigger ADB's Environmental safeguards policy principles 8 and 11; as such, the project is a category "A". Under the VNC rehabilitation works, nine anicuts extend within the Tungabhadra Otter Conservation Reserve and five canals have been identified within the UNESCO Hampi Heritage Area. An EIA report and EMP were prepared for Project 2 in June 2018.

86. The EIA outlines the potential environmental impacts and mitigation measures for Project 2. Project 2 will not have adverse impacts to wildlife or physical cultural resources, however, increased noise levels and reduction in air quality is anticipated as a result of anicuts and canal rehabilitation. The EMP outlines how the project will manage environmental impacts; particular emphasis is placed on two sensitive environment sites within the project corridor of impact; the Tungabhadra Otter Conservation Reserve and Hampi World Heritage area. Environmental and

¹⁹ Initial Environmental Examination is in [Appendix 11 of the Report and Recommendation of the President \(MFF 0085-IND\)](#).

wildlife clearances are required from the corresponding government departments as a condition for working in the two environmentally sensitive areas of the project.

87. The Tungabhadra Otter Conservation Reserve is a protected otter habitat that includes nine anicuts located within Project 2. Under SPS, the Tungabhadra Otter Conservation Reserve is identified as a “critical habitat”. As such, the VNC physical works cannot enter the Tungabhadra Otter Conservation Reserve unless the Project can assure that adverse impacts to habitats and wildlife will be avoided and necessary environmental and wildlife clearances are sought from the corresponding government department. In May 2018 the PSC completed a rapid biodiversity assessment (RBA) of the Tungabhadra Otter Conservation Reserve. The RBA identified environmental risks and mitigation measures that the PIU will follow to guarantee protection of this critical habitat. Furthermore, KNNL is seeking environmental clearances from the National Board for Wildlife (NBWL) and State Board for Wildlife (SBWL) to initiate works in the nine anicuts. As of June 2018, Range Forest Officers (RFO) and Assistant Conservator of Forest (ACF) had inspected seven of the nine anicuts. The RFO and ACF will complete inspection of the Ramsagar and Kampli anicuts; the SBWL will then make recommendations from the inspection report prior to the approval of the NBWL.

88. There are five canals of the VNC project that pass through the Hampi World Heritage Area. An environmental clearance was sought and issued by the Hampi World Heritage Area Management Authority (HWHAMA) in May 2018. The clearance provides the PIU permission to undertake canal rehabilitation in Basavanna, Raya, Khalghatta, Turtha and Anegundi canals. The HWHAMA has outlined terms and conditions that must be adhered to during the project implementation. These terms and conditions are included within the Project 2 EMP.

C. Social Safeguards

1. Social Safeguards: Involuntary Resettlement

a. Project Scope and Resettlement Impacts

89. An assessment for involuntary resettlement impacts has been made for Gondi, VNC and TLBC irrigation subprojects. Project 1 and Project 2 are categorized as C for involuntary resettlement. This is since: (i) the project involves modernization of an existing canal system within the existing right-of-way; (ii) the WRD owns the right of way of 5 meters alongside the canals for facilitating maintenance works and for access; and (iii) a due diligence of the entire canal system was undertaken to confirm that works under Project 2 will not cause involuntary displacement to any non-titled holders. There will be no land acquisition or other losses resulting in involuntary resettlement under Project 1 and Project 2. A resettlement framework has been prepared for the entire investment program pursuant to SPS that will be followed for the subsequent Projects.²⁰

D. Indigenous Peoples (IP)

90. SPS defines IPs as “a distinct, vulnerable, social and cultural group possessing the following characteristics in varying degrees: self-identification as members of a distinct indigenous cultural group and recognition of this identity by others; collective attachment to geographically distinct habitats or ancestral territories in the project area and to the natural resources in these habitats and territories; customary cultural, economic, social, or political institutions that are

²⁰ Resettlement Framework is in [Appendix 13 of the Report and Recommendation of the President \(MFF 0085-IND\)](#).

separate from those of the dominant society and culture; and a distinct language, often different from the official language of the country or region.”

91. Project 1 and Project 2 are categorized as C for the indigenous people’s safeguard. There are no indigenous peoples as defined by SPS in any of the areas of Project 1, Project 2, and Project 3 of the investment program. As these would also be categorized as C, an Indigenous Peoples Planning Framework was not prepared.

E. Grievance Redress Mechanism

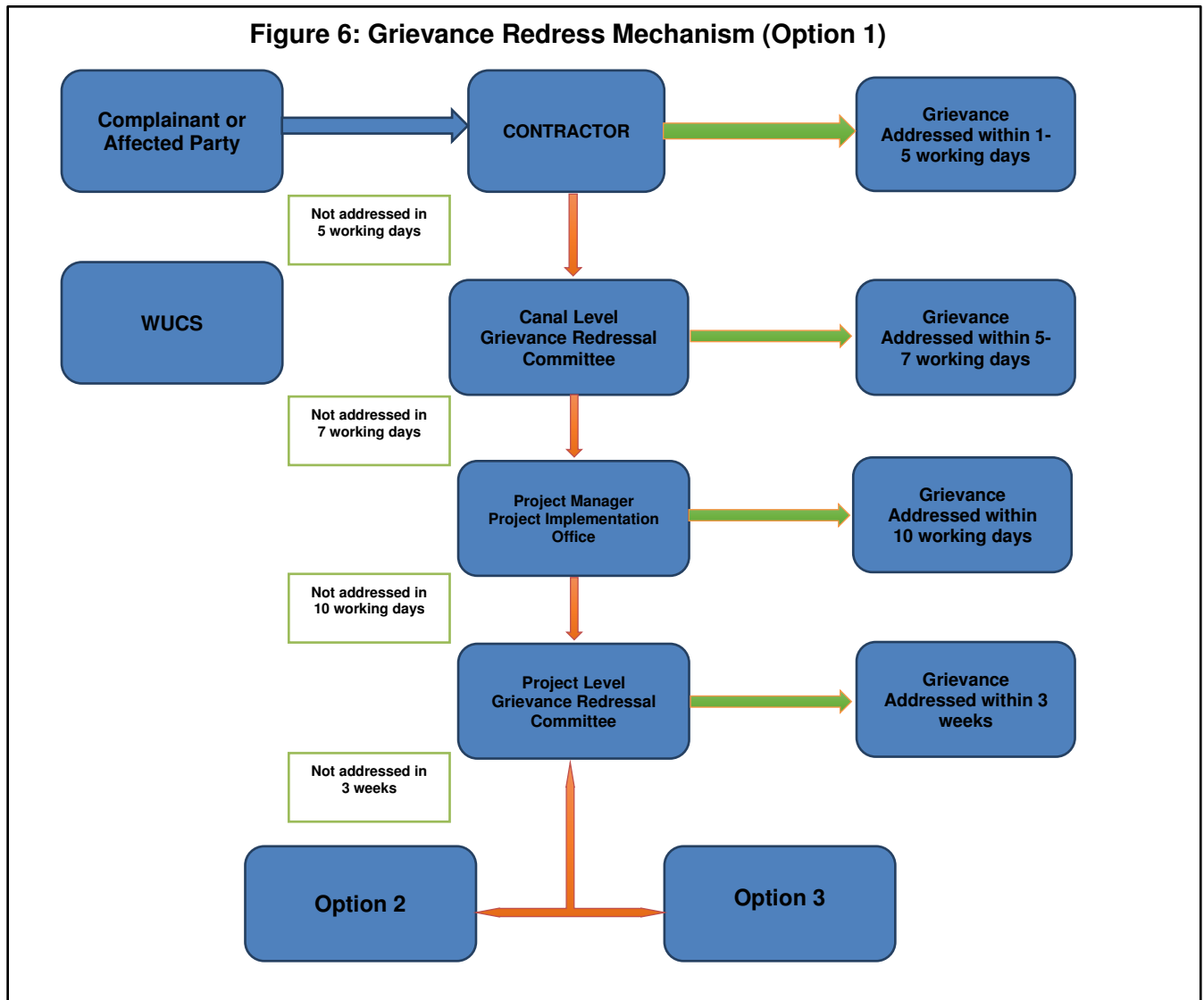
92. The contractor shall outline a detailed procedure for community complaint and grievance redressal mechanism (GRM). Every affected person shall have three options to get the grievance redressed. Option 1 is established under the Project, option 2 is accessing through the country’s legal system, and option 3 is for the affected person to access through the ADB Accountability Mechanism.

93. **Option 1:** Option 1, which is the Project-level GRM shall consist of the following stages:

- (i) **First Stage:** At the first stage, the person with any form of grievance would approach the Contractor (proposed as the agency responsible for implementation of the EMP). A copy of the grievance may also be provided by the affected person to the local WUCS office. The Contractor shall make efforts to resolve the grievance within 1-5 working days at that level in a consultative manner. A Grievance Redressal Register must be maintained by the contractor and WUCS for all such complaints.
- (ii) **Second Stage:** If the affected person is not satisfied or the grievance is not redressed within 5 working days, the contractor will be responsible for assisting the concerned person for getting the grievance registered with the Canal-Level Grievance Redressal Committee (CL-GRC) who shall comprise representatives from the Contractor, WUCS, KNNL and GP. The CL-GRC shall make efforts to resolve the grievance of the complainant within 5-7 working days after the matter is brought to the Committee notice. A Grievance Redressal Register shall be maintained by the contractor for all complaints. The contractor shall share the information on such complaints with the executing agency on a monthly basis. Additionally, the local WUCS office shall be instructed to maintain a Complaint Register.
- (iii) **Third Stage:** If the affected person is not satisfied or the grievance is not redressed within 7 working days, the contractor shall assist the affected person to register the complaint with the project manager at the KNNL PIO. At the third stage, the project manager will ensure that the aggrieved person is heard, and the grievance redressed in the best possible manner in a consultative manner within 10 working days from the date of registering the grievance.
- (iv) **Fourth Stage:** If the affected person is not satisfied or the grievance is not redressed within 10 working days, the project manager will be responsible for getting the grievance registered for the hearing by the Project-Level Grievance Redressal Committee (PL-GRC). The PL-GRC is comprised of the PIO chief engineer as chairman, one member from the Revenue and Agriculture Departments, a representative of KNNL, contractor, panchayat, a representative from the WUCS Federation, and members and representatives of affected persons, including women and vulnerable people. The PL-GRC will conduct a hearing on the grievance within 3 weeks from the date of registration of grievance. Other than disputes relating to ownership rights under the court of law, the PL-GRC will review grievances involving all environmental and social impacts arising from the project implementation. All costs incurred in resolving the complaints will be borne by the project. A comprehensive

record will be maintained by the executing agency for all grievance proceedings organized at different stages and reported within the safeguard monitoring report, submitted to ADB on a bi-annual basis.

94. The flow chart of the option 1 GRM process is provided in Figure 6.



95. **Option 2:** An aggrieved person is free to access the country's legal system and that this is not conditional upon the perceived unsatisfactory outcome of the CL- or PL-GRM.

96. **Option 3:** In the event that the established GRM is not in a position to resolve the issue the affected person can also use the ADB Accountability Mechanism by directly contacting (in writing) the complaint receiving officer (CRO) at ADB headquarters or ADB India Resident Mission (INRM). The complaint can be submitted in any of the official languages of ADB's developing member countries. Before submitting a complaint to the Accountability Mechanism, it is recommended that affected people make a good faith effort to resolve their problems by working with the concerned ADB operations department (in this case, the resident mission). Only after doing that, and if they are still dissatisfied, they could approach the Accountability Mechanism.

97. The ADB Accountability Mechanism information will be included in the project-relevant information to be distributed to the affected communities, as part of the project GRM.

98. **Record Keeping.** Records of all grievances received, including contact details of complainant, date the complaint was received, nature of grievance, status and agreed corrective actions will be kept by the contractor, PIO (with the support of PSC) will collect the data from the contractor and submit to PMU (state level). This information will be reported to ADB by the PMU in bi-annual safeguard monitoring reports.

99. **Information Dissemination Methods of the Grievance Redress Mechanism.** The PIU, assisted by PSC will be responsible for information dissemination to affected persons and general public in the project area on the options for filing a grievance; including easy to understand detail of the first option. A public awareness campaign will be conducted at the individual canal-level, at the start of the project to ensure awareness on the project and its grievance redress procedures. The campaign will ensure that the poor, vulnerable (women) and others are made aware of grievance redress procedures. The information campaign will also provide details on whom to contact and when, where/ how to register grievance, various stages of grievance redress process, time likely to be taken for redress of minor and major grievances, etc. Grievances received, and responses provided will be documented and reported back to the affected persons. The number of grievances recorded and resolved, and the outcomes will be displayed/disclosed in the PIO offices, WUCS, GP offices, as well as reported in the semi-annual safeguard monitoring reports to be submitted to ADB. Contractors must display details of the CL-GRM on their notice board at every site.

100. **Periodic Review and Documentation of Lessons Learned.** The PMU will periodically review the functioning of the GRM and record information on the effectiveness of the mechanism, especially on the PIU's ability to prevent and address grievances.

101. **Costs.** All costs involved in resolving the complaints (meetings, consultations, communication and reporting/information dissemination) will be borne by the respective PIU. Cost estimates for grievance redress are included in resettlement cost estimates.

VIII. GENDER AND SOCIAL DIMENSIONS

A. Summary Poverty Reduction and Social Strategy (SPRSS)

102. The poverty reduction and social strategy under the investment program is aligned with the Eleventh Five Year Plan (2007–2012) and draft Twelfth Five Year Plan (2012–2017), the Planning Commission, Government of India, and ADB Strategy 2030 on promoting inclusive growth and achieving rural poverty reduction through strengthening the rural infrastructure services.²¹ The program focuses on adopting pro-poor and gender-inclusive mechanism within the overall design. It aims at providing enhanced water accessibility and equity to water users within a sub-basin and specifically for agricultural and household purposes. The investment program will improve equity of water distribution to directly benefit tail-end farmers, marginal, small and landless farmers, and the poor and vulnerable women.

103. Investment program interventions will have positive impacts on the women and socially excluded groups in terms of (i) strengthened grassroots institutions; (ii) improved irrigation infrastructure and practice; (iii) improved and equitable access to water by small, marginal and landless farmers; (iv) improved farm incomes; (v) enhanced off-farm livelihood opportunities (vi) sustainable and innovative irrigation and agricultural practices, including drip and micro-irrigation and crop diversification; (vii) strengthened water resources and environmental management; (viii) social inclusion and in water governance and decision-making processes; and (ix) awareness generation, community mobilization and participatory irrigation management.

B. Gender Action Plan (GAP)

104. The investment program is classified as Category II: effective gender mainstreaming (EGM) following ADB's gender categorization of projects.²² The gender strategy will mainstream women in the context of irrigation management. Specific activities for women will include: (i) provision of capacity building for technical staff of WRD in IWRM topics. The program will ensure training of at least 600 WRD staff of which at least 30% (200) will be women; (ii) WUCS management boards will have at least 30% representation of women (from current 20%) which will increase the role of women in decision making process pertaining to water resources management; and (iii) gender specific capacity building in women in water management, women's leadership and role in WUCS. The PSC for each project should include a social development and gender specialist, and each field support services team for WUCS strengthening should have one female community organizer. A GAP is provided in Table 31 for the VNC subproject. The GAP of Trance 2 will build women's technical skills and leadership capacity to engage in the agricultural sector, primarily through the WUCS. While Tranche 1 GAP activities are ongoing, monitoring data highlights that the project is on track to meet gender quotas for training and representation in key decision-making bodies. Furthermore, project interventions have increased women's income and livelihood standards through improved agricultural opportunities including crop diversification. Tranche 2 will continue and expand the scope of activities.

²¹ ADB. 2018. *Strategy 2030: Achieving a Prosperous, Inclusive, Resilient, and Sustainable Asia and the Pacific*. Manila

²² ADB's Policy on Gender and Development (1998) and ADB's Gender Mainstreaming Criteria Guidelines (2010).

Table 31: Gender Action Plan for Project 2

Activity	Indicators/Targets	Responsibility	Time Schedule
Output 1. State and Basin Institutions Strengthened for IWRM			
Develop the capacities of WRD staff including women in IWRM	At least 600 WRD staff (of which 200 are women staff) are certified in IWRM (2018 baseline: 150 staff are certified of which 40 are women)	WRD/PMU/PIOs	Year 2
Output 2. Irrigation system infrastructure and management modernized			
Build awareness of women's equal rights and participation in water resources management and livelihood improvement	30 sensitization workshops focusing on gender equality and social inclusion (GAP) approaches and women's role and participation in water resources management and farm/off farm livelihood activities conducted for all WUCS in the VNC subproject area (T: 30 workshops conducted with at least 60% men's participation)	PMU/PIOs	Year 2–3
	At least 30% of WUCS management board members are women (T: M– 150; F–50)	PMU/PIOs	Year 1–2
Improve the knowledge of relevant government departments on GAP approaches in agriculture, irrigation and water resources management	5 gender sensitization workshops provided to WRD and KNNL staff on GAP concepts and in the design and implementation of water resources management, irrigation and agriculture programs for vulnerable populations, including women and use of sex-disaggregated data for decision making (T: F–30; M–70)	WRD/PMU/PIOs	Year 1–2
Output 3: Program and management systems operational			
Monitor progress of gender and socially inclusive focused activities in water resources management	Project management information system with gender indicators	PMU/PIOs	Year 1–4
	Sex-disaggregated data collected, analyzed and reported in quarterly progress reports and safeguard monitoring reports	PMU/PIOs	Year 1-4

F = female, GAP = gender action plan, IWRM = integrated water resources management, M = male, NGO = nongovernment organization, PIM = participatory irrigation management, PIO = project implementation office, PIU = project implementation unit, PMU = project management unit, T = target, WRD = Water Resources Department, WUCS = Water Users Cooperative Society

Note: The Gender Action Plan is subject to further revisions post consultations with Asian Development Bank and State Government of Karnataka.

IX. PERFORMANCE MONITORING, EVALUATION, REPORTING AND COMMUNICATION

A. Program/Project Performance Monitoring

105. During the course of the investment program implementation, monitoring will be carried out for: (i) periodic progress reporting; (ii) safeguard monitoring; (iii) benefit monitoring and evaluation; and (iv) financial management monitoring. The DMFs for the investment program, Project 1, and Project 2 will be the basis for monitoring progress.

106. **Periodic progress monitoring.** The PMU will prepare periodic progress reports based on loan implementation. Specifically, the following reports will be required:

- (i) Inception report (draft submitted within 2 months of mobilization and finalized within 3 months of mobilization);
- (ii) Quarterly progress reports (QPR) (within 30 days of the end of each quarter)
- (iii) Annual report (within 30 days of the end of each calendar year)
- (iv) Midterm report (middle of year 2 for Project 1, 3rd quarter of 2021 for Project 2)
- (v) Project completion reports (PCR) (draft submitted 3 months prior to Project completion and finalized within 3 months after Project completion)

107. **Compliance and Safeguard monitoring.** Monitoring of compliance (including the monitoring of policy, legal, financial, economic, and other covenants) and safeguard issues (including the monitoring of social and poverty action plans), will be undertaken by the PIO and reported in the MIS database. The main aim will be for the PIO to ensure that the projects are implemented with due concern for safeguards and quality and specifically to ensure that these issues are adequately addressed to the requirements of the state and ADB. This will include: (i) environmental quality monitoring of air and water quality and noise levels parameters (as referred to in the EIA and EARF) which will be undertaken periodically by a specialist firm and reported to the PMU through PIO; (ii) gender mainstreaming; (iii) management of temporary land access and potential resettlement impacts like compensation for loss of income; (iii) quality of work necessary to ensure sustainable outputs and; (iv) grievances.

108. **Gender and social dimensions monitoring.** Monitoring and quarterly reporting of gender and social dimensions will be the responsibility of the social development specialist of the PSC. The program GAP sets out the gender monitoring indicators and targets for each project component. Further, all Project data on staffing and recruitment, participation, consultation, wages, training, safeguards implementation and compensation, and impacts will be sex disaggregated. GAP implementation progress, in accordance with the GAP progress update format will be included and submit to ADB as part of the quarterly progress reports. An MIS was established accordingly at Project-1 start up to ensure data is stored and managed well for monitoring. The MIS designed for Project 1 will be adopted for Project 2 at the start up to ensure data is stored and managed well for monitoring.

109. **Effects Monitoring.** During the project preparatory technical assistance, a socioeconomic baseline survey was undertaken for Gondi subproject. Approximately 3.6% of sample households (equivalent of 100 households) to the total households of the command area were taken up for the study. This may be expanded to a larger sample as part of the comprehensive baseline survey which will be used for monitoring project outcome and impact and periodic monitoring. Similarly, a baseline survey was undertaken as part of the feasibility study for Project 2. Final effects monitoring for Project 1 will be undertaken during Project 2. Similarly, prior to the investment program completion another effects monitoring and evaluation will be undertaken

(in October 2022) for Project 2. The PSC under Project 2 will engage a specialist entity (including academic institute) for expansion of the baseline survey and effects monitoring and evaluation at various stages of program implementation (see Attachment 7 for draft terms of reference).

110. As part of the program performance management system (PPMS), Project 1 developed a comprehensive MIS database which incorporated Output 1 parameters for IWRM with those of Output 2 for infrastructure modernization and WUCS strengthening. The same firm will also undertake effects monitoring and evaluation and provide periodic progress updates based on the MIS and baseline data. Project 2 will develop a similar comprehensive MIS database for effects monitoring and evaluation and provide periodic progress updates based on the MIS and baseline data.

111. A separate MIS database will be established within the PMU for financial management monitoring.

112. PSC MIS specialist will establish an MIS database for program monitoring including key indicators for implementation progress and relating to program outcomes. Capacity building of staff and facilitate/review period updating. This will not be overly cumbersome and will collect poverty and gender-disaggregated data. Within 6 months of loan effectiveness of each project, the PMU will submit a detailed implementation monitoring plan to ADB for review and concurrence by ADB. Financial performance monitoring and review, including spot checks, will be conducted from time to time and will be carried out by ADB loan administration missions.

113. During the course of program implementation, the PPMS will be further fine-tuned as needs are recognized. Project performance monitoring will be undertaken by the Project Director with the support of the PSC. The PIO will provide the PMO with quarterly reports on their: (i) activities; (ii) compliance with safeguard requirements; and (iii) progress toward output targets. The PMU will consolidate these in a QPR that will be finalized and circulated to ADB, WRD, and through WRD to the state IWRM and program steering committees. QPRs will be circulated within one month of the end of the reporting quarter.

B. Evaluation

114. The PD will facilitate (at least) six-monthly reviews by ADB and the government. The reviews will (i) examine the continuing appropriateness of implementation arrangements; (ii) examine activity schedules with particular concern for progress towards IWRM targets and institutional strengthening parameters like capacity building and; (ii) construction standards and quality of works implementation; (iii) review compliance with agreed procurement procedures; (iv) analyze progress and success of capacity building and training programs; and (v) monitor the effectiveness of safeguard procedures. A comprehensive mid-term review will be undertaken within 24 months of loan effectiveness to identify any problems and constraints encountered and assess any need for modification of Project's scope and financing and/or implementation arrangements.

115. Three months before completion of each project, the PSC will produce a draft PCR for each project which will be finalized during the last month of the Project. This will be following the receipt of comments from the State and ADB received by the start of that final month. The PCR will detail the lessons learned from each project implementation, for guidance of similar future activities. Within 6 months of physical completion of the Projects, KNNL will submit a draft PCR to ADB and DEA who would review and submit their comments for incorporation into the final report.

C. Reporting

116. PMU will provide ADB with: (i) QPRs in a format consistent with ADB's PPMS; (ii) consolidated annual reports including (a) progress achieved by output as measured through the indicator's performance targets, (b) key implementation issues and solutions; (c) updated procurement plan and (d) updated implementation plan for next 12 months; and (iii) a PCR within 6 months of physical completion of each Project in the investment program. To ensure projects continue to be both viable and sustainable, project financial statements and the executing agency's audited financial statements, together with the associated auditor's report, should be adequately reviewed.

D. Stakeholder Communication Strategy

117. A comprehensive stakeholder analysis and strategy has been prepared for the program (see Attachment 10). The Stakeholder Communications Strategy is based on a thorough stakeholder analysis and consultations with stakeholder groups during project preparation.

118. The strategy is two-pronged: (i) at headquarters: ACIWRM and KNNL will establish program websites to disseminate information regarding the program objectives, progress and any events etc. ACIWRM has a comprehensive communications strategy as part of the IWRM road map and this will be followed; and (ii) at subproject level: PIO will establish a subproject specific information site which may also be used for addressing complaints as part of the grievance redress mechanism, For WUCS strengthening, communications will be rigorous as part of the social mobilization process and will include, workshop, training, focus group discussions etc.

119. One-day workshops will be held annually at each of the projects' subproject sites where work is planned, ongoing or recently completed. These workshops will be organized by the PIOs with support of PSC and PMU.

120. Public notification of subproject milestones and safeguards, as well as progress towards these milestones and measures taken to impose safeguards will be posted on both KNNL/PIO and ADB websites and will be displayed prominently in hardcopy at a central location within the subproject.

X. ANTICORRUPTION POLICY

121. ADB reserves the right to investigate, directly or through its agents, any violations of the Anticorruption Policy relating to the Project.²³ All contracts financed by ADB shall include provisions specifying the right of ADB to audit and examine the records and accounts of the Executing Agency and all Project contractors, suppliers, consultants and other service providers. Individuals/entities on ADB's anticorruption debarment list are ineligible to participate in ADB-financed activity and may not be awarded any contracts under the project.²⁴

122. The government, the state, KNNL, and implementing agencies are advised of ADB's Anticorruption Policy (1998, as amended to date). Consistent with its commitment to good governance, accountability and transparency, implementation of the projects under the investment program shall adhere to ADB's Anticorruption Policy. ADB reserves the right to review and examine, directly or through its agents, any alleged corrupt, fraudulent, collusive, or coercive practices relating to the Projects under the investment program. In this regard, investigation of government officials, if any would be requested by ADB to be undertaken by the government. To support these efforts, relevant provisions of ADB's Anticorruption Policy are included in the loan regulations, loan agreement and the bidding documents. In particular, all contracts financed by ADB shall include provisions specifying the right of ADB to audit and examine the records and accounts of the project executing agency and implementing agencies, and all contractors, suppliers, consultants, and other service providers as they relate to projects under the investment program. Individuals/entities on ADB's anticorruption debarment list are ineligible to participate in ADB-financed activity and may not be awarded any contracts under the projects.

123. The state will ensure that (i) KNNL conducts periodic monitoring inspections on all contractors' activities related to fund withdrawals and settlements; and (ii) all contracts financed by ADB in connection with the project include provisions specifying the right of ADB to audit and examine the records and accounts of KNNL and all contractors, suppliers, consultants and other service providers as they relate to the project.

124. ADB's Anticorruption Policy designates the Office of Anticorruption and Integrity (OAI) as the point of contact to report allegations of fraud or corruption among ADB-financed projects or its staff. OAI is responsible for all matters related to allegations of fraud and corruption. For a more detailed explanation refer to the Anticorruption Policy and Procedures. Anyone coming across evidence of corruption associated with the project may contact the Anticorruption Unit by telephone, facsimile, mail, or email at the following numbers/addresses:

by email at integrity@adb.org or anticorruption@adb.org

by phone at +63 2 632 5004

by fax to +63 2 636 2152

by mail at the following address (Please mark correspondence Strictly Confidential):

Office of Anticorruption and Integrity

Asian Development Bank

6 ADB Avenue Mandaluyong City

1550 Metro Manila, Philippines

²³ Available at: <http://www.adb.org/Documents/Policies/Anticorruption-Integrity/Policies-Strategies.pdf>

²⁴ ADB's Integrity Office website is at: <http://www.adb.org/integrity/unit.asp>

XI. ACCOUNTABILITY MECHANISM

125. People who are, or may in the future be, adversely affected by the project may submit complaints to ADB's Accountability Mechanism. The Accountability Mechanism provides an independent forum and process whereby people adversely affected by ADB-assisted projects can voice, and seek a resolution of their problems, as well as report alleged violations of ADB's operational policies and procedures. Before submitting a complaint to the Accountability Mechanism, affected people should make a good faith effort to solve their problems by working with the concerned ADB operations department. Only after doing that, and if they are still dissatisfied, should they approach the Accountability Mechanism.²⁵

²⁵ For further information see: <http://www.adb.org/Accountability-Mechanism/default.asp>.

XII. RECORD OF FAM CHANGES

126. The first draft of FAM has been prepared and agreed upon at the loan fact-finding. All revisions/updates during the course of implementation will be retained in this Section to provide a chronological history of changes to implementation arrangements recorded in the FAM.

- (i) The FAM has been revised considerably to incorporate implementation arrangements for Project 2.
- (ii) Changes to Project 1 implementation are detailed in the succeeding items.
- (iii) The role of UNESCO-IHE revised as reflected in Table 8, under institutional strengthening, as follows:
 - From: "The ACIWRM will be accredited by UNESCO-IHE as regional institute for professional training in IWRM."
 - To: "UNESCO-IHE will contribute training peer-review, advisors, and possibly suggestions for improvement at the end of the first cycle."
- (iv) Period of implementation of Project 1 revised from 2014–2018 to 2015–2019.
- (v) Charts on program and project implementation updated to include Project 2.
- (vi) Table on project readiness for Project 2 added.
- (vii) Program Coordination Committee (PCC) established on 27 November 2015 under the chairmanship of Additional Chief Secretary/Principal Secretary.
- (viii) Members of the PCC updated, removing Chief Engineers of Hydrology and Central Mechanical Organization.
- (ix) Director, SAER and Mission Leader updated.
- (x) Cost table (page 25) updated based on amendment to the loan agreement on reallocation and change in financing percentages dated 27 July 2016, and further updated to incorporate the reallocation of loan proceeds following the partial cancellation of \$5 million (effective 6 May 2019).
- (xi) Expected contract awards updated reflecting actual amounts awarded, and ADB share based on amendment to the loan agreement on reallocation and change in financing percentages dated 27 July 2016.
- (xii) S-Curve updated based on actualized contract awards and disbursements from effectiveness date to December 2017.
- (xiii) Financing plan for Project 2 (cost tables, cost allocation table, S-Curve added.
- (xiv) Section on financial management updated based on the financial management assessment done as part of Project 2 processing.
- (xv) Fund flow diagram updated with each of KNNL and ACIWRM arrangements shown separately in two diagrams.
- (xvi) Fund flow arrangements for ACIWRM updated such that the Registrar (not Chief Accounts Officer) in ACIWRM shall submit utilization certificates for the expenditure duly authorized by the Accounts Superintendent/ Audit Officer to PMU for the later to consolidate the expenditure claims for submission to CAAA-ADB for reimbursement.
- (xvii) Description on financial management of ACIWRM updated.
- (xviii) Terms of reference (TOR) of the Project Support Consultants (PSC 1) updated to align with the signed contract and approved contract variations.
- (xix) TOR for PSC 2 added.
- (xx) Effects monitoring for Project 1 to be undertaken under Project 2.
- (xxi) Number of water user cooperative societies to be covered by the Project 1 increased from 9 to 11.
- (xxii) Removal of \$100,000 ceiling limit under the statement of expenditure disbursement procedure.

(xxiii) Two Gender and Action Plan indicators deleted as agreed at the ADB midterm review mission as such were deemed too ambitious unachievable within the remaining timeframe:

- a. Partnerships with nongovernment organizations developed to conduct sensitization and community mobilization campaigns on women's participation in water users' cooperative societies (WUCS), participatory irrigation, crop diversification and management, and agriculture extension services
- b. 50 WUCS (including women and men) strengthened and training and marketing support provided to microenterprises and livelihood clusters [farm/off-farm] (Target: M-1,000; F-1,000)

Date	Original	Revised	Remarks
29 July 2016	Para. 10: "... (iv) operationalizing a program management information systems database within 3 months after commencement of project..."	Para. 10: "... (iv) operationalizing a program management information systems database by 31 July 2017 ..."	
29 July 2016	Para. 13: "... Project-1 will be implemented between 2014 and 2018 and will be of 4 years in duration. It is expected that Project-2 is expected to be approved in 2015 for duration of 6 years until September 2021 when the implementation of the MFF will be completed...."	Project-1 will be implemented between 2015 and 2019. The Project-1 loan is effective on 13 July 2015. The activities under Project 1 are expected to be completed by September 2018, and loan closing is scheduled for 31 March 2019. It is expected that Project-2 will commence in late 2017 until September 2021 when the implementation of the MFF will be completed.	
29 July 2016	Overall Program Implementation Plan (chart, page 7)	Dates updated.	
29 July 2016	Implementation Plan for Project-1 (chart, page 8)	Schedule updated from 2014–2018 to 2014–2019	
29 July 2016	Program Coordination Committee, page 9: "The Program coordination committee (PCC) will be established under the chairmanship of Principal Secretary."	"The program coordination committee (PCC) has been established on 27 November 2015 under the chairmanship of Addl. Chief Secretary/Principal Secretary..."	
29 July 2016	Members of the PCC, page 10: "(vi) Chief Engineers of Water Resources Development Organization and Hydrology	(vi) Chief Engineers of Water Resources Development Organization	

Date	Original	Revised	Remarks
	and Central Mechanical Organization,"		
29 July 2016	ACIWRM (Implementing Agency), page 14: "Consulting services will provide technical support to ACIWRM, specifically the team leader of consulting services who will be the IWRM policy specialist."	"...Consulting services provide technical support to ACIWRM, specifically the Chief Technical Advisor (CTA) who is the IWRM policy specialist..."	
29 July 2016	Mission Leader, page 17: Yasmin Siddiqi Tel.: +632 632-5723 Email: ysiddiqi@adb.org	Lance Gore Tel.: +632 632-5954 Email: lgore@adb.org	
29 July 2016	B. Detailed Cost Estimates by Expenditure Category for MFF (page 23)	Updated based on amendment to the loan agreement on reallocation and change in financing percentages dated 27 July 2016	
29 July 2016	C. Detailed Cost Estimates by Expenditure Category for Project 1 (page 24)		
29 July 2016	D. Allocation and Withdrawal of Loan Proceeds for Project-1 (page 25)		
29 July 2016	E. Detailed Cost Estimates by Financier for Project-1 (page 26)		
29 July 2016	F. Detailed Cost Estimates by Financier for MFF (page 27)		
29 July 2016	G. Detailed Cost Estimates by Outputs/Components for Project-1 (page 28)		
29 July 2016	H. Detailed Cost Estimates by Outputs/Components for MFF (page 29)		
29 July 2016	I. Detailed Cost Estimates by Year for Project-1 (page 30)		
29 July 2016	J. Detailed Cost Estimates by Year for MFF (page 31)		
29 July 2016	Budget Line Item in the State Budget (para. 36, page 37): "The project will be pre-financed by the state through a dedicated budget head, separately identifiable line item for the ADB project in the Demand for Grants of the State Water Resources Department (WRD) for allocation of funds for the program starting Financial Year 2013-14 based on the expenditure for the program for FY2013–2014..."	The project has been-financed by the state through a dedicated budget head, separately identifiable line item for the ADB project in the Demand for Grants of the State Water Resources Department (WRD) for allocation of funds for the program starting Financial Year 2015–2016 based on the expenditure for the program for FY2015–2016	

Date	Original	Revised	Remarks
29 July 2016	Para. 39, page 37: "...Similarly, the CAO in ACIWRM shall submit utilisation certificates for the expenditure duly authorized by the Accounts Superintendent to PMU for the later to consolidate the expenditure claims for submission to CAAA-ADB for reimbursement..."	Para. 39: Similarly, the Registrar in ACIWRM shall submit utilisation certificates for the expenditure duly authorized by the Accounts Superintendent/ Audit Officer to PMU for the later to consolidate the expenditure claims for submission to CAAA-ADB for reimbursement.	
29 July 2016	Existing Fund Flow Arrangements for ACIWRM, para. 43, page 39: "As a standard procedure for state owned agencies in Karnataka, the financial management of ACIWRM is governed by the Public Works Department codal provisions, departmental codes of accounting, procedures and statutory requirements under the Karnataka Societies Act, KTPP Act, Karnataka Finance Code and Manual of Contingent Expenditures, and other circulars /guidelines issued by the State from time to time..."	As a standard procedure for state owned agencies in Karnataka, the financial management of ACIWRM is governed by the procedures and statutory requirements under the Karnataka Societies Act, KTPP Act, Karnataka Finance Code and Manual of Contingent Expenditures, and other circulars /guidelines issued by the State from time to time.	
29 July 2016	Consultant's Terms of Reference, para. 90, page 47: "...The PSC will be engaged for a period of 4 years for Project-1 and a separate PSC package for Project-2..."	The PSC has been engaged for a period of 42 Months for Project-1 and a separate PSC package for Project-2	
5 October 2016	Consultant's Terms of Reference, para. 92, page 48: "...The total inputs required for PSC for Project-1 will be 230 person-months (p-m), including 20 p-m of international and 210 p-m of national consultants. Firms will be selected using QCBS with 90:10 (quality: cost ratio)"	The total key expert inputs required for PSC for Project-1 will be 328 person-months (p-m), including 20 p-m of international and 308 p-m of national consultants.	
5 October 2016	Table 8: Program Support Consultancy - Summary Terms of Reference (page 49) • Participatory Irrigation Management Specialist - 48 person-months	• Participatory Irrigation Management Specialist: 42 p-m • Water Institutions Specialist: 28 p-m	Support Services Team for WUCS integrated into the PSC contract through a contract variation.

Date	Original	Revised	Remarks
	• Total national person-months: 210 • Total person-months (international and national): 230	• Additional positions for WUCS Support Services Team: ◦ Agro-Economist: 4 p-m ◦ Community Organizers: 36 p-m ◦ Water Management Extension Agent: 36 p-m • Total national person-months: 308 • Total person-months (international and national): 328	
9 February 2018	Para. 53: Statement of Expenditure (SOE) Procedure. The SOE procedure will be used to reimburse/liquidate eligible expenditures not exceeding \$100,000 equivalent (as stated in ADB Loan Disbursement Handbook, 2012) for any individual payment. For all individual payments exceeding \$100,000 equivalent, documented claim procedure¹⁵ will be followed. Withdrawal applications for payments made to contractors/consultants in currencies other than Indian Rupees if any, will be submitted for reimbursement to CAAA-ADB will follow documented claim procedure and will be in the currency paid to contractors/consultants. Claims in the SOE sheets must carry on the corresponding Procurement Contract Summary Sheet (PCSS) number provided for each contract under the Program.	Para. 53: Statement of Expenditure (SOE) Procedure. The SOE procedure will be used to reimburse/liquidate eligible expenditures using ADB's Client Portal Disbursement (CPD) at https://cpd.adb.org/. All claims beginning 2 February 2018 may use SOE without any limit. Previously, SOE was used only for claims not exceeding \$100,000 equivalent (as stated in ADB Loan Disbursement Handbook, 2012) for any individual payment and for all individual payments exceeding \$100,000 equivalent, documented claim procedure²⁶ was followed. KNNL and ACIWRM has been provided with their CPD user accounts and are encouraged to use the CPD in all withdrawal applications. Manual submission is permitted when there are technical issues or for cases requiring detailed summary sheets and remarks such when any refund or recovery on disbursements is	

²⁶ All supporting approved invoices, bills, receipts of payment

Date	Original	Revised	Remarks
		necessary. Withdrawal applications for payments made to contractors/consultants in currencies other than Indian Rupees if any, will be submitted for reimbursement to CAAA-ADB will follow documented claim procedure and will be in the currency paid to contractors/consultants. Claims in the SOE sheets must carry on the corresponding Procurement Contract Summary Sheet (PCSS) number provided for each contract under the Program.	
[7 March 2018]	Para. 29: The graph below shows the contract awards and disbursements curves for ADB's portion of PFR1 based on the procurement plan and implementation schedule. Total projected contract awards and disbursements only included allocated portion of \$31 million. A total of \$2.14 million remains unallocated. Within fourth quarter of 2014, expected contract awards amount to \$25.229 million for Gondhi civil works for a total \$16 million (out of which ADB's portion is \$11.83 million), telemetry system, and program support consultants. Succeeding quarters up to fourth quarter of 2017 would show contract awards for other consultant's services (strengthening consultants, environmental monitoring and effects monitoring consultants), and command area development works in multiple lots. Disbursements follow a progressive curve based on mobilization fee requirements and progress payments under the various	Para. 29: The graph below shows the contract awards and disbursements curves for ADB's portion of PFR1, actualized to 2017. Total contract awards and disbursements only include allocated portion of \$31 million. The Gondhi civil works contract was awarded for \$18.7 million (equivalent) in Q1 2016, the Project Support Consultant (PSC) for \$3.5 million (equivalent) and telemetry for \$2.7 million (equivalent) in Q4 2015. Community participation packages have been awarded to 11 WUCS in 2017 totaling \$2.5 (ADB allocation).	S curve on contract awards and disbursements updated showing actualized data up to end 2017 and projections from 2018 to Q1 2019.

Date	Original	Revised	Remarks
	contracts and implementation schedule.		
[7 March 2018]		Para. 5: ACIWRM may directly submit withdrawal applications to CAAA-ADB upon authorization of the State for Superintending Engineer, ACIWRM to be a signatory to withdrawal applications in addition to Program Director, KNNL.	Sentence inserted in para. 55.
[7 March 2018]	Para. 112: A final effects monitoring will be undertaken for Project-1 (in January 2019) prior to its completion.	A final effects monitoring for Project-1 will be undertaken during Project-2.	
	Effects Monitoring, para. 113, page 56: "...An impact evaluation study is planned to demonstrate outcomes and impacts of the planned investment, for which a baseline study will be undertaken in November 2014..."	An impact evaluation study is planned to demonstrate outcomes and impacts of the planned investment, for which a baseline study will be undertaken in October 2016.	
[7 March 2018]	Executing Agency, page 17: Mr. Sri R. Rudraiah	Mr. Mallikarjuna B Gunge	
[7 March 2018]	ADB, page 17: Takashi Matsuo Director, SAER Tel: +632 632-5579 Email: tmatsuo@adb.org	Mio Oka Director, SAER Tel: +632 632-6834 Email: moka@adb.org	
[7 March 2018]	Para. 14 (ii) Also, there will be capacity building of 9 WUCS and increasing women representation in the WUCS management boards.	Also, there will be capacity building of 11 WUCS and increasing women representation in the WUCS management boards.	
[7 March 2018]	Para. 26: It is expected that the second tranche will commence in 2015 for 6 years until September 2021.	Para. 26: It is expected that the second tranche will commence in 2019.	

ATTACHMENT 1: INTEGRATED WATER RESOURCES MANAGEMENT ROADMAP

Key IWRM Theme	Definition and Activities	Program Objectives	Responsibilities	Timing
1. Integrated River Basin Management	Activities to improve water access and services for users, including the environment, in river basins based on the principles of integrated water resources management. Required activities include: (i) Establishment of effective river basin management arrangements including financing arrangements; (ii) Preparation of State River Basin Inventory; (iii) Development and implementation of river basin plans in selected river basins including participation of stakeholders, updating of water resource inventory, inter-sectoral planning of water development and management; groundwater development and management plans; water quality and quantity planning and management systems to maintain river health; and plan financing; (iv) Preparation of pilot, community based, integrated Land and Water Management Plans in selected sub-basins and/or irrigation areas; (v) Operational management systems to implement basin water plans and where possible to include flow monitoring, volumetric measurement of abstractions, flow control and enforcement, and sharing of water; and (vi) Implementation of public awareness, information education and communication systems and campaigns on relevant water sector issues.	(i) Access for all people in the basin to adequate water supply and sanitation (ii) Water consumption is within the sustainable limits of water availability in river basins and sub-basins. Sources of water for domestic, irrigation, industrial, hydropower, aquaculture, leisure, and other uses are developed and managed consistent with water availability and sustainability (iii) River basin planning is transparent, integrative, consultative, effective, and uses information-based systems including modelling and spatial data (iv) River basin planning and management integrates rainwater, surface water, groundwater; water quantity, water quality; water and land use; and in-stream flow needs with all relevant sectors involved (v) Agreed policies, rules, and operational procedures including monitoring to implement river basin plans and water use by different sectors are implemented effectively	AC – IWRM WRD WRDO Directly involved stakeholders: CWC, WRA, TB, KNNL, other Nigams, DMI, CADA, DA, DE, DF, DMG, WD, KUWSDB, WUCS; local government, civil society	August 2021

Key IWRM Theme	Definition and Activities	Program Objectives	Responsibilities	Timing
2. Institutions and Policies for IWRM	"Institutions" is used in its broadest sense including organizations, legislation, policies and other protocols that define the relations among water sector organizations and their clients. Planning for IWRM (in this context, river basin planning) is seen as a mechanism for promulgating and implementing policies of government. Activities required include: (i) Organizational alignment for IWRM processes with interim arrangements initially for selected river basins (ii) Organizational capacity building for IWRM processes, including career development and reward systems (iii) Alignment and implementation of legislative framework (iv) Preparation of a State IWRM Strategy to and development of water resources policies aligned with economic, social and environmental objectives (v) Regulation (such as sustainable water utilization, wastewater discharge and water tariff approaches) (vi) Institutional development for participatory irrigation management (PIM) including formation and strengthening water user committees (vii) Capacity building of staff of agencies including IWRM Certification Program	(i) ACIWRM recognized as an international standard think tank providing advice to the State (ii) Appropriate IWRM institutional frameworks within the State with clearly defined responsibilities and working partnerships with stakeholders in Karnataka leading to coordinated water resource management in the State (iii) National and international IWRM research and policy partnerships established with ACIWRM providing access to state-of-art knowledge (iv) State IWRM Strategy in place and regularly updated and guiding strengthening of water sector governance (v) Policies, procedures and responsibilities suitable for Karnataka piloted and established to sustainably manage water resources (surface water, groundwater, quantity and quality) and involving major water users including the environment (vi) Decision makers, advisers, technical experts, and other key stakeholders effectively carry out their responsibilities with regard to water resources policy, planning and management (vii) Policy makers' and other stakeholders' knowledge and willingness raised to undertake necessary reform measures	ACIWRM WRD CADA Directorate Directly involved stakeholders: CWC, WRA,	August 2021

Key IWRM Theme	Definition and Activities	Program Objectives	Responsibilities	Timing
		(viii) Capacity of WRD and water related agency staff raised to ensure long term IWRM		
3. Data, Information, and Knowledge Management	Data, information and knowledge are fundamental to all aspects of decision making in water resource planning and management. IKM theme will include: (i) Preparation of a Strategy and Plan for Strengthening the Water Resources Information System; (ii) Data collection on surface and groundwater quantity and quality and collation of other natural resource data, such as land cover, ecosystems, etc., as well as such socioeconomic data such as population, poverty, and land use; for priority river basins; (iii) Hydrological monitoring in the K-8 and other priority river basins; (iv) Data archiving and management, including the collation of data from various sources, validation, computerization, and so on; (v) Data sharing and dissemination among government agencies, research establishments, and providing public access to data; (vi) Identification, development and implementation of appropriate decision-support tools, including geographic information systems, hydrologic and hydraulic models, and other analytical tools; and (vii) Research to increase knowledge in such fields as catchment processes, demography, etc., as well as new technologies for water conservation and environmental protection.	(i) IWRM and river basin planning and policies are based on strong data and information sources and decision support systems (ii) A comprehensive State database on land and water resources in place and in a form that is accessible to all who need it to facilitate sustainable management of the State's water resources (iii) Agencies concerned with water management monitor and record water resources to agreed standards and according to their responsibilities (iv) Effective data-sharing arrangements in place among agencies and stakeholders in the basin and with central agencies including data sharing agreements (v) Research programs are providing highly relevant information to support IWRM	ACIWRM WRDO KNNL, other Nigams, Directly involved stakeholders: CWC, DPS, DITB; DeG; DMI, DA, DE, DF, DMG, WD, KUWSDB	June 2020

Key IWRM Theme	Definition and Activities	Program Objectives	Responsibilities	Timing
4. IWRM Based Irrigation Management	Modernization of irrigation services, consistent with water management policies including infrastructure for irrigation water supply, integration with tank storages, command area works, water measurement and control systems, and improved operation and maintenance of the infrastructure developed in the process. It includes: (i) Participatory project planning, including irrigation modernisation planning (infrastructure and management development distinct from broader basin planning); matching cropping patterns to available water and increased income, water distribution plans, responsive to stakeholders; (ii) Modernising systems including canal lining, structure repair / replacement; incorporating tanks and offline storage, command area improvement including pipeline systems, drip irrigation, improved drainage systems to eliminate waterlogging; (iii) Automatic flow measurement and transmission of flow data for the main distribution system to monitor and enable volumetric measurement for each outlet to enable water control and bulk water charging for each WUCS; (iv) Operating and maintaining infrastructure including timely and volumetrically based water supply, asset management and full recovery of at least O&M costs; (v) Capacity building of both system operations staff and water user organisations; and	(i) All water supply infrastructure capable of operating at design capacity (ii) Sustainable asset-management practices in place for all water-related infrastructure in the basin (iii) Institutions managing the infrastructure, volumetric supply of irrigation water to users and cost recovery of O&M at government, water authority and WUCS levels (iv) Reliable delivery of irrigation services within the constraints of available water resources (v) Commitment of involvement of water users in planning and management of irrigation systems	KNNL and other Nigams CADA Directorate Directly involved stakeholders: DA, WUCS	August 2021

Key IWRM Theme	Definition and Activities	Program Objectives	Responsibilities	Timing
	(vi) Establishment of effective self-managed WUCS institutions for participatory irrigation management.			
5. Water Use Efficiency and Water Productivity	Increasing water use efficiency (WUE) and water use productivity (WUP) is a high government priority in order to save water that can be used for other purposes including use by industry, urban areas, and irrigation. However, by focusing on single irrigation systems impacts at the larger basin level on downstream users and the environment are often overlooked. Also, a focus on agricultural production does not necessarily translate into increased incomes to farmers. This theme will include: (i) Development of the concepts and understanding of water productivity, water use efficiency, crop and agricultural productivity and their practical application to IWRM (ii) Basin scale assessments of water use efficiency and water productivity including by remote sensing to identify the actual scope for making 'real' water savings as well as to support river basin planning and management. (iii) Improving WUE and Irrigation Systems by undertaking comprehensive assessments of irrigation systems using the FAO MASSCOTE tools, the assessments would also support Irrigation modernisation and Land and Water Management Plan activities	(i) River basin and water resources planning treats the water resource as finite (ii) Increased provision of water for irrigation or industry expansion is based on real water savings. (iii) Modernisation of irrigation systems is based on comprehensive assessments of system configuration and conditions (iv) The interests of farmers of increased profitability is recognised in irrigation and water resources planning	ACIWRM KNNL and other Nigams CADA Directorate Directly involved stakeholders: DA, DITB	February 2021
6. Stakeholder involvement and Capacity Building	Engagement of relevant stakeholders in planning and implementation, monitoring and evaluation of water resources management is a key "foundation" of IWRM. Stakeholders include the communities, businesses and government	(i) Communities in selected basins are aware of and actively participate in the conservation, utilization, and protection of natural resources	ACIWRM KNNL and other Nigams	August 2021

Key IWRM Theme	Definition and Activities	Program Objectives	Responsibilities	Timing
	agencies active in the basins. Building the capacity of stakeholders for IWRM is important to sustainable water resources management. Stakeholder participation will include: (i) Education, awareness raising, and capacity building of communities and individuals on water management issues; and (ii) Community involvement in river basin planning and management, irrigation modernisation, and Land and Water Management Plan programs.	(ii) Communities are actively involved in deciding levels of services and water tariffs by water management utilities (iii) Local communities are actively participating in forums for planning and managing basin water resources (iv) Government commitment to stakeholder and community participation in river basin planning, water resources management and irrigation system management		

ACIWRM = Advanced Centre of IWRM, CADA = Command Area Development Authority, CWC = Central Water Commission, DA = Department of Agriculture, DE = Department of Ecology and Environment, DeG = Department of e-Governance, DF = Department of Fisheries, DITB = Department of Information Technology and Biotechnology, DMG = Department of Mines and Geology, DMI = Department of Minor Irrigation, DPS = Department of Planning and Statistics, KNNL = Karnataka Neeravari Nigam Limited, KUWSDB = Karnataka Urban Water Supply and Sewerage Board, TB = Tungabhadra Board, WD = Watershed Development Department, WRA = Water Resources Authority, WRD = Water Resources Department, WRDO = Water Resources Development Organization, WUCS = Water User Cooperative Societies.

ATTACHMENT 2: DESIGN AND MONITORING FRAMEWORK FOR THE INVESTMENT PROGRAM

Design Summary	Performance Targets and Indicators with Baselines	Data Sources and Reporting Mechanisms	Assumptions and Risks
Impact Improved sustainable water security in selected river basins in Karnataka	By 2025: Non-agricultural use increases from 16% (in 2000) to 27% of total water use State gross value of annual agricultural production increases from Rs620.6 billion to Rs931.0 billion	WRD and/or KNNL annual reports and scheme monitoring reports Economic Survey of Karnataka, FY2013	Assumption Interstate water agreements are maintained Risk Climate change impacts exceed projections
Outcome IWRM successfully implemented in selected river basins in Karnataka	By 2024: State water resources management strategies institutionalized Systems for quantitative water use by sectors is established (from 2018) 99 out of 110 WUCSs confirm receipt of water service (baseline = 0)	WRD and/or KNNL annual reports and scheme monitoring reports WRA annual reports ACIWRM annual report and state annual budget report KNNL annual and scheme monitoring reports	Assumption Continued political and community and stakeholder support at the national level to sustain and proceed with reforms
Outputs 1. State and basin institutions strengthened for IWRM	By 2024: At least three river basin plans are developed (baseline = 0) At least 600 WRD staff (of which 200 are women staff) are certified in IWRM (baseline = 0) State water resources information systems established (baseline = 0)	River basin management meeting minutes WRD and KNNL reports and minutes ACIWRM and WRA annual reports	Assumption Certified WRD staff remain in office
2. Irrigation system infrastructure and management modernized	By 2024: 300 km of irrigation canal upgraded (baseline = 0)	WRD and/or KNNL annual reports and scheme monitoring reports	Assumption State government continues to support irrigation management

Design Summary	Performance Targets and Indicators with Baselines	Data Sources and Reporting Mechanisms	Assumptions and Risks								
	Asset management plans implemented for main system infrastructure on three selected systems (baseline = 0) More than 82 of 110 WUCSs operate and maintain transferred infrastructure, including fee collection (baseline = 0) At least 30% of WUCS management boards are women	CADA annual report	transfer, and WUCSs collecting irrigation service fees								
3. Program management systems operational	Project outputs delivered on time and within budget	Quarterly disbursement and contract award projections and financial reports prepared by WRD and/or KNNL									
Activities with Milestones Confirmation of aide memoire from Secretary to DEA (WRD, 13 May 2016) Output 1: State and basin institutions strengthened for IWRM - Submit draft contract documents for River Basin Modeller and Hydrologist to ADB for prior approval (ACIWRM, 15 June 2016) - Mobilize outstanding contractual staff (ACIWRM, 30 June 2016) - Submit draft contract document for Policy and Institutions Consultant to ADB for prior approval (ACIWRM,15 June 2016) - Mobilize Policy and Institutions Consultant (ACIWRM, 30 June 2016) - Publish website (ACIWRM, 31 May 2016) - Submit Submission 0's to ADB for prior approval for the Climate Change Specialist and Monitoring and Evaluation Expert (ACIWRM, 31 July 2016) - Complete design and accreditation of IWRM training program (ACIWRM, 31 December 2016) Output 2: Irrigation system infrastructure and management modernized - Submit draft SST subcontract to ADB for review and prior approval (KNNL, 7 May 2016) - Submit first draft CAD works package to ADB for review and prior approval (KNNL, 30 June 2016) - Award of at least 30 CAD packages to WUCS (KNNL, 31 December 2016) - Submit first Safeguards Monitoring Report to ADB (KNNL, 31 July 2016) - Obtain CWC's final approval of the VNC and TLBC DPR (KNNL, 30 September 2016) - Submission of draft VNC feasibility study (PSC, 30 June 2016) - Submission of draft VNC bid documents to ADB for review a prior approval (KNNL, 31 December 2016) - Submission of draft TLBC feasibility study (PSC, 30 November 2016)			Inputs (\$ million) <table><tr><td>ADB</td><td>150.0</td></tr><tr><td>Government</td><td>70.0</td></tr><tr><td>Beneficiaries</td><td>5.0</td></tr><tr><td>Total</td><td>225.0</td></tr></table>	ADB	150.0	Government	70.0	Beneficiaries	5.0	Total	225.0
ADB	150.0										
Government	70.0										
Beneficiaries	5.0										
Total	225.0										

Design Summary	Performance Targets and Indicators with Baselines	Data Sources and Reporting Mechanisms	Assumptions and Risks
- Initiate civil works for Vijaynagara and TLBC irrigation subprojects (June 2017) - Prepare main system asset management and operation and maintenance plan for all irrigation subprojects (May 2020) - Install telemetry systems throughout selected subbasins (May 2021) - Complete capacity building of 110 WUCSs and sign service agreement with KNNL at Vijaynagara and TLBC irrigation subprojects (August 2021) Output 3: Project and management systems operational - Loan amendment to be signed and made effective (KNNL, WRD, DEA, 30 June 2016) - PCC meetings held every quarter and minutes included within QPR (PCC, Quarterly) - QPR to be submitted one month after reporting period; first QPR for 2016 to be submitted to ADB (ACIWRM, KNNL, 15 May 2016) - PPMS and PMIS made operational (KNNL, 30 September 2016) - Recruitment and mobilization of MIS and GIS Communication Expert (KNNL, 30 June 2016) - Timely submissions of withdrawal applications to CAAA, one month prior to quarter end (KNNL, Quarterly) - Second withdrawal application submitted to CAAA (KNNL, 7 May 2016) - APFS for 2015-2016 submitted to ADB (KNNL, 30 September 2016) - Updated Facility Administration Manual submitted to ADB (KNNL, 31 May 2016) - Complete midterm effects monitoring (January 2018) and investment program effects monitoring (May 2021) - Project 2 fact finding mission (February 2017) - Government submits to ADB the Period Financing Request (April 2017) - Safeguard approvals by ADB (April 2017) - Loan negotiations (May 2017) - ADB Management approval (July 2017) - Loan signing (August 2017)			

ACIWRM = Advanced Centre for Integrated Water Resources Management, ADB = Asian Development Bank, CADA = Command Area Development Authority, FY = fiscal year, IWRM = integrated water resources management, km = kilometer, KNNL = Karnataka Neeravari Nigam Limited, MIS = management information system, PIO = project implementation office, PMU = project management unit, PSC = project support consultant, TLBC = Tungabhadra Left Bank Canal, WRA = Water Resources Regulatory Agency, WRD = Water Resources Department, WUCS = water users cooperative society.

Note: Baseline data supplied by project preparatory technical assistance for Karnataka Integrated and Sustainable Water Resources Management Investment Program (TA 7954-IND).

Source: Asian Development Bank.

ATTACHMENT 3: DESIGN AND MONITORING FRAMEWORK FOR PROJECT 1

Design Summary	Performance Targets/Indicators	Data Sources/ Reporting Mechanisms	Assumptions and Risks
Impact Integrated water resources management in selected river basins in Karnataka successfully implemented	By 2021: River basin management and State IWRM strategies established Quantitative water use by sectors is reported annually (from 2018) ACIWRM fully resourced with appropriately qualified staff 99 out of 110 WUCS confirm receipt of water service (baseline = 0)	WRD/KNNL annual reports and scheme monitoring reports WRA annual reports ACIWRM annual report and State annual budget report WRD/KNNL annual reports and scheme monitoring reports	Assumption Continued political and community/stakeholder support to sustain and proceed with reforms
Outcome River basin plan for Krishna-8 (K-8) subbasin successfully implemented	By 2018: Institutional arrangements for river basin management in the K-8 subbasin are established (currently none)	River basin management meeting minutes	Assumption Interstate water agreements are fully implemented.
Outputs 1. State and basin institutions strengthened for IWRM	By 2018: At least 200 WRD staff (of which 75 are women staff) are certified in IWRM (baseline = 0)	WRD/KNNL reports and minutes	Assumption Certified WRD staff remain in office
2. Irrigation system infrastructure and management modernized	By 2018: 123 km of irrigation canal upgraded at Gondi irrigation subproject Gondi irrigation subproject main-system asset management (operation and maintenance) plan developed (baseline = 0)	WRD/KNNL annual reports and scheme monitoring reports WRD/KNNL annual reports and scheme monitoring reports WRD/KNNL annual report	Assumption State government continues to support irrigation management transfer and WUCS collecting irrigation service fees

Design Summary	Performance Targets/Indicators	Data Sources/ Reporting Mechanisms	Assumptions and Risks
	Telemetry systems operational for at least 100 locations within the sub-basin At least 7 WUCS operating and maintaining transferred systems (baseline = 0) At least 7 WUCS have 30% representation of women on management boards	CADA annual report CADA annual report	
3. Project and management systems operational	KNNL approves annual work plan within 30 days of receipt PMU is operational Contract award and disbursement project versus actual	KNNL annual report Government notification of PMU establishment and staffing Quarterly disbursement and contract award projections and financial reports prepared by WRD/KNNL	Assumption Counterpart staff appointment to the project is retained
Activities with Milestones Confirmation of aide memoire from Secretary to DEA (WRD, 13 May 2016) Output 1: State and basin institutions strengthened for IWRM - Submit draft contract documents for River Basin Modeller and Hydrologist to ADB for prior approval (ACIWRM, 15 June 2016) - Mobilize outstanding contractual staff (ACIWRM, 30 June 2016) - Submit draft contract document for Policy and Institutions Consultant to ADB for prior approval (ACIWRM,15 June 2016) - Mobilize Policy and Institutions Consultant (ACIWRM, 30 June 2016) - Publish website (ACIWRM, 31 May 2016) - Submit Submission 0's to ADB for prior approval for the Climate Change Specialist and Monitoring and Evaluation Expert (ACIWRM, 31 July 2016) - Complete design and accreditation of IWRM training program (ACIWRM, 31 December 2016) Output 2: Irrigation system infrastructure and management modernized - Submit draft SST subcontract to ADB for review and prior approval (KNNL, 7 May 2016) - Submit first draft CAD works package to ADB for review and prior approval (KNNL, 30 June 2016) - Award of at least 30 CAD packages to WUCS (KNNL, 31 December 2016) - Submit first Safeguards Monitoring Report to ADB (KNNL, 31 July 2016) Output 3: Project and management systems operational			Inputs (\$ million) ADB 31.0 Government 16.6 Beneficiaries 0.4 Total 48.0

Design Summary	Performance Targets/Indicators	Data Sources/ Reporting Mechanisms	Assumptions and Risks
- Loan amendment to be signed and made effective (KNNL, WRD, DEA, 30 June 2016) - PCC meetings held every quarter and minutes included within QPR (PCC, Quarterly) - QPR to be submitted one month after reporting period; first QPR for 2016 to be submitted to ADB (ACIWRM, KNNL, 15 May 2016) - PPMS and PMIS made operational (KNNL, 30 September 2016) - Recruitment and mobilization of MIS and GIS Communication Expert (KNNL, 30 June 2016) - Timely submissions of withdrawal applications to CAAA, one month prior to quarter end (KNNL, Quarterly) - Second withdrawal application submitted to CAAA (KNNL, 7 May 2016) - APFS for 2015–2016 submitted to ADB (KNNL, 30 September 2016) - Updated Facility Administration Manual submitted to ADB (KNNL, 31 May 2016) Preparation of Project 2 - Obtain CWC's final approval of the VNC and TLBC DPR (KNNL, 30 September 2016) - Submission of draft VNC feasibility study (PSC, 30 June 2016) - Submission of draft VNC bid documents to ADB for review a prior approval (KNNL, 31 December 2016) - Submission of draft TLBC feasibility study (PSC, 30 November 2016)			

ACIWRM = Advanced Centre for Integrated Water Resources Management, ADB = Asian Development Bank, CADA = Command Area Development Authority, IWRM = integrated water resources management, km = 85kilometre, KNNL = Karnataka Neeravari Nigam Limited, MIS = management information system, PD = Project Director, PIO = project implementation office, PMU = project management unit, PSC = Project Support Consultant, WRD = Water Resources Department, WUCS = Water User Cooperative Society.

Note: Baseline data supplied by project preparatory technical assistance (TA 7954-IND: Karnataka Integrated and Sustainable Water Resources Management Investment Program).

Source: Asian Development Bank.

ATTACHMENT 4: DESIGN AND MONITORING FRAMEWORK FOR PROJECT 2

Impact the Project is Aligned with			
(i) Sustainable water security in selected river basins in Karnataka improved (program defined) ^a (ii) Water use efficiency in irrigated agriculture improved (program defined) ^a			
Results Chain	Performance Indicators with Targets and Baselines	Data Sources and Reporting	Risks
Outcome IWRM successfully implemented in selected river basins in Karnataka	By 2024: a. River basin plan for the Middle Krishna Basin (K2, K3, K4 sub-basins) endorsed by State IWRM Committee (2019 baseline: n.a.) b. 30 VNC WUCS confirmed receipt of improved water delivery service (2019 baseline: n.a.) c. VNC tertiary irrigation system operated and maintained by the VNC WUCS (2019 baseline: n.a.)	a. WRD and/or ACIWRM annual reports and scheme monitoring reports b–c. KNNL and/or CADA annual and scheme monitoring reports	Loss of political, community and stakeholder support at the state level to sustain and proceed with reforms
Outputs 1. State and basin institutions for IWRM strengthened	By 2023: 1a. River basin plan for the Middle Krishna Basin (K2, K3, K4 sub-basins) developed (2019 baseline: n.a.) 1b. At least 600 WRD staff (of which 200 are women staff) certified in IWRM (2019 baseline: 150 staff are certified under Project 1, of which 40 are women) 1c. State water resources information system expanded to include analysis and reporting modules (2019 baseline: n.a.)	1a. ACIWRM annual reports 1b. WRD, KNNL and ACIWRM progress reports 1c. ACIWRM annual reports	Trained WRD staff do not remain in office
2. VNC irrigation system infrastructure and management modernized	By 2023: 2a. 12 anicuts, intake structures, and 442 km of irrigation canals covering 11,150 ha upgraded (2019 baseline: n.a.) (RFI A) 2b. Asset management plan for the main VNC system infrastructure prepared (2019 baseline: n.a.)	2a–b. KNNL annual reports and scheme monitoring reports	State government does not operationalize policy for irrigation management transfer and WUCS collection of irrigation service fees

Results Chain	Performance Indicators with Targets and Baselines	Data Sources and Reporting	Risks
	2c. 30 WUCS established and trained on operation and maintenance of tertiary irrigation system and improved on-farm water application methods (with 30% women representation in management boards and 40% attendance in training) (2019 baseline: 0) 2d. At least 50% of all WUCS officers and members (with at least 60% men's participation) reported increased knowledge on the application of gender concepts and principles to water resource management. (2019 baseline: n.a.) 2e. At least 50% of KNNL staff (including 30% women) reported increased knowledge on the application of gender concepts and principles to water resource management. (2019 baseline: n.a.)	2c. KNNL and CADA annual reports with sex-disaggregated data. 2d-e. Pre- and post-training assessment.	
3. Program management systems operational	By 2023 3a. Project outputs delivered on time and within budget (2019 baseline: n.a.) 3b. Project MIS expanded to include VNC activities with gender indicators (2019 baseline: n.a.) 3c. Quarterly progress reports and safeguard monitoring reports submitted on time with sex-disaggregated data and analysis (2019 baseline: n.a.) 3d. TLBC periodic financing request and associated due diligence documents prepared and advance procurement actions initiated (2019 baseline: n.a.)	3a. Quarterly disbursement and contract award projections and financial reports prepared by KNNL and ACIWRM 3b-c. KNNL and ACIWRM quarterly progress reports and semi-annual safeguard monitoring reports 3d. ADB mission aide memoire	

Results Chain	Performance Indicators with Targets and Baselines	Data Sources and Reporting	Risks
Key Activities with Milestones			
1. State and basin institutions for IWRM strengthened			
1.1 Initiate preparation of profile and basin plan for Middle Krishna River basins (August 2019).			
1.2 Continue developing the Karnataka Water Resources Information System (August 2019)			
1.3 Continue providing IWRM training to WRD staff (August 2019)			
2. VNC irrigation system infrastructure and management modernized			
2.1 Initiate civil works for VNC subproject (July 2019)			
2.2 Establish VNC WUCS and initiate strengthening activities, including on-farm extension services (January 2020)			
2.3 Initiate community participation packages with VNC WUCS (January 2020)			
2.4 Complete all works and WUCS activities (July 2023)			
2.5 Conduct gender sensitization workshops for KNNL staff and 30 VNC WUCS (July 2019)			
2.6 Finalize feasibility studies, detailed project report, detailed designs, and bid documents for TLBC subproject (June 2020)			
3. Program management systems operational			
3.1 Fully staffed PIO by loan effectiveness (August 2019)			
3.2 Update MIS with gender indicators (August 2019)			
3.3 KNNL prepares PFR for Tranche 3 to ADB (July 2020)			
3.4 Project 2 outputs delivered on time and within budget (July 2023)			
Inputs			
ADB: \$91 million (regular ordinary capital resources loan)			
Government: \$39 million			
Assumptions for Partner Financing			
Not Applicable			

ACIWRM = Advanced Center for Integrated Water Resources Management, ADB = Asian Development Bank, CADA = Command Area Development Authority, GAP = gender action plan, ha = hectare, IWRM = integrated water resources management, km = kilometer, KNNL = Karnataka Neeravari Nigam Limited, MIS = management information system, n.a. = not applicable, PFR = periodic financing request, PIO = project implementation office, TLBC = Tungabhadra Left Bank Canal, VNC = Vijayanagara Channels, WRA = Water Resources Regulatory Agency, WRD = Water Resources Department, WUCS = water users cooperative society

^a This impact was formulated following the previous design and monitoring framework guidelines. However, it remains aligned with: NITI Aayog. 2017. *India Three Year Action Agenda 2017–18 to 2019–20*. New Delhi.

Contribution to ADB Results Framework

RFI A: Land improved through irrigation, drainage, and/or flood management (hectares). Target: 11,150 ha.

Source: Asian Development Bank.

ATTACHMENT 5: SELECTION CRITERIA AND APPROVAL PROCESS FOR SUBPROJECTS

Summary Subproject Selection Criteria and Approval Process

1. **Selection Criteria.** Under the investment program, the following criteria will apply in selecting subprojects intended for financing under the Investment Program.

- (i) The subprojects will be located within selected sub basins in Karnataka which will be identified by the Water Resources Department in consultation with ACIWRM. The sub-basins will be suitably representative of the typical constraints faced in water resources management, like inter sector water demands between various users, have identifiable investment opportunities which meet state government and ADB financing criteria and any development interventions do not adversely impact on inter-state water agreements.
- (ii) They should be consistent with IWRM principles, which are: (i) water is a finite and vulnerable resource, essential to sustain life, development and the environment; (ii) water development and management should be based on a participatory approach, involving users, planners and policy makers at all levels; (iii) women play a central part in the provision, management and safeguarding of water; and (iv) water has an economic value in all its competing uses and should be recognized as an economic good.
- (iii) There will be no duplication of activities on subprojects with other on-going programs which are financing the same activities in the same area.
- (iv) The subprojects will involve modernization of the existing irrigation and associated infrastructure. Subprojects will either be major (having a command area of over 10,000 hectares [ha]) or medium-sized (between 1,000 ha and 10,000 ha).
- (v) For each subproject, a participatory feasibility study (as further detailed below) and necessary due diligence will have been prepared, covering technical design, economic and financial viability, institutional arrangements, social assessments including social safeguards plans as applicable, environmental analysis, and an implementation plan.
- (vi) All necessary central and state government approvals will be in place.
- (vii) For individual subproject feasibility assessments, the following criteria will apply:
 - (a) technical feasibility, with no significant technical risks that would undermine efficacy, economic return, safety, or sustainability;
 - (b) reliable water availability and quality, with no significant negative impact on the other users of the same source or the ecosystem downstream;
 - (c) financial and economic viability with an economic internal rate of return of over 12% with robustness under sensitivity and risk analysis;
 - (d) social and environmental soundness with no significant negative impacts;
 - (e) consideration of gender mainstreaming principles;
 - (f) fulfillment of safeguards requirements (involuntary resettlement, indigenous peoples, and environment) of the Government and ADB, including the preparation of required safeguard planning documents prepared following the Environmental Assessment and Review Framework and the Resettlement Framework, and efforts to minimize land acquisition and resettlement in the planning and design process; and
 - (g) a record of consultation and participation of the stakeholders prior to and during the feasibility study, with concurrence of the concerned water user cooperative societies (WUCS) on the basic design, including farmer contribution (in cash or in kind) equivalent to 10% of the contract value (as

per the Government norms which is currently under practice), for minor facilities and command area development works like Field Irrigation Channels and Land Reclamation (in accordance with prevailing Government norms and practice), and operation and maintenance (O&M) of the minor facilities within the WUCS boundary.

2. **Procedures.** For the purpose of the first periodic financing request (PFR) under the investment program, an appraisal process has been undertaken for one medium irrigation subproject (Gondi).

3. For all irrigation subprojects to be proposed for inclusion in the subsequent PFRs, each subproject will be processed in accordance with the following procedures:

- (i) A consultative process will be adopted and fully documented demonstrating the involvement of stakeholders in the subproject identification and development process. This will include a communication plan for the subproject.
- (ii) Feasibility studies will be conducted by the executing agency (EA) including provision of subproject development cost estimates. The EA will also prepare an initial poverty and social assessment and fill out checklists for involuntary resettlement, indigenous peoples, and environment; and submit the same to ADB for review and categorization.
- (iii) Based on the categorization and the feasibility studies, the EA will prepare the draft appraisal reports for all subprojects to be considered under the respective PFRs following the selection criteria, together with any required attachments, i.e., draft resettlement plan (RP),¹ draft indigenous peoples development plan, and draft environmental assessment.²
- (iv) During the above process, the EA, with the assistance of the program support consultants, will also assess the progress and performance of the implementation of the preceding loans under the Investment Program, with necessary actions to enhance their effectiveness.
- (v) If a proposed subproject is not likely to satisfy the eligibility criteria and/or the agreed procedures, ADB will advise the EA to either (a) modify the subproject proposal in a manner that will make it eligible, or (b) reject the subproject. In the latter case, the EA may propose a replacement subproject under the loan.
- (vi) Upon completing the feasibility assessments of the concerned subprojects and progress and performance review of the preceding loans under the Investment Program, the EA will submit these documents with the PFR to ADB for approval. The EA will ensure that the necessary approvals of the Government of India or the State have been obtained prior to submission of the PFR.

¹ The EA will translate the RP into the local language and disclose it to the affected people and incorporate the results of the consultation. The final RP will also be disclosed on the ADB website and the website of the EA.

² If any of the subprojects are categorized as environmental category A or B sensitive in any subsequent loan, compliance is required with the 120-day advance disclosure rule. The summary environmental impact assessment or summary initial environmental examination must be circulated to the ADB Board of Directors and made available to the general public 120 days before the PFR is submitted to ADB.

ATTACHMENT 6: SUBPROJECT IMPLEMENTATION METHODOLOGY

Irrigation Planning and Implementation—Main Activities and Responsibilities

	Activity	Main responsibility	Follow-up
1	Awareness Campaign		
	Information dissemination throughout Project target area	PIO, SST with WUCS support	
	Preparation of water awareness materials including modernization under IWRM approach in Kannada	PSC, PIO, SST with WUCS support	
	Preparation for folk play (Bayalu Nataka)	PSC, PIO with SST	
	Enactment of open theater play	PIO to engage and monitor and SST to facilitate	
	Materials on sub project and sub basin in Kannada	PSC and PIO and SST to assist	
	Village consultation meetings	SST	
	Main system walks through/feedback survey (to be undertaken before construction and during construction)	PIO/PSC, SST and WUCS	PMU/PSC on technical design components
2	Training Designs		
	Preparation of WUCS Benchmark Formats	PIO, SST with WUCS support and with inputs from PSC	
	Benchmark survey	SST with WUCS support	To submit filled in formats to PIO
	Preparation of training modules and methods on various items like WUCS roles, responsibilities, financial management based on Karnataka Irrigation Act, Karnataka Command Area Act & Karnataka Cooperative Act, its rules and regulations MOU and handing over,	PSC, PIO and SST with the involvement of WUCS	
	Irrigation Management	PSC, PIO and SST with the involvement of WUCS	
	Water Management in different cropping systems	PSC, PIO and SST with the involvement of WUCS	
	SRI Paddy, Farm Mechanizations and field visit a. SRI (Paddy) method of cultivation b. Farm mechanization c. Exposure visits	PIO and SST with the involvement of WUCS and inputs from PSC	
	Improved Cropping Systems a. Crop Intensification Soil testing and its importance, b. INPM, Bio-pesticides and Bio-fertilizers, d. Organic farming	PIO and SST with the involvement of WUCS	
	WUCS as multi-purpose organization	PSC, PIO and SST with the involvement of WUCS	
	Training modules checked & approved	PMU	PMU reviews and endorse training modules
	Publication of training modules and other support materials	PIO	
3	Community Mobilization	SST with WUCS support	PIO to monitor
4	Management Information System of Project (Kannada and English) a) Project b) Canal Network, c) Organizational structure of KNNL/CADA/WUCS Agriculture profile	PIO and SST with WUCS support and inputs from PSC	
5	MIS of WUC (Kannada and English)	PIO and SST with WUCS support	
6	Records/Bookkeeping (Kannada)	PIO and SST with WUCS support	

	Activity	Main responsibility	Follow-up
7	WorkBook Formats Preparation and Printing	PIO and SST with WUCS support and inputs from PSC	
8	Performance Indicators and PSA Charts in Kannada	PSC, PIO and SST with WUCS support	
9	Materials on Measuring Devices in Kannada and English	PSC, PIO and SST with WUCS support	
10	Compendium of Government Orders related to Irrigation and WUCS in Kannada	PMU	
11	Flood Management Guidelines	PSC, PIO and SST with WUCS support	
12	Ground water mapping and Conjunctive Use	PMU, PIO and SST	
13	Identification of CAD works a. Walkthrough survey on canals to identify works b. Prioritization of works and approve in GB meeting c. Preparation of estimates	PIO and SST with WUCS support	PMU to make periodic progress
14	Social audit boards	PIO and SST with WUCS support	
15	Construction of CAD works with standard designs		
	Technical training to WUCS on sustainable quality and quantity of construction works	PSC, PIO with SST	On site regular visits by SST
	Identification and prioritization of works through participatory walk through survey	WUCS with support from PIO and SST with inputs from PSC	
	Preparation for works during pre-closure and closure	WUCS with support from PIO and SST with inputs from PSC	
	Facilitating identification of materials	PIO with SST	
	Carrying out construction of CAD works by WUCS	WUCS Training by PIO and SST and inputs from PSC	
	Recording Measurement Book	PIO with SST	
	Financial Management	PIO with SST and inputs from PSC	
16	Operation and maintenance		
	WUCS engaging their own secretaries and <i>laskhars</i> / <i>neergantis</i>	WUCS Training by PIO and SST	
	Micro plan preparation including area irrigated, water charges and water rates and collection (WorkBook Exercise)	Community Training provided by Project	PIO tests different models with WUCS
	Works contract with WUCS	PIO	
	MOU for water management with WUCS	PIO and SST staff with WUCS support	
	Water rates prepared according to volumetric supply and timely supplied to WUCS	PIO and SST with WUCS support	
	Collection of water charges and remittance through WUCS	Community mobilization by SST and WUCS	Comparative progress
17	Monitoring and evaluation		
	Following construction, review of scheme implementation in order to identify problems and improve procedures for future subprojects	PIO with SST with inputs from PSC	PIO with SST
	Spot checks of subprojects during implementation	Safeguards and Quality Monitoring Cell	Identifies problems and solutions, provides feedback to PIO

	Activity	Main responsibility	Follow-up
	Socio-economic evaluation of 20% of subprojects about 1 year after construction	Safeguards and Quality Monitoring Cell with inputs from PSC	Identifies problems and solutions, provides feedback to PIO
18	Formation of Main System WUCS Federation		
	Fulfill the eligibility requirement to form federation at all the primary level WUCS	PIO and SST with WUCS	
	Facilitate identification of chief promoter to form federation as required by Cooperative Act	PIO and SST with WUCS	
	Preparation of Bylaws of WUCS Federation	PIO and SST with WUCS	Wetted by Registrar of Cooperatives

CAD = command area development, INPM = integrated nursery pest management, KNNL = Karnataka Neeravari Nigma Limited, MOU = memorandum of understanding, PIO = Project Implementation Office, PMU = Project Management Unit, PSC = Project Support Consultant, SRI = system of rice intensification, SST = Support Services Team, WUCS = water user cooperative societies.

ATTACHMENT 7: UPDATED DRAFT TERMS OF REFERENCE FOR PROJECT 2 SUPPORT CONSULTANTS AND ACIWRM INDIVIDUAL CONSULTANTS (JULY 2019)

A. Project 2 Support Consultants

1. The Project 2 Support Consultants (PSC 2) shall review and build upon the knowledge, experience and systems already developed under Tranche 1. These include the water user cooperative societies' (WUCS) capacity building strategy, training activities, guidelines, procedures, and manuals, communication strategy and training activities, the online performance monitoring/management information system, and the GIS-based Asset Management Plan and the O&M plan. The principal activities of PSC 2 include:

1. Overall Coordination and Implementation

- (i) Provide overall technical, monitoring and management support to the PMU and PIOs for successful implementation of the Project;
- (ii) Provide support to PMU, PIOs and other concerned agencies for program implementation-specifically for Outputs 2 and 3 (with KNNL) and coordinate with PIO (ACIWRM) for Output 1;
- (iii) Prepare a program implementation plan (PIP) and annual implementation plans in consultation with the PMU and PIOs;
- (iv) Coordinate with relevant agencies implementing similar works in the sector for improved agricultural production;
- (v) Provide support and training to KNNL, PIO and contractors on the themes of quality control, construction management and contract administration, and
- (vi) Support KNNL for progress monitoring of works using latest software for analyzing output such as MS Office. S-curves may also be used to represent the physical and financial progress on a monthly basis and included in the periodic reports.

2. Capacity Building

- (i) Organize training to PMU, PIO and CADA in ADB procurement procedures, financial management and project management;
- (ii) Organize training in ADB communications policy and safeguard procedures;
- (iii) Prepare project implementation guidelines, procedures and appraisal manuals for PIOs;
- (iv) As part of long-term capacity building and training needs of KNNL, WRD and CADA, prepare a training plan including estimated costs and submit to KNNL. The training needs should specifically include social and gender, and environmental safeguard aspects;
- (v) Work with State Water Institutions and work for dovetailing/convergence of training activities;
- (vi) Prepare training manuals for WUCS as needed in basic construction techniques and financial management (bookkeeping); draw upon Tranche 1 Gondi best practices while preparing the manuals;
- (vii) Prepare and implement a GIS-based Asset Management Plan and an O&M Plan for VNC project; draw upon Tranche 1 Gondi experience;
- (viii) Implement the Gender Action Plan (GAP) for enhanced women's participation in the VNC project, and
- (ix) Provide technical guidance and expertise for resettlement planning and implementation, if necessary.

3. Program Monitoring Management Systems

- (i) Support PMU, KNNL and ACIWRM, and other consulting agencies (such as for environmental monitoring, and monitoring and evaluation) to determine the optimal approach for data gathering, storage and use for planning and monitoring. This may require one MIS database and a number of other topic-specific (e.g., safeguard monitoring, environment) databases. The custodian of a centralized MIS is to be decided by KNNL;
- (ii) Develop MIS database for physical and financial monitoring as part of Project Performance Monitoring System (PPMS) along with PMIS;
- (iii) Develop content and share with KNNL to host on KNNL website reflecting, among others, the physical and financial progress and indicators;

4. Project and Financial Management

- (i) Provide technical support to PMU and PIOs in monitoring physical and financial progress including in the preparation of periodic reports (quarterly progress reports, mid-term report and project completion report) for submission to ADB. PMU will hire a Financial Management consultant to ensure timely submission of withdrawal applications, timely payments, cash flow management and audit.

2. Total international inputs will be for 8 person-months (p-m), and 308 p-m for national consultants for the VNC project. Computer literacy in using latest software (such as MS Office) will be mandatory for all specialists recruited.

Table 1: Estimated Person Months for Specialists and Non-Key Specialists

Position	Person-months
International	
International Irrigation Specialist	8
Total International Key Specialists	8
National	
Team Leader/ PIM Specialist	30
Irrigation Specialist	18
Operation & Maintenance Specialist	20
Water Institutions Specialist / DTL	33
Environment Specialist	15
Cultural Heritage Specialist	4
Bio-diversity Specialist	8
Agriculture Specialist	22
Agriculture Economist	7
Social Development & Gender Specialist	26
Construction Management Specialist 1	20
Construction Management Specialist 2	13
Communication & Documentation Specialist	26
MIS/GIS Specialist	9
Procurement Specialist	5
Training Coordinator 1	26
Training Coordinator 2	26
Sub-Total (National Experts)	308
Total (International + National Experts)	316
Support Services Team for Tranche 2-VNC and TLBC*	
Community Organizer (Female-3 persons, 33 months; 2 prsns@12 months/prsn)	123

Position	Person-months
Community Organizer (Male-3 persons, 33 months; 2 prsns@12 months/prsn)	123
Water Management Extension Associate (3 persons, 38 months)	90
Agriculture Extension Associate (3 persons, 38 months)	90
Total (Support Services Team)	427
Non-Key Experts	
PSC (includes SST 1) in Hospet/Munirabad	
Construction Management Engineer (2 persons @30 months)	50
Site cum Design Engineer (5 persons @24 months)	120
MIS/IT executive	28
Office Superintendent	26
Accountant	33
Computer/ Data Entry Operator (2 persons @30 months)	52
Sub-Total (Non-Key Experts – PSC+SST 1)	309
SST 2 Office (in Sindhanur)	
Office Manager-cum-Accountant	30
Computer / Data Entry Operator	30
Sub-total (Non-key Specialists – SST 2)	60
Total (Non-key Experts)	369

* 3 SST for Tranche 2; one more team with 4 members (4 Community Organizers-2 male, 2 female) for preparatory activities of TLBC as described in the Scope of Services.

3. The Terms of Reference of new PSC 2 specialist positions are given below.

5. Irrigation Specialist (international, 8 person months)

4. The International Irrigation Specialist/ (IIS) will be required to work closely with PMU, and with AC-IWRM where required, for Output 1 of the Program. The IIS/ will either be a chartered professional engineer or have a master's degree in civil or agricultural engineering or Water Resources engineering or Irrigation Engineering. S/he will have at least 15 years of project implementation experience, out of which at least 7 years shall be in irrigation/water resources development with major irrigation systems in Asia, preferably South Asia/India.

5. The IIS will be directly reporting to the Team Leader and will provide support to PSU in program implementation. Specifically, the IIS will:

- (i) Guide, coordinate and support PSC activities and provide overall guidance and direction and ensure that the PSC works in harmony with AC-IWRM, PIO and KNNL;
- (ii) Assist in preparation of Inception Report including methodology and detailed work plan for PSC 2 Team that conforms to project implementation work plan;
- (iii) Support in planning and coordinating physical interventions under Outputs 2 and 3 as necessary;
- (iv) Assist the Project Director in planning, management, monitoring and reporting of all project activities including progress monitoring systems;
- (v) Assist PIOs in establishing robust operational practices for participatory irrigation management system;
- (vi) Develop indicators for monitoring equitable distribution of canal supply based on flow monitoring system implemented under the Project;
- (vii) Guide/prepare training materials on technical aspects (including for project management and information systems) and participatory irrigation management (PIM);

- (viii) Support development of a Management Information System (MIS) for the Program to address, among other things, project operational performance, operation and maintenance, social and environmental safeguard monitoring;
- (ix) Contribute to preparation of quarterly, half yearly, annual, end of assignment and other periodic reports listed under 'Deliverables' as per ADB and KNNL requirements spelt out in contract agreement;
- (x) Coordinate with WRD, KNNL and AC-IWRM on any relevant activities related to river basin planning.

6. Team Leader / Participatory Irrigation Management Specialist (national, 30 person-months)

6. The Team Leader/Participatory Irrigation Management (PIM) Specialist will have relevant background in water resources management, natural resource management, social sciences or other associated subjects with at least a master's degree. The specialist shall have at least 15 years of experience and associated with participatory processes. S/he shall have a working knowledge and implementation experience of similar irrigation/rural development projects. Working knowledge of Kannada language is desirable.

7. The TL/PIM Specialist will establish systems for promoting participatory irrigation management and command area development, impart training and support the PIM process. He will assume responsibility for day-to-day activities of the PSC 2. Specifically, the TL will:

- (i) Guide, coordinate and support PSC activities and provide overall guidance and direction and ensure that the PSC works in harmony with AC-IWRM, PIO and KNNL;
- (ii) Prepare an Inception Report including methodology and detailed work plan for the PSC 2 Team that conforms to project implementation work plan;
- (iii) Assist the PIO in planning and coordinating physical interventions under outputs 2 and 3;
- (iv) Assist the Project Director in planning, management, monitoring and reporting of all project activities including progress monitoring systems;
- (v) Support development of a Management Information System (MIS) for the Program to address, among other things, project operational performance, operation and maintenance, social and environmental safeguard monitoring;
- (vi) Coordinate with PIO and provide guidance to WUCS and CADA for promoting/strengthening PIM and participatory CAD activities;
- (vii) Prepare quarterly, half yearly, annual, end of project reports;
- (viii) Work with specialists to ensure that their terms of reference are achieved in sync with the PIP;
- (ix) Assist CADA and KNNL in developing and implementing Action Plans for PIM and WUCS;
- (x) Plan and operationalize at Project level methods, procedures, and systems for institutional strengthening of WUCS along with WIS;
- (xi) Guide the communications and documentation specialist to implement the stakeholder communication strategy and develop communication tools and materials for WUCS, CADA, KNNL and process related documents.

7. Water Institutions Specialist

8. The Water Institutions Specialist (WIS) shall have at least a master's degree in civil engineering, with at least 15 years of experience in water resources management, institutional and

human resource management, community development or other associated areas. The WIS will be directly reporting to the TL/DTL and will undertake the following specific tasks:

- (i) Prepare and develop a Capacity Building Strategy and Action Plan for WUCS on Institutional Strengthening and O&M.
- (ii) Build capacities of SSTs to enable them to provide effective support to WUCS in collaboration with KNNL and CADA.
- (iii) Implement the Capacity Building Strategy and Action Plan in coordination with KNNL and CADA, and other state training institutions as appropriate.
- (iv) Work with SST and the Training Specialist/Training Coordinator to conduct training programs as per timeline.
- (v) Improve irrigation scheduling and volumetric based supply towards increased water use efficiency through KNNL/CADA and WUCS Federation interactive meetings.
- (vi) Work with PIM specialist for improving coordination between KNNL, CADA and WUCS for establishing effective participatory irrigation management systems.
- (vii) Provide need-based support to concerned state training institutions such as Regional Institute of Cooperative Management (RICM), Water and Land Management Institute (WALMI) and Karnataka Engineering Research Station (KERS) for improving training curriculum, pedagogy, upgrading their training infrastructure, and faculty improvement (based on budget available in this sub-project).

8. Irrigation Specialist (national, 18 person-months)

9. The Irrigation Specialist (IS) will have a master's degree in water resources engineering or irrigation or hydraulics or a related discipline, and preferably have 10 years of experience in integrated water resources management. The IS will be directly reporting to the TL. Specific responsibilities include the following:

- (i) Coordinate with KNNL engineers to facilitate effective execution of works by contractor on the main canals and distributaries;
- (ii) Interact with Water Users Cooperative Societies to understand their requirement at field level and help in planning the CAD works;
- (iii) Work with CADA and WUCS in the execution of CAD works on FICs;
- (iv) Monitor quality of works both by contractor and WUCS, identify issues and facilitate addressing by KNNL and CADA in time;
- (v) Prepare progress reports periodically for meetings with KNNL and CADA and PMU, and for periodic reports to be submitted by PSC to PMU/ADB;
- (vi) Work with Agricultural Specialist to improve water use efficiency as well as promote appropriate crops and cropping pattern as per feasibility study and PPTA reports;
- (vii) Promote micro irrigation and conjunctive irrigation wherever possible, and
- (viii) Work with KNNL towards improving irrigation scheduling.

9. Cultural Heritage Specialist (national, 4 person-months)

10. The Cultural Heritage Specialist (CHS) shall have at least a Degree in Architecture, Planning, or a related discipline with a minimum of 10 years of experience in working in heritage sites. S/he must have a demonstrated ability to analyze site management problems, develop indicators, devise solutions, and design conservation and heritage components and must have been associated with civil construction works in Heritage areas. The CHS reports directly to the TL and will play a leading role in the tasks described below:

- (i) In consultation with PMU/PIO, review the design suggested in the DPR and its fit within the overall site;

- (ii) Assess and, where necessary, mitigate possible adverse impacts of the project's physical and non-physical interventions on the affected monuments and their surrounding buffer zones;
- (iii) Oversee implementation of modernization works within the conservation area, restoration, removal, addition, rearrangement and other interventions that may be required so as not to disturb the existing Heritage structures due to the proposed canal rehabilitation works;
- (iv) In close collaboration with the O&M Expert, Environment Specialist and Social and Gender Development Specialists, devise protocols and systems for on-site monitoring and predictive modelling and reporting of the impact of tourism, if any, on the site, their buffer zones, and the surrounding environment;
- (v) Ensure implementation of the Environmental Management Plan for modernization works in heritage areas as provided in the EIA chapter of the VNC Feasibility Study Report;
- (vi) Prepare an Operations Manual for implementation of the aforementioned protocols and systems and conduct workshop to train site managers in the application of the manual;
- (vii) Collection of evidence and preparation of reports towards compliance related to conservation and mitigation efforts implemented as per the guidelines in the EIA chapter in the VNC Feasibility Study Report;
- (viii) Contribute to the preparation of environmental monitoring reports for submission to ADB.

10. Bio-diversity Specialist (national, 8 person-months)

11. The Bio-diversity Specialist (BS) shall have at least a master's degree in Environmental Sciences, Life Sciences, or related discipline with a minimum of 10 years of experience in projects dealing with Biodiversity and Ecological conservation. S/he should have experience in monitoring, evaluation and documentation of bio-diversity conservation measures required in infrastructure development projects in and around ecologically sensitive areas for flora and fauna. The Specialist will support the smooth running of the Project in accordance with project work plans and develop strategy for preservation of the ecological balance. The BS will report directly to TL and shall undertake the following key responsibilities:

- (i) Develop a project implementation strategy and plan for an ecologically safe construction schedule to minimize disturbance to ecologically fragile species of the region;
- (ii) Surveys of flora and fauna before and during construction, and structural inspections regarding potential roosting of the local fauna;
- (iii) Integration of solutions to nature conservation issues into the construction program;
- (iv) Early detection of adverse impacts with respect to flora and fauna and implementing mitigation measures;
- (v) Coordination with relevant authorities as needed;
- (vi) Sensitize the project staff (both Engineers and Contractors) and Water User Cooperative Societies about protecting local biodiversity as well as conservation aspects;
- (vii) Collection of evidence and preparation of reports towards compliance related to conservation and mitigation efforts implemented as per the guidelines in the Biodiversity and EIA chapters in the VNC Feasibility Study Report.

11. Agriculture Economist (national, 7 person-months)

12. If an institution or a University is willing to provide this service, PSC will enter into an MoU with clearly agreed terms of reference when and as needed for specific tasks. The CV of the key expert will be shared with KNNL/ADB for approval.

13. The Agriculture Economist should have at least a post graduate degree in Agriculture Economics or similar with a work experience of 15 years. He should have carried out economic and financial analysis of small and medium level irrigation projects and possess practical understanding of field level issues. Specific tasks include:

- (i) Carry out project benefit and cost analysis using with-project and without-project scenarios. The basic criteria for assessing project economic viability will be in terms of economic parameters such as economic net present value and economic internal rate of return;
- (ii) Collect/compile/collate data needed for economic and financial analysis and use the same in agricultural economic analysis;
- (iii) With the help of the Agriculture Specialist, prepare production budgets as necessary, and quantitatively assess physical production responses (e.g., crop yields) due to project intervention, and future scenarios;
- (iv) Carry out sustainability analysis to explain if net benefits sustain through the life span of the project at a level sufficient to meet the economic viability criteria;
- (v) Undertake sensitivity and risk analysis.

12. Construction Management Engineer (2 positions – 25 person-months each)

14. The Construction Management Engineer (CME) shall have at least a bachelor's degree in Civil Engineering or Irrigation Engineering or a related discipline with at least 8 years of experience in construction projects preferably Irrigation Projects and having worked with minimum supervision. The CME will be directly reporting to TL and will undertake following tasks:

- (i) Together with CMS, monitor the progress of the Contractor's Construction works as per project deadlines and prepare progress reports (both Physical and Financial related to the civil works);
- (ii) Coordinate with KNNL for main and distributary works, and with CADA and WUCS for CAD works in ensuring quality, identifying site issues and work for redressal through appropriate divisions;
- (iii) Coordinate with KNNL offices at PIO and PMU levels in collecting and collating data for progress reporting;
- (iv) Monitor the work of the Site cum Design Engineers with regard to progress and quality of CAD works;
- (v) Any other duties assigned by Team Leader/ Construction Management Specialist, relevant to the project.

13. Support Services Team (427 person months)

15. Specific tasks of the SST members, expected outputs, implementation arrangements and others remain the same as in the main ToR. In addition to the 3 Support Service Team for VNC subproject, there will be one (SST) team comprising four members-two Community Organizers (Female) and two Community Organizers (Male) to carry out preparatory activities for TLBC in terms of initiating mobilization of WUCS and organizing them as defined in the Scope of Services. Two Training Coordinators (placed in the two SST offices) will together be responsible for executing the capacity building action plan for the 30 WUCS spread across three districts.

16. **Program/Project Performance Monitoring.** During the course of Tranche 2 implementation, monitoring and reporting will be carried out for as in Tranche 1: (i) periodic progress reporting; (ii) safeguard monitoring; and (iii) physical and financial monitoring. The VNC project and program DMF indicators shall be the basis for monitoring and reporting. The results of monitoring will be placed on the website of KNNL as part of PPMS and PMIS.

17. **GIS-based Asset Management Plan and O&M Plan.** As done for Tranche 1, the GIS activity will be further improved and updated to cater to Tranche 2 requirements.

18. **Compliance and Safeguard Monitoring.** The compliance and monitoring (policy, legal, financial, economic and other covenants) and safeguard aspects (social and poverty action plans) for Tranche 2 will be carried out as specified in the Tranche 1 ToR. Further, the PIO will ensure that the Project is implemented with due concern for safeguards and quality, and specifically, to ensure that safeguard issues are adequately addressed to meet the requirements of GoK and ADB. This will include: (i) environmental quality monitoring of water and soil quality parameters (as referred to in the Internal Environmental Examination/EIA) periodically and reported; (ii) potential resettlement impacts (if any) such as compensation for loss of income; (iii) quality of work necessary to ensure sustainable outputs; (iv) biodiversity monitoring; (v) monitoring of compensatory afforestation, and (vi) guiding and monitoring of implementation of the Contractor's EMP both in OCR and Hampi heritage area.

19. PSC shall be carrying out safeguard monitoring and submit a half yearly Safeguard Monitoring Report to KNNL, which after review will be submitted to ADB.

20. **Physical and Financial Monitoring.** PSC shall monitor overall progress of the project, on monthly basis, using latest software for analyzing output such as MSOffice; further s-curves should be prepared every month for monitoring physical and financial progress, and include in the quarterly progress reports and the KNNL website.

21. **Other Review and Monitoring Mechanisms.** The Project Director will facilitate (at least) six monthly reviews by the Government and ADB. At the government level, the reviews will be undertaken by the State Steering Committee, and the Program Coordination Committee. In addition, the Project Director may also conduct frequent reviews to ensure smooth progress by addressing emerging issues. The reviews will: (i) examine activity schedules with particular focus on IWRM targets and institutional strengthening parameters such as capacity building; (ii) examine the continuing appropriateness and effectiveness of implementation arrangements; (ii) construction standards and quality of works implementation; (iii) review compliance with agreed procurement procedures; (iv) analyze overall physical and financial progress; (iv) review of capacity building and training programs; and (v) review of effectiveness of safeguard procedures. ADB Missions will review once in every six months.

22. **Online Project Monitoring System.** The PPMS, PMIS and website developed during Tranche 1 will be modified/upgraded for VNC project as required. The PMU will monitor online the project progress for activities, compliance with safeguard requirements, and progress toward output targets from time to time.

23. **Gender and Social Dimensions monitoring.** The program shall adopt effective gender mainstreaming and work for Gender Equality and Social Inclusion (GESI) approach. It sets out gender monitoring indicators and targets for each project component. All data on KNNL and CADA staffing and recruitment, participation, consultation, wages, training, safeguards implementation and

compensation, and impacts will be sex disaggregated. A similar sex-aggregated data will be collected for WUCS as well as the federations. The Management Information System (MIS) will be established accordingly for Tranche 2 right from start up to ensure that data is stored and managed well for various purposes. The Social Development and Gender Specialist shall be responsible for this task of monitoring of the GAP.

24. **Progress Reports to ADB by KNNL.** PMU shall submit to ADB the following reports: (i) QPRs in a format consistent with ADB's PPMS; (ii) consolidated annual progress reports including: (a) progress achieved by output as measured through indicators and performance targets; (b) key implementation issues and solutions; (c) updated procurement plan; (d) updated implementation plan for ensuing 12 months; (iii) Mid Term Report, and (iv) a Project Completion Report (PCR) for the VNC project. KNNL shall organize for adequate review of project accounts and executing agency's audited financial statements, together with the associated auditor's report for the projects to continue to be both viable and sustainable.

25. **Stakeholder Communication Strategy.** Based on experience gained from Tranche 1, an updated stakeholder Communication Strategy is presented below for Tranche 2.

Table 2: Stakeholder Communication Strategy

Written	Verbal	Visual
Awareness/ Training Material	Radio episode series/ Television/ Folk Theatre/ Folk Art Form/Street plays/ Puppetry/ Ventriloquism	Film/ Electronic Media Campaign
Pamphlet/Brochure/Newsletter/ Poster/Wall Paintings/Presentation	Public Meeting/ Campaign/ Public Briefing	Display/ Poster/Wall paintings-project display board
Website/WhatsApp/SMS/ Group Email /Social Network Sites	Groups Discussion/ Stakeholder Interview / Personal Contact	Exhibition/ Fair
Case Study/ Working Paper/ Technical Note/Newspaper Articles	Orientation/Workshop/ Seminar/ Consultation/Interactive Meetings	Video clips
Press Release/ Advertisement	Media Briefing	Short duration video films
Site Announcement Sign	Site Visit / Exposure Visit	Photo albums
Letter / Email / Memo / Notice		
Policy / Legislation / Government Order		
Agreement/ Contract/ Memorandum of Understanding		
Action Taken Report		

26. The Communication Strategy adopted in Tranche 1 will further be improved and adopted. ACIWRM and KNNL may continue to use the websites developed during Tranche1 to disseminate information about Tranche 2 for Output 1 and post the progress from time to time.

27. PIO shall also establish a grievance redressal mechanism to address issues arising if any from implementation of the Tranche 2. The communication tools and techniques adopted in Tranche 1 will be continued and improved based on scope and learnings.

28. For WUCS strengthening, a multilevel, multi-dimensional communication strategy will be adopted to strengthen the social mobilization process and will include, among others, large group interactions, workshops, trainings, orientation programs, focus group discussions, and exposure visits.

29. Public notification of project milestones and safeguards, as well as progress towards these milestones and measures taken to impose safeguards will be posted on both KNNL website and copies available at Project office for examination or verification.

B. ACIWRM Individual Consultants

1. International Chief Technical Advisor (integrated river basin management)

30. The Chief Technical Advisor (CTA) will have professional experience of more than 15 years in integrated water resources management (IWRM) and qualifications in water resources engineering or equivalent, preferably at post graduate level. The specialist is preferred to have at least 10 years' experience of working in government water and river basin agencies, including river basin management and policy experience, and preferably with experience in community-based land and water management planning at sub-basin and irrigation levels. The consultant should be experienced working as a CTA and Team Leader with multi-disciplinary and multi-cultural teams for similar projects and with experience of more than 10 years in developing countries. The consultant should have good interpersonal and communication skills (written and oral) and a demonstrated ability to draft reports. Strong experience in working in developing countries with counterpart government water resource and river basin agencies and local and international consultants, preferably in India, is needed.

31. The CTA will provide specialist advice and input in regard to IWRM approaches and river basin management in the selected sub-basin. The CTA will work with the ACIWRM in assisting and synergizing and syncing its IWRM approaches with:

- (i) Planning and conducting the project including preparing project inception reports, project plans and schedules, terms of reference, project reports, coordinating inputs from international and national consultants, and in managing the project and team activities so that practical project outputs are fully achieved.
- (ii) Preparation of the IRBM Strategy for selected sub-basin including appraising of international policy approaches to IWRM and river basin management; international visits to evaluate alternative approaches; assessment of current approaches in India and Karnataka; evaluation of alternative approaches and in drafting of options papers; consultations; and, once agreed implementation of the Strategy such as drafting of regulations, guidelines, etc. as appropriate.
- (iii) River basin management including alternative arrangements and approaches for river basin organizations (RBO), preparation of river basin plans, after agreement on sub-basins, including preparation of guidelines, technical assessments and information systems; stakeholder participation; financing, establishing RBO offices and management mechanisms, and building an overall strong approach to river basin management.

- (iv) Preparing terms of reference and guiding international and national specialists and other providers in preparing IWRM and river basin approaches; assessing, preparing and delivering water resource information systems; training programs; irrigation and water use efficiency investigations; remote sensing of consumptive water use, and communications and participation.
- (v) Making linkages and arrangements for partnerships with relevant organizations (e.g., government water departments, river basin organizations, research organizations, etc.) in other countries, particularly where there is a strong 'best practice' approach to water resources and river basin management where water is scarce. This could include investigatory study visits, work placements and staff exchange, training courses, etc.
- (vi) Planning and assisting with delivery of training programs.
- (vii) Undertaking project communications and promotional activities.

2. MASSCOTE Specialist

32. The MASSCOTE specialist will have qualifications in irrigation engineering, water resources engineering or related field, at post graduate level. The expert will have (i) at least 20 years of practical experience in irrigation modernization; (ii) experience in conducting MASSCOTE training workshops and studies; (iii) experience in government agencies, multilateral organizations, research organizations in countries where irrigation modernization is implemented; and (iv) extensive knowledge and experience in FAO's MASSCOTE methodology, rapid appraisal procedure (RAP), benchmarking of irrigation projects, training staff in irrigation modernization in developing countries.

33. The international consultant will support ACIWRM and KNNL staff in building the capacity of WRD engineers and other project stakeholders in using MASSCOTE. He/she will have the following deliverables:

- (i) Training courses on MASSCOTE to the engineers of Nigams, WRDO engineers & others. Each training course is of 2 weeks duration
- (ii) MASSCOTE reports on selected irrigation systems with short term, medium term and long- term recommendations which can be used further to develop an overall modernization investment plan for the selected systems.

3. Hydrology Specialist

34. The specialist will have qualifications in hydrology, water resources engineering or related field, preferably at post graduate level, and have at least 15 years of practical experience in the design, installation, calibration, monitoring, data collection including via remote means, processing, storage, quality assurance, and management of hydrological data. The specialist will have extensive knowledge and preferably experience in surface water hydrology and hydrological monitoring, hydrological monitoring techniques, analysis, investigation and applications, hydrological data processing, databases and data management systems, flow measurement and telemetry, capacity building of staff.

35. The hydrologist will support the Water Resource Information System (WRIS) and hydrology inputs for the KISWRMIP. The consultant's role will principally involve supporting the assessment and upgrading of the hydrological monitoring and data management system, including its goals and applications, for project river basins and the State's WRIS. He/she will also provide relevant advice for the State IWRM Strategy, the WRIS design, and the capacity building program of engineers and

other specialists. He/she will work closely with the international water resources modeller. Under the supervision of the Director-ACIWRM, the international hydrologist will support the national specialist staff, local consultants, partners and the international consultants by assisting and advising on:

- (i) the strategic direction for hydrology, hydrological assessment, monitoring, and the State WRIS so that it is consistent with international best practice.
- (ii) end user and management needs of hydrological data, including lessons and SWOT from the World Bank Hydrology Project (HP) and recommendations for the useful and utilized State hydrological monitoring management system
- (iii) the hydrological monitoring network and data in priority river basins and its upgrading and linkage to the State water resource information system to support river basin management and IWRM
- (iv) applications of the data and its suitability including for preparation of the state water resources inventory, flood management and warning, drought management, river basin management, and managing water allocation and irrigation water deliveries
- (v) preparation of a state-wide water resources inventory
- (vi) the use of telemetry and other technologies to strengthen hydrological monitoring consistent with national and international developments as well as for the modernization of irrigation systems, particularly regarding the monitoring.
- (vii) training, advising and assisting staff in data management and processing, preparation of the state water resources inventory and the state hydrological database.
- (viii) preparing terms of reference for the procurement of services. Preparing specifications for the procurement of equipment

4. Irrigation Modernization Specialist

36. The irrigation modernization specialist will have at least 15 years of experience with irrigation modernization projects including scoping, design, evaluation, management. The specialist will also have experience in (a) evaluation, design, operation and maintenance of modernization projects in developed countries; (b) evaluation, design, operation and maintenance of modernization projects with an appropriate level of technology in developing countries (preferably India); (c) full range of irrigation management projects from historic gravity operated systems, SCADA managed systems, to fully automated gravity canal systems and associated piped systems; and (d) water user association approaches to the management of secondary/tertiary canal systems. The assignment will mainly comprise two elements:

- (i) assessment of broad options for the modernization of selected irrigation systems; and
- (ii) training of Nigam engineers.

5. River Basin Modeller

37. The river basin modeler (RBM) will have qualifications in hydrology, water resources engineering or related field, preferably to post graduate level. He/she will have at least 15 years of practical experience in water resources and river basin modelling and its practical applications to river basin management. The specialist will have experience in government agencies, river basin organizations or water authorities in developed countries responsible for water resources management and government policy, and experience with training staff in river basin modelling preferably in developing countries and preferably in India. He/she will have extensive knowledge and preferably experience in:

- hydrological river basin models and their application in developing river basin plans and water sharing arrangements
- data collection and generation from diverse sources, where local data is limited, to support river basin modelling and planning
- hydrological modelling at different scales including sub-basins/catchments and irrigation command areas including for local land and water management plans
- water balance studies at multiple scales, groundwater modelling and the conjunctive use of surface water and groundwater,
- climate change studies and of global climate models,
- preparation of river basin and State level water accounts,
- use of the eWater 'SOURCE' river basin model which is being used in associated national initiatives. In the absence of this experience, there should be a willingness and capacity to learn and apply the model.

38. The consultant will support the national specialist staff, local consultants, partners and the international CTA by advising and providing support on:

- (i) the strategic directions to build State capacity in water resources modelling and approaches to use of models to support development of government policy reflective of international best practice water resource management
- (ii) hydrological and basin modelling and scenario analysis to support river basin, sub-basin and land and water management planning
- (iii) methodologies and guidelines for water balance studies to be undertaken as part of Detailed Project Reports for irrigation development/modernization projects
- (iv) studies of climate change impacts, conjunctive surface water and groundwater use and required management approaches
- (v) preparation of a state-wide water resources inventory, database for hydrological data for modelling purposes and water accounts
- (vi) capacity building of ACIWRM staff, national consultants and other participants (for example from universities and the private sector) in order to build the modelling capacity in the State

39. The terms of reference of six other international consultants to support ACIWRM, namely water productivity specialist, agricultural (water management) specialist, water law specialist, river health and environment specialist, water outlook specialist, and asset management and *nigam* strengthening, will be developed during project implementation. There will also be 17 national consultants to be engaged to support the institutional strengthening of ACIWRM.

ATTACHMENT 8: PROCUREMENT PLAN FOR PROJECT 1

Basic Data

Project Name: Karnataka Integrated and Sustainable Water Resources Management Investment Program - Project 1	
Project Number: 43253-023	Approval Number: 3172
Country: India	Executing Agency: Karnataka Neeravari Nigam Limited
Project Procurement Classification: Category B	Implementing Agency: Advanced Centre for Integrated Water Resources Management
Project Procurement Risk: Medium	
Project Financing Amount: US\$ 48,000,000 ADB Financing: US\$ 31,000,000 Cofinancing (ADB Administered): Non-ADB Financing: US\$ 17,000,000	Project Closing Date: 30 September 2018
Date of First Procurement Plan: 17 October 2014	Date of this Procurement Plan: 13 July 2018

A. Methods, Thresholds, Review and 18-Month Procurement Plan

1. Procurement and Consulting Methods and Thresholds

Except as the Asian Development Bank (ADB) may otherwise agree, the following process thresholds shall apply to procurement of goods and works.

Procurement of Goods and Works		
Method	Threshold	Comments
International Competitive Bidding for Goods	US\$ 1,000,000 and Above	
National Competitive Bidding for Goods	Between US\$ 100,000 and US\$ 999,999	The first NCB is subject to prior review, thereafter post review.
Shopping for Goods	Up to US\$ 99,999	
International Competitive Bidding for Works	US\$ 40,000,000 and Above	
National Competitive Bidding for Works	Up to US\$ 39,999,999	Below that stated for ICB, works other than WUCS works contracts.
Community Participation in Procurement for Works	Up to US\$ 30,000	

Consulting Services	
Method	Comments
Quality- and Cost-Based Selection for Consulting Firm	Prior; 90:10 weighting for technical and financial scores respectively for PSC
Individual Consultant Selection for Individual Consultant	Prior

2. Goods and Works Contracts Estimated to Cost \$1 Million or More

The following table lists goods and works contracts for which the procurement activity is either ongoing or expected to commence within the next 18 months.

Package Number	General Description	Estimated Value	Procurement Method	Review (Prior/Post)	Bidding Procedure	Advertisement Date (quarter/year)	Comments
None							

3. Consulting Services Contracts Estimated to Cost \$100,000 or More

The following table lists consulting services contracts for which the recruitment activity is either ongoing or expected to commence within the next 18 months.

Package Number	General Description	Estimated Value	Recruitment Method	Review (Prior/ Post)	Advertisement Date (quarter/year)	Type of Proposal	Comments
ENVM-4	Environmental Monitoring	480,000.00	QCBS	Prior	Q4 / 2017	STP	Assignment: National Quality-Cost Ratio: 90:10 Comments: As per FAM

4. Goods and Works Contracts Estimated to Cost Less than \$1 Million and Consulting Services Contracts Less than \$100,000 (Smaller Value Contracts)

The following table lists smaller-value goods, works and consulting services contracts for which the activity is either ongoing or expected to commence within the next 18 months.

Goods and Works								
Package Number	General Description	Estimated Value	Number of Contracts	Procurement Method	Review (Prior/ Post)	Bidding Procedure	Advertisement Date (quarter/year)	Comments
Atlas	Micro-watershed Atlas of Karnataka	16,000.00	1	SHOPPING	Post		Q2 / 2018	
Gondi – DC-multiple (additional)	Command area development works	1,980,000.00	66	CPP	Post		Q2 / 2018	Comments: 66 CPPs
IWRM Curriculum Material	Curriculum material for IWRM training	40,000.00	1	SHOPPING	Post		Q2 / 2018	
Land & Water Management	Pilot Land & Water Management Plans	70,000.00	3	SHOPPING	Prior		Q2 / 2018	
LULC & CCA Maps	Land Use, Land Cover & Command Area maps	199,998.00	2	SHOPPING	Post		Q2 / 2017	Comments: Purchase of maps
Multiple-1	Hardware and software (various in multiple lots)	50,000.00	2	SHOPPING	Post		Q1 / 2015	Comments: No. of contracts to be determined; various in multiple lots
Multiple-2	Office equipment	55,000.00	2	SHOPPING	Post		Q4 / 2015	Comments: No. of contracts to be determined; various in multiple lots
Software	Groundwater software with GIS interface	99,000.00	1	SHOPPING	Post		Q2 / 2018	

Consulting Services								
Package Number	General Description	Estimated Value	Number of Contracts	Recruitment Method	Review (Prior/Post)	Advertisement Date (quarter/year)	Type of Proposal	Comments
AC-IWRM-09	National Irrigation Modernization Expert	36,000.00	1	ICS	Prior	Q3 / 2017		Assignment: National Expertise: irrigation modernization Comments: Per Aide Memoire of May 2017 Review Mission
AC-IWRM-10	TIDEDA Software Expert	30,000.00	1	ICS	Prior	Q3 / 2017		Assignment: International Expertise: TIDEDA Software Comments: Per Aide Memoire of May 2017 Review Mission
AC-IWRM-16	Environment (River Health) Specialist	31,000.00	1	ICS	Prior	Q4 / 2017		Assignment: National Expertise: environment
AC-IWRM-17	Gender Expert	28,000.00	1	ICS	Prior	Q3 / 2017		Assignment: National Expertise: gender
AC-IWRM-19	Graphics & Visual Expert	18,000.00	1	ICS	Prior	Q4 / 2017		Assignment: National Expertise: visuals development Comments: Per Aide Memoire of May 2017 Review Mission
AC-IWRM-21	Data Management Expert	40,000.00	1	ICS	Prior	Q3 / 2018		Assignment: National Expertise: data management

Consulting Services								
Package Number	General Description	Estimated Value	Number of Contracts	Recruitment Method	Review (Prior/Post)	Advertisement Date (quarter/year)	Type of Proposal	Comments
								Comments: Per Aide Memoire of May 2017 Review Mission
AC-IWRM-22	MASSCOTE Specialist	40,000.00	1	ICS	Prior	Q2 / 2018		Assignment: International Expertise: salinity, drainage
AC-IWRM-23	Communication Expert	40,000.00	1	ICS	Prior	Q2 / 2018		Assignment: National Expertise: climate change variability
AC-IWRM-24	Folk Art Communication Expert	40,000.00	1	ICS	Prior	Q2 / 2018		Assignment: National Expertise: folk art Communication
AC-IWRM-25	Curriculum Development Review Expert	25,000.00	1	ICS	Prior	Q2 / 2018		Assignment: National Expertise: curriculum development

B. Indicative List of Packages Required Under the Project

The following table provides an indicative list of goods, works and consulting services contracts over the life of the project, other than those mentioned in previous sections (i.e., those expected beyond the current period).

Goods and Works							
Package Number	General Description	Estimated Value (cumulative)	Estimated Number of Contracts	Procurement Method	Review (Prior/Post)	Bidding Procedure	Comments
None							

Consulting Services							
Package Number	General Description	Estimated Value	Number of Contracts	Recruitment Method	Review (Prior/Post)	Type of Proposal	Comments
None							

C. List of Awarded and On-going, and Completed Contracts

The following tables list the awarded and on-going contracts, and completed contracts.

1. Awarded and Ongoing Contracts

Goods and Works							
Package Number	General Description	Estimated Value	Awarded Contract Value	Procurement Method	Advertisement Date (quarter/year)	Date of ADB Approval of Contract Award	Comments
FMT- ICB-2	Flow measurement and telemetry system – supply and installation	3,500,000.00	2,549,194.74	ICB	Q1 / 2015	13-NOV-15	
Multiple-2	Office equipment	45,000.00	40,679.46	SHOPPING	Q4 / 2015		
Multiple-1	Hardware and software (various in multiple lots)	260,000.00	233,302.02	SHOPPING	Q1 / 2015		
TIDEDA Software	Hydrology software	10,000.00	13,200.00	SHOPPING	Q4 / 2017	08-FEB-18	
LULC & CCA Maps	Land Use, Land Cover & Command Area maps	66,000.00	43,094.70	SHOPPING	Q2 / 2017		
Printing	Printing of publications & communication materials	99,000.00	58,741.46	SHOPPING	Q2 / 2018	28-JUN-18	
WRIS	Water Resource Information System	174,000.00	203,198.36	SHOPPING	Q1 / 2017		
Gondi- NCB-1	Main canal and distributaries earthwork, lining, structures and flow measurement devices	16,000,000.00	16,665,632.30	NCB	Q4 / 2015	18-FEB-16	
Gondi – DC- multiple lots	Command area development works (minimum number of 10 packages per WUCS)	3,700,000.00	3,481,616.00	CPP	Q2 / 2016		

Consulting Services							
Package Number	General Description	Estimated Value	Awarded Contract Value	Recruitment Method	Advertisement Date (quarter/year)	Date of ADB Approval of Contract Award	Comments
PSC-1	Program Support Consultants (including WUCS strengthening support)	3,700,000.00	2,632,613.28	QCBS	Q4 / 2013	30-JUN-15	
AC-IWRM-1	AC-IWRM Strengthening Consultants (Chief Technical Advisor)	500,000.00	523,677.00	ICS	Q4 / 2013	05-MAY-15	
AC-IWRM-02	Hydrologist	60,939.00	54,775.00	ICS	Q4 / 2015	27-JUN-16	
AC-IWRM-08	International Irrigation Modernization Expert	90,000.00	115,364.00	ICS	Q3 / 2017	10-AUG-17	

Consulting Services							
Package Number	General Description	Estimated Value	Awarded Contract Value	Recruitment Method	Advertisement Date (quarter/year)	Date of ADB Approval of Contract Award	Comments
AC-IWRM-03	River Basin Modeller	67,389.00	66,780.00	ICS	Q4 / 2015	27-JUN-16	
AC-IWRM-15	Agricultural Specialists	56,000.00	46,665.75	ICS	Q3 / 2017	03-NOV-17	
AC-IWRM-18	Economist/Livelihoods Expert	55,000.00	51,132.84	ICS	Q4 / 2017	03-NOV-17	
AC-IWRM-05.	MASSCOTE Expert	34,188.00	33,390.00	ICS	Q3 / 2016	11-AUG-16	
AC-IWRM-06	Training & Capacity Building Specialist	30,000.00	9,983.70	ICS	Q2 / 2017	25-SEP-17	
AC-IWRM-07	Videographer	50,000.00	19,633.17	ICS	Q2 / 2017	31-AUG-17	
AC-IWRM-13	Water Quality Specialist	65,000.00	35,952.96	ICS	Q3 / 2017	15-SEP-17	
AC-IWRM-04	Water Resources Policy and Institutions Specialist	28,989.00	21,821.37	ICS	Q4 / 2015	27-JUN-16	

ATTACHMENT 9: PROCUREMENT PLAN FOR PROJECT 2**PROCUREMENT PLAN****Basic Data**

Project Name: Karnataka Integrated and Sustainable Water Resources Management Investment Program - Project 2		
Project Number: 43253-026	Approval Number:	
Country: India	Executing Agency: Karnataka Neeravari Nigam Limited	
Project Procurement Classification: Category B	Implementing Agency: Advanced Centre for Integrated Water Resources Management, Karnataka Neeravari Nigam Limited	
Project Procurement Risk: Medium		
Project Financing Amount: US\$ 130,000,000 ADB Financing: US\$ 91,000,000 Cofinancing (ADB Administered): Non-ADB Financing: US\$ 39,000,000	Project Closing Date: 31 March 2024	
Date of First Procurement Plan: 28 May 2019	Date of this Procurement Plan: 28 May 2019	
Procurement Plan Duration (in months): 18	Advance Contracting: Yes	e-GP: Yes https://eproc.karnataka.gov.in/

A. Methods, Review and Procurement Plan

Except as the Asian Development Bank (ADB) may otherwise agree, the following methods shall apply to procurement of goods, works, and consulting services.

Procurement of Goods and Works	
Method	Comments
Open Competitive Bidding (OCB) for Goods	
Request For Quotation for Goods	
Open Competitive Bidding (OCB) for Works	
Request For Quotation for Works	
Community Participation in Procurement for Works	

Consulting Services	
Method	Comments
Quality- and Cost-Based Selection for Consulting Firm	Prior; 90:10 weighting for technical and financial scores respectively
Competitive for Individual Consultant	
Direct Contracting for Individual Consultant	

B. Lists of Active Procurement Packages (Contracts)

The following table lists goods, works, and consulting services contracts for which the procurement activity is either ongoing or expected to commence within the procurement plan duration.

Karnataka Neeravari Nigam Limited

Goods and Works							
Package Number	General Description	Estimated Value (in US\$)	Procurement Method	Review	Bidding Procedure	Advertisement Date (quarter/year)	Comments
Equipment 2	Water quality monitoring equipment and installation	180,000.00	OCB	Prior	1S1E	Q4 / 2020	Advertising: National No. Of Contracts: 1 Prequalification of Bidders: No Domestic Preference Applicable: No Advance Contracting: No Bidding Document: Goods e-GP: Yes
Equipment 5	KNNL office furniture, equipment, software, and vehicles	1,120,000.00	RFQ	Prior		Q1 / 2020	No. Of Contracts: 12 Advance Contracting: No e-GP: No Comments: Each contract not exceeding \$100,000.
VNC 1	Modernization of Vijayanagar Channels: Anicuts, main canals and distributaries works outside Hampi Heritage Area and Tungabhadra Otter Conservation Area	65,000,000.00	OCB	Prior	1S2E	Q1 / 2018	Advertising: International No. Of Contracts: 1 Prequalification of Bidders: No Domestic Preference Applicable: No

							Advance Contracting: Yes Bidding Document: Large Works e-GP: Yes Comments: Contract awarded on 08-Mar- 19 for \$52.645 million
VNC 2	Modernization of Vijayanagar Channels: Anicuts, main canals and distributaries works inside Hampi Heritage Area and Tungabhadra Otter Conservation Area	15,000,000.00	OCB	Prior	1S2E	Q4 / 2019	Advertising: International No. Of Contracts: 1 Prequalification of Bidders: No Domestic Preference Applicable: No Advance Contracting: No Bidding Document: Small Works e-GP: Yes
VNC-CPP- multiple lots	Command area development works (minimum number of 10 packages per WUCS)	17,250,000.00	CPP	Prior		Q4 / 2020	No. Of Contracts: 575 Advance Contracting: No e-GP: No Comments: At least

							10 contracts each for 25 water user groups, each contract costing no more than \$30,000
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Advanced Centre for Integrated Water Resources Management

Goods and Works							
Package Number	General Description	Estimated Value (in US\$)	Procurement Method	Review	Bidding Procedure	Advertisement Date (quarter/year)	Comments
Equipment 1	Hydrometeorology equipment and installation	4,000,000.00	OCB	Prior	1S1E	Q3 / 2020	Advertising: National No. Of Contracts: 1 Prequalification of Bidders: No Domestic Preference Applicable: No Advance Contracting: No Bidding Document: Goods e-GP: Yes
Equipment 3 & 4	ACIWRM office furniture, equipment, software, and vehicles	260,000.00	RFQ	Prior		Q1 / 2020	No. Of Contracts: 5 Advance Contracting: No e-GP: No Comments: Each contract not exceeding \$65,000.
Print 1	Printing	130,000.00	RFQ	Prior		Q4 / 2019	No. Of Contracts: 2 Advance Contracting: No e-GP: No

							Comments: Each contract not exceeding \$65,000.
Studies 1 to 12	Various technical studies	1,010,000.00	RFQ	Prior		Q3 / 2020	No. Of Contracts: 12 Advance Contracting: No e-GP: No Comments: Each contract not exceeding \$100,000.
SW 1 & 2	Water Resource Information System- software and data	190,000.00	RFQ	Prior		Q1 / 2020	No. Of Contracts: 2 Advance Contracting: No e-GP: No Comments: Each contract not exceeding \$100,000.

Advanced Centre for Integrated Water Resources Management

Consulting Services							
Package Number	General Description	Estimated Value (in US\$)	Selection Method	Review	Type of Proposal	Advertisement Date (quarter/year)	Comments
ICS-01	ACIWRM Chief Technical Advisor (CTA)	200,000.00	SSS	Prior		Q3 / 2019	Type: Individual Assignment: International Expertise: integrated water resources management Advance Contracting: Yes e-GP: No
ICS-02	ACIWRM Water	105,000.00	Competitive	Prior		Q3 / 2019	Type: Individual

	Productivity Specialist						Assignment: International Expertise: water resources management Advance Contracting: Yes e-GP: No
ICS-03	ACIWRM MASSCOTE Specialist	83,810.00	Competitive	Prior		Q3 / 2019	Type: Individual Assignment: International Expertise: MASSCOTE Advance Contracting: Yes e-GP: No
ICS-04	ACIWRM Agricultural (WM) Specialist	57,143.00	Competitive	Prior		Q3 / 2019	Type: Individual Assignment: International Expertise: agriculture - water management Advance Contracting: Yes e-GP: No
ICS-05	Hydrology Specialist	77,150.00	SSS	Prior		Q3 / 2019	Type: Individual Assignment: International Expertise: hydrology Advance Contracting: Yes

							e-GP: No
ICS-06	ACIWRM Water Law Specialist	206,000.00	Competitive	Prior		Q3 / 2019	Type: Individual Assignment: International Expertise: water law Advance Contracting: Yes e-GP: No
ICS-07	ACIWRM Irrigation Modernization Specialist	210,000.00	SSS	Prior		Q3 / 2019	Type: Individual Assignment: International Expertise: irrigation modernization Advance Contracting: Yes e-GP: No
ICS-08	ACIWRM River Health & Envi Specialist	43,584.00	Competitive	Prior		Q3 / 2019	Type: Individual Assignment: International Expertise: river health, environmental management Advance Contracting: Yes e-GP: No
ICS-09	ACIWRM River Basin Modeller	370,000.00	SSS	Prior		Q3 / 2019	Type: Individual Assignment: International

							Expertise: river basin modelling Advance Contracting: Yes e-GP: No
ICS-10	ACIWRM Water Outlook Specialist	62,000.00	Competitive	Prior		Q3 / 2019	Type: Individual Assignment: International Expertise: water outlook Advance Contracting: Yes e-GP: No
ICS-11	ACIWRM Asset Management and Nigam Strengthening	70,000.00	Competitive	Prior		Q3 / 2019	Type: Individual Assignment: International Expertise: asset management and strengthening of "nigams" Advance Contracting: Yes e-GP: No
ICS-National	ACIWRM Various individual consultants	945,000.00	Competitive	Prior		Q3 / 2019	Type: Individual Assignment: National Expertise: various Advance Contracting: Yes

							e-GP: No Comments: 17 national experts
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Karnataka Neeravari Nigam Limited

Consulting Services							
Package Number	General Description	Estimated Value (in US\$)	Selection Method	Review	Type of Proposal	Advertisement Date (quarter/year)	Comments
PSC-2	Project Support Consultants	6,400,000.00	QCBS	Prior	FTP	Q2 / 2019	Type: Firm Assignment: International Quality-Cost Ratio: 90:10 Advance Contracting: Yes e-GP: No

C. List of Indicative Packages (Contracts) Required Under the Project

The following table lists goods, works, and consulting services contracts for which procurement activity is expected to commence beyond the procurement plan duration and over the life of the project (i.e., those expected beyond the current procurement plan duration).

Goods and Works						
Package Number	General Description	Estimated Value (in US\$)	Procurement Method	Review	Bidding Procedure	Comments

Consulting Services						
Package Number	General Description	Estimated Value (in US\$)	Selection Method	Review	Type of Proposal	Comments
None						

ATTACHMENT 10: STAKEHOLDER PARTICIPATION & CONSULTATION

A. Project Introduction

1. The State Government of Karnataka (the State) has requested financing from the Asian Development Bank (ADB) to enhance water security in selected basins where there is increasing water stress due to rapid economic growth and future competing needs for water supply and industry. An integrated water resources management (IWRM) approach is required to meet increasing water demand while promoting its judicious use and protecting human water needs and river environments. Greater efforts will be required to improve efficiency and sustainability in water use, particularly for irrigation (which uses 85% of abstracted water), urban water supply and sanitation (WSS), industry and the environment. The State, Government of India and ADB have agreed on technical assistance for preparing the Karnataka Integrated and Sustainable Water Resources Management Investment Program (KISWRMIP). The expected impact of the program is enhanced water security, sustainability and river environment for rural and urban population in selected water-scarce river basins, which will be measured through improved and sustainable access and incomes.

B. Need for Stakeholder's Participation and Consultations (P&C)

2. ADB's Safeguard Policy stresses that the Project executing agency (EA) will conduct meaningful stakeholders consultation with affected persons, and civil society for every project identified as having multiple dimensions of project impact. However, it is also important to know that there may also be issues where consultation is not appropriate, i.e. when a) the room for maneuver is so limited, either by statutory; or b) budgetary restrictions, that the consultation is meaningless; c) where we have already consulted on a similar issue; d) where consultation fatigue' may set in; and e) where the issue should or could be resolved without needing a process of consultation.

3. Stakeholder participation is a process whereby stakeholders—those with rights (and therefore responsibilities) and interests—play an active role in decision-making and in the consequent activities which affect them. Stakeholder participation goes much further than, and has to be distinguished from, stakeholder consultation, which refers to one aspect of participation only. Stakeholder consultation is providing a forum for stakeholders to speak up and voice opinions only. For projects like KISWRMIP, which address environmental, social and economic impacts, consultation will not be a single conversation but a series of opportunities to create understanding about the project among those it will likely affect or interest, and to learn how these external parties view the project and its attendant risks, impacts, opportunities, and mitigation measures. Listening to stakeholder concerns and feedback can be a valuable source of information that can improve project design and outcomes and help ADB to identify and control external risks. It can also form the basis for future collaboration and partnerships. For stakeholders, ADB initiated consultation process is an opportunity to get information, as well as to learn about the local context in which a project will take place, to raise issues and concerns, ask questions, and potentially help shape the project by making suggestions for the ADB to consider and respond to. Further, P & C process help increase the level of support for the Project and related activities from a range of stakeholders, which can speed up processing and reduce challenges during implementation. P & C is also expected to improve the effectiveness, relevance, and sustainability of development activities in the long run.

4. Whether a project proposal is straightforward or complex, effective stakeholder P&C process, particularly prior to initiating the formal approvals process, is a key factor in achieving timely completion of the government approvals process.
5. Appropriate P&C Process will enable proponents to:
 - (i) Identify the key issues involved with the project and the likely impacts as an early identification of these matters allows the opportunity to consider alternative approaches with less impact.
 - (ii) Obtain information as to how best to manage any adverse impacts. Early discussion on management approaches will reduce the time required for negotiation on these matters during the formal assessment.
 - (iii) Be aware of information that will need to be provided in support of the formal approvals that will be required to allow the project to proceed. An early understanding of information requirements will help ensure that information gathering processes do not cause delay.
 - (iv) Establish two-way communications channels is essential
6. The nature and level of P&C Process that is appropriate will differ depending on the particular proposal, including such factors as the scale of the project and its potential social or environmental impacts.

C. Key Objectives of Stakeholder P&C

7. The broad objective of the stakeholder engagement in P&C process is to provide authorities, as well as interested and affected parties with the opportunity to identify issues, concerns, and opportunities regarding the proposed Project and to address key stakeholder concerns during the preparation of the various safeguard intervention measures. The specific objectives shall cover the following:
 - (i) To provide information about the project and to tap stakeholder information on key issues /problems and concerns affecting their livelihood
 - (ii) To elicit environmental and social and economic baseline information in the project areas;
 - (iii) To provide opportunities to stakeholders to discuss their opinions and concerns;
 - (iv) To manage expectations and misconceptions regarding the project;
 - (v) To discuss potential impacts and verify significant or major environmental, social and economic impacts identified; and Acquire suggestions of the community for mitigating any anticipated adverse environmental and social impacts and increasing expected benefits of the Project
 - (vi) To inform the process of developing appropriate mitigation and management measures as well as institutional arrangements for effective implementation.
 - (vii) Promote participation of the local people, local level government stakeholders, elected representatives and other community representatives to create opportunity to play a role and express their views.
 - (viii) Facilitate the community to be prepared with enhanced confidence and capacity for meaningful engagement in the project
8. The prerequisites to achieve the objectives:
 - (i) To lay down the guiding principles for P&C.
 - (ii) Key stakeholder's identification.
 - (iii) Share information on P&C processes undertaken so far and the outcome of such consultation process thereof.

- (iv) Sharing information on resources and responsibilities planned for P&C implementation.
- (v) Development of framework for monitoring and reporting on the effectiveness of the P&C process

D. Guiding Principles for Stakeholder P&C

- (i) P&C should aim to seek informed public opinion and not just instant reaction.
- (ii) The public should have a say in decisions about actions that could affect their lives.
- (iii) P&C includes the promise that the public's contribution will influence the decision and that it promotes sustainable decisions by recognizing and communicating the needs and interests of all participants, including decision makers.
- (iv) P&C seeks out and facilitates the involvement of those potentially affected by or interested in a decision.
- (v) P&C process help seek input from participants in designing how they participate.
- (vi) P&C should provide participants with the information they need to participate in a meaningful way.
- (vii) P&C shall help inform participants how their input affected the decision.
- (viii) Consultation needs to be carried out throughout the project cycle and timely disclosure of relevant and adequate information has to be undertaken.
- (ix) In P&C Process, all relevant views of affected people and other stakeholders into decision making, such as project design, mitigation measures, the sharing of development benefits and opportunities, and implementation issue also need to be incorporated environmental quality.
- (x) P&C Process should ensure consultation is undertaken in an atmosphere free of intimidation or coercion and is particularly inclusive of and responsive to the needs of the disadvantaged and vulnerable groups especially those below the poverty line, the landless, the elderly, female headed households, women and children, Indigenous Peoples, and those without legal title to land.
- (xi) P&C Process shall facilitate access to information as access to information provides an important guard against arbitrariness in public decision-making and also helps to enhance transparency and accountability. Information dissemination at all levels will hence be a critical pre-requisite for meaningful consultations.
- (xii) P&C process should establish feedback mechanism and such feedback needs to be incorporated into policy design and decision-making processes.
- (xiii) In the P&C process, all relevant views of affected people and other stakeholders into decision making, such as project design, mitigation measures, the sharing of development benefits and opportunities, and implementation issue also need to be incorporated.

E. Key Stakeholders in the Project

9. Stakeholders are people/communities who may - directly or indirectly, positively or negatively affect or be affected by the outcomes of projects or programs.

- (i) **Primary stakeholders** are the beneficiaries of a development intervention or those directly affected (positively or negatively) by it. They include local populations (individuals and community-based organizations) in the project/program area, in particular, poor and marginalized groups who have traditionally been excluded from participating in development efforts.
- (ii) **Secondary stakeholders** are those who influence a development intervention or are indirectly affected by it. They include the borrowing government, line ministry and project staff, implementing agencies, local governments, civil society

organizations, private sector firms, service providers, the financial institutions and other development agencies/ institutions/ organisations.

F. Summary of Stakeholder Consultations Undertake

10. Many water and land related discussions arose during stakeholder consultation process and in Focused Group Discussions though relevant and important but were not necessarily 'within scope' of the proposed project and therefore not appropriate to address those emerging issues in detail in this report. In particular, only those issues surrounding enhanced water security, sustainability and river environment for rural and urban population in selected water-scarce river basins as measured through improved and sustainable access, incomes, and water related parameters are considered as important. The project undertook consultations with Focused Group Discussions and with Water Users Cooperative Society (WUCS).

11. The timeline and the profile of the participants are given in the tables 1 to 8. The stakeholder profile as indicated in cols (1-7) respectively corresponds to 1) State Govt 2) Villagers 3) Local Govt/ Gram Panchayat/ WUCs /Aanganwadi members 4) SOE (KNNL, CADA) 5) CSO 6) Female and 7) Consultants

Table 1: FGD - Stake holder's consultations and profile of participants in TLBC Project

Place	Date	Stakeholders Profile						
		1	2	3	4	5	6	7
Madhya and Durga Camp	23/03/12		15	5	3	2	2	3
Budihaal	23/03/12		20	6	2	2	9	3
Nelahaal	25/04/12		20	5	2	2	8	3
Kareguda	04/06/12	01	18	02			05	
Mustoor	04/06/12	01	17	03			05	
Hirekotankal	05/06/12	02	10	02	1		02	3
Tadakaal	05/06/12	02	23	04	1		1	
Chikaltarvi	08/06/12		15	03			07	
Devipur	08/06/12	02	28	03			06	
Sujebal	12/06/12		15	01	2	1	03	
Hanwal	14/06/12	02	24	04			10	
Hagedal	15/06/12	01	10	02		02	06	
Hebbal	15/06/12	01	19	03			08	
Acharnarsapur	16/06/12		20	01			06	
Gundur	18/06/12		15	02		01	04	
Challur	19/06/12		15	03			05	
Turvihal	21/06/12		19	02			04	
Kholabal	23/06/12		22	01	1	2	03	
TOTAL		12	325	52	12	12	94	12

Table 2: FGD - Stakeholders consultation and profile of participants in Bhadra Project

Place	Dates	Stakeholder Profile						
		1	2	3	4	5	6	7
Shingrihalli	01/06/12		15	10	5		8	3
Doddabathi	16/06/12	02	10			2	01	4
Nalluru	16/06/12		19				03	2
Kondajji	18/06/12	02	12	06			05	3
Basavanahalu	18/06/12	01	12				04	3
Sathur	21/06/12	01	30	02		3	7	3
Kumbalur	21/06/12		9	02		2	04	3
Kulambi	21/06/12	01	17	7			06	3
Thadasa	22/06/12	01	15	03			07	01
Arahatholalu	22/06/12		11				04	2
TOTAL		8	150	30	5	7	49	27

Table 3: FGD - Stakeholders consultation and profile of participants in Gondi Project

Place	Dates	Stakeholder Profile						
		1	2	3	4	5	6	7
Thallikatte	06/08/12		10	5	1	1		2
Dasarakallahalli	07/08/12		10	4	2			4
Dayanakpura	08/08/12		15	4	2	2		
Barandoor	08/08/12	1	15	6	2	3	2	
Korekoddegahalli	09/08/12		21	4	1	2	5	
Doddagopenhalli	28/08/12		20	4	1	1	3	4
TOTAL		1	91	27	9	9	10	10

Table 4: FGD - Stakeholders consultations and profile of participants in Vijayanagara Channel

Place	Date	Stakeholder Profile						
		1	2	3	4	5	6	7
Rama Sagara	09/08/12		15	8	3	4	3	3
Anegundi	07/09/12	01	19	04	2	1	02	3
Ibrahimpura	09/09/12	01	20	01			04	
Hulagi	11/09/12		25	03		2	06	
Nagenahalli	12/09/12		15	3	2	1		2
TOTAL		2	94	19	7	8	15	8

Table 5: WUCS - Stakeholders consultation and profile of participants in TLBC

Place	Date	Stakeholders Profile						
		1	2	3	4	5	6	7
Krishi Vigyan Kendra Gangavathi	11/04/12			6			1	3
Rice Research Station – Gangavathi	12/04/12			7				3
Agriculture Department, Gangavathi	12/04/12	8						3
Basapatna	15/05/12		3	2	3	51	2	1
Mushtoor	18/05/12		2		3	59	4	1
Budugumpa	23/05/12			2	3	40	1	1
Hirekotnakal	25/05/12			3	4	32	2	
Lakkamdinee	29/05/12		2	2	3		2	1
Hunshihall Hudda	04/06/12		3	1	3	30	2	
TOTAL		8	10	23	19	212	14	13

Table 6: WUCS - Stakeholders consultation and profile of participants in Bhadra Project

Name of WUCS and Place	Stakeholder Profile							
	Date	1	2	3	4	5	6	7
Beernahalli, Shivamogga Taluk	13/05/12		2	2	3	34	3	1
Nellisara, Badravathi Taluk	17/05/12		1	1	3	32	1	1
Yalavatti, Shivamogga Taluk	21/05/12		2		3	28	1	
Hosabelavanuru, Davanagere Taluk	23/05/12		1	2	3	23	2	1
Ganganarasi, Harihara Taluk	26/05/12		1	1	3	18	1	1
Kattalagere, Chennagiri Taluk	29/06/12		3	1	3	30	2	
Kulagatti-2, Honnali Taluk.	03/06/12		1	2	3	31	3	
Mudugodu, Tarikeri Taluk	07/06/12		1	2	3	13	1	
Ragimasalavada, Harapanahali Taluk	12/06/12		2	1	3	21	1	
TOTAL		0	14	12	27	230	15	4

Table 7: WUCS - Stakeholders consultation and profile of participants in Gondi Project

Place	Date	Stakeholder Profile						
		1	2	3	4	5	6	7
Kagekodamage	06/8/12		30	3	1	4	3	1
Dasarakallahalli	07/8/12		10	4	2			4
Koppa Doddakere	07/8/12							
Bharandur	16/9/12		23		1	3		1
Shimoga	11/9/12				7	3		1
BRP Dam Site	28/9/12				12			9
TOTAL		0	63	7	23	10	3	16

Table 8: WUCS - Stakeholders consultations and profile of participants in Vijayanagara Channel

Place	Date	Stakeholder Profile						
		1	2	3	4	5	6	7
CADA-Munirabad	08/6/12				6			6
Tungbhadra Board -Hospet	09/6/12				4			6
Thurtha Channel, Venkatapura	09/8/12		30	3	1	4	3	1
Ramasagara Channel - Ramasagara	09/8/12		25	8	3	4	3	3
Anegundi	13/9/12		40		2	3	2	1
Raya Channel - Nagenahalli	14/9/12					3	4	
HWHAMA -Hospet	14/9/12		2		3	2		1
Upper Gangavathi – Ayodhya	15/9/12		16	2	2	3		9
Siruguppa and Channel - Siruguppa	17/9/12		15	1	1	3		1
Bichal Channel - Rajoli	19/9/12		41	2	2	3	2	
TOTAL		0	169	16	24	25	14	28

G. CSO Consultation Workshop

12. A consultation workshop with Civil Society Organization representatives¹ on Program outputs and projects was held on 14 December 2012 in Bengaluru. The topics discussed during the workshop were:

¹ Involved organisations were Action for Food Production, Ashoka Trust for Research in Ecology and the Environment, Adavi Sidheshwara Rural Development Society, Consulting Engineering Services (I) Pvt Ltd, CIVIC, Dhanya, Gram Vikas, Institute for Studies on Agriculture & Rural Development, JalaSpandana – South India Farmers Organisation for Water Management, Public Affairs Centre, SECON Pvt Ltd, Technology Informatics Design Endeavour (TIDE)

- (i) Advanced Center for IWRM – Innovative Initiative in Karnataka
- (ii) IWRM based Water Management in Tungabhadra Sub Basin
- (iii) Capacity Building of Water Users Cooperative Societies; and
- (iv) Institutional Framework of the Program

13. The learning from the CSO consultation workshop was as follows:

Workshop Topic	Issues Raised
Advanced Center for IWRM – Innovative Initiative in Karnataka	- On Public Private Partnership concern was raised on: - How it works - Who are all the partners - What is the water allocation for the PPP - How will private partners recover the investment - Detail study required on the actual allocation & availability of water in the state - Policy for the total IWRM solution should be formed - All IWRM actions and information should be transparent and shared with the stakeholders
IWRM based Water Management in Tungabhadra Sub Basin	- Is existing and future industrial development being considered and what is their water requirement - Power, chemical and mining industries which are emerging in the area need to be considered during the basin planning and allocation. - Requirements from the industries are to be consider during Basin Planning - How will this effect agriculture water supply
IWRM based Water Management in Tungabhadra Sub Basin	- Social and technical issues to be shared and discussed with the WUCS before finalizing project interventions - Social and technical aspects should be integrated to the dimension of the project - Sustainable management should be considered from the beginning of project designing and planning - Alternate technologies such as precast semi pipe water channels for CAD work
Capacity Building of Water Users Cooperative Societies	- WUCS strengthening is critical for modernization planning, implementation and O&M - CSO experience and learning should be considered while planning WUCS interventions - The gaps between the planning and actual requirement should be filled for the successful implementation of project - Requisite power and authority should be provided for the WUCS to allow them to function effectively - Land salinity problems should be addressed - Social processes (people's participation) should be given equal importance as technical aspects. - Both the social and technical aspects should run in parallel in project implementation - Different / new farming system has to be considered in consultation with farmers
Institutional Framework of the Program	- Role of PRI should be included in implementation design - Innovative framework must be considered for implementation - The Program PMU should be located in the project field area

Workshop Topic	Issues Raised
	• IT systems should be used for monitoring and evaluation of the project • Involvement of the Agriculture Department will be important for project success • Good documentation process should be adopted by the project covering study of project process, outcomes and impacts

H. Key Responses by Stakeholders

14. Emerging response from stakeholder consultation process is indicative of the policy and practices of water resources development in the basin over a period of time. What could be inferred from such consultations is that there is need for a more holistic perspective for water resource management. The issue of political economy has of course the least of the place. Skills in basin planning, hydrology, water management, financial planning, O&M institutions and sustainable development issues are totally lacking in the water domain. What could be inferred is that various forms of articulation/responses by stakeholders during the consultation process broadly (if not specifically) relates or falls into to the following key areas requiring attention-action and support.

i. Policy and Institutional Arrangements for IWRM²

15. There are no institutional arrangements at the State level to consider sectoral water demands, plan and manage water resources. Responsibilities of water management are fragmented between departments without formal mechanism to ensure co-ordination. The current water law/acts/policies are inadequate to facilitate successful implementation of a basin organization in the Tungabhadra Sub-basin, let alone improve overall integrated water management in the State. The lack of a coordinated and modern water law, embracing all aspects of water usufruct rights and allocation, trading, pricing etc., and the plethora of departments involved in water management goes against rational and sustainable planning and management of water in Karnataka. The issue is to create a high level or State-wide framework that will clearly develop or guide the essential water policies and strategies right across Karnataka and in so doing will lead to improved water management in the inter-State river basins as well. With increasing water scarcity and conflicts between various competing sectors and at various levels - macro as well as micro level of distributaries, communities and individual users, the formal water rights as a mechanism for water allocations and accountability has gained importance. Currently, the water rights are governed by the Water Policy and Irrigation Acts, of the respective states. However, there is no clear legal framework specifying water rights, even though various acts have some basis for defining some form of rights", Competing Water Uses in Tungabhadra Sub-basin, India. The concept of IWRM is the integration and coordination of (i) management of surface water, groundwater and rainwater (ii) government and private sectors that are separately managing water resources; (iii) policy, resource management and service delivery of water; (iv) management, conservation, maintenance, scientific use of water resources; and integration of (v) economic, social and environmental outcomes; and (vi) issues related to food and energy security. The proposed interventions address some of the above-mentioned issues.

² More for details see CDTA C1 Component 1 Report.

ii. River Basin Plan and Organisations³

16. The objective of the river basin sub-component of the project is to improve water resource conditions and management in selected, pilot river basins in Karnataka by taking an IWRM approach which will include cross sectoral river basin organisations, preparing integrated river basin plans and implementing plan elements as agreed. There are a number of plans that impact on the water and land resources of the sub-basin. Each sub-sector (irrigation, industry, urban water supply and sanitation, and so on) has a plan of some description developed by the relevant agency. But these are prepared in isolation and do not adequately take into account cross-sectoral water resource management aspects. An effective river basin plan provides the foundation and overall policies directions for water resource management at basin level and the development of such a plan aims to provide a comprehensive guide for the integrated development and management of its water resources in a river basin, the hydrologic planning unit generally accepted to be most appropriate for water resource planning. Such a plan should be the primary instrument for integrated water resources management and improved coordination within the individual basin, guided by policies and plans implemented at national and state level. Having clearly defined and robust institutional arrangements for water resource management in place are a fundamental requirement for good river basin management. In this sense, “institutions” refers to the various laws and policies in place, the organisations that have been established by government to implement them, and the mechanisms for interaction among those organisations.

iii. Irrigation Modernization⁴

17. Irrigation Modernisation initiative is viewed as an opportunity to comprehensively examine the operations of the Irrigation services including layout and function of the irrigation water supply system. The aim is to identify changes that can be made to the supply system itself and its operation in order to strengthen viability and efficiency in the face of a future with less water. Modernisation planning promotes challenging the status quo within which an organisation has been operating and questioning existing biases and testing boundaries are critical elements of this challenge.

iv. Land and Water Management Plan

18. In any basin or sub-basin, there are close linkages among different uses of water and land resources, and planning each in isolation leads to inefficiencies and “side impacts” such as water shortage and environmental degradation. The government has a State Water Plan, but this does not seem to play a significant role in water resource management and appears to have done little to integrate planning and management across the sub-sectors and across the State. In addition, while irrigation development planning is to some extent done at the basin level, at the sub-basin level, no integrated planning process is in place, and this is needed.

³ Source: CDTA Component 2 of Upper Tunga Framework Plan

⁴ Stakeholder consultation in TLBC voiced their view on Irrigation modernization and according to them the Provision of parallel field canal for efficient water management to tail end command of the particular pipe outlet. b) Use of regenerated water for feeding to tail end command by provide lift irrigation scheme lifting water from Nala and feed to suffering command in the tail end reach wherever possible. c) Projects like construction of lift irrigation schemes and pickups with a focus on irrigate lands both localized and non-localized command d) filling of pit wells drinking water, ponds and balancing reservoirs. e) existing tunnel design discharge of 3100 cusecs in the jurisdiction of Vaddarahatti division which is bottleneck for drawing more than the required design discharge so as to accommodate developed localized and non-localized command, to be taken up as activities as part of Irrigation modernization.

v. Planning and Execution of CAD and OFD Works

19. The primary objective of CADA was to improve water use efficiency through integrated area development in the irrigation command, including modernisation of the distribution system, the provision of drainage and the maintenance and operation of both the distribution and drainage systems. On-Farm Development⁵ (OFD) is an integrated process of levelling and shaping the land for smooth flow of water, constructing field irrigation channels, providing drainage facilities, forming the farm roads and realigning the field boundaries through appropriate consolidation of holdings. Thus, OFD⁶ includes works, such as field channels, land levelling, field drains and conjunctive use of surface and groundwater sources; introduction of rotational system of water distribution (RWS) to ensure equitable and timely supply of water to each and every farm holding; and evolving and propagating crop patterns and water management practices appropriate to each command area.

vi. Formation and Strengthening of WUCS

20. WUCS at different levels are expected to be actively involved in: (i) maintenance of irrigation system in their area of operation; (ii) distribution of irrigation water to the beneficiary farmers as per the roster system; (iii) preparation of water demand and collection of water charges; (iv) resolve disputes among the members and WUA; (v) monitoring flow of water in the irrigation systems. In order that the WUCS at different levels are able to resolve conflict along irrigation systems in which they have a role it will be necessary that selected representatives have a greater understanding of such matters as irrigation scheduling, water usufructs and seasonal quotas to complement the conflict management techniques which will also be necessary. Training will have to be designed and provided in order to strengthen the WUCS. Farmers have been expressing the need for capacity building initiatives in order to discharge their functions. Also articulated their concerns for strengthening WUCS by addressing the problems confronted by them in terms of (a) lack of legal back up and policy changes; (b) system deficiency; (c) uncertainty of water availability; (d) fear of financial viability; (e) lack of technical knowledge; (f) lack of leadership; (g) lack of publicity and training.

vii. Water Resource Information System (WARIS)

21. Water Resource Information is the broad class of information needed by managers to manage not only operations and maintenance, but also initial water used on farm. It includes information for planning, implementation, monitoring, review and evaluation. It is vital for planning, directing and controlling operation and maintenance activities. The availability of reliable information on all aspects of water resources management is fundamental to the successful implementation of efficient water management strategies. Planning is the first step in managing operation and maintenance of irrigation scheme. Further, the knowledge and monitoring of water resources, uses, demands and risks related to water are indispensable basic elements to ensure a good water management. This function of knowledge and monitoring is developed in several stages which are the collection of basic data, their validation and stocking, their processing and the diffusion of information obtained on water situations.

⁵ As defined by Central Water Commission, Government of India

⁶ Comment: On-Farm Development (OFD) is considered as one of the strategies to improve water use efficiency and to reduce adverse effects on soil fertility. But its successful implementation at the field level seems to be far from satisfactory. Therefore, it is necessary to address both technical and institutional factors influencing the adoption or otherwise of OFD by the farmers and its impact on productivity of crops

viii. Capacity Building of Water Institutions and Agricultural Staff

22. Building capacity at all levels is vital at different levels within government institutions, as also within local communities, in order to improve the provision and maintenance of service delivery. These capacities do not only relate (as is commonly understood) to technical issues, but more importantly to a range of social, managerial and institutional issues from organizing effective community participation, building solidarity, vision and a sense of purpose within communities, and starting and running efficient community-based organizations' to dealing with government procedures and legal requirements and conflict resolution within the Community Based Organisations-all in conformity with the IWRM principles.

I. Public Consultation and Participation during Project Implementation

23. The P&C process is aimed at informing the stakeholders of the project plans and activities in an ongoing manner. Consultations & communication will be continuous agenda and an integral part of the project. To ensure the same the Program Communication Plan⁷ will be an instrument to carry forward the P&C process.

24. The Program Communications Plan covers a range of communication needs of the project. First, it delineates the stakeholder consultation process to be adopted by project management to facilitate creating awareness and participation for effective implementation and safeguard. Second, it addresses the need to introduce and explain the project to a wide range of audiences who can influence and affect an enabling environment for the project. Finally, it also informs the general public in the State and outside about the IWRM based water management actions being initiated by the government to elicit their support, knowledge and participation.

25. Consequently, the Communication Plan has two components:

- i. Communication about project actions to stakeholders and other audience implemented by the PMU and effective during the lifetime of the project
 - a. Under the Right to Information Act
 - b. Under the Project Management System
- ii. Information and knowledge communication related to IWRM based water management actions in the State that will be implemented by the ACIWRM that transcends the life of the project.

26. The range of communication medium and tools that could be used are:

Written	Verbal	Visual
1. Education/Training Material	1. Folk Theatre/Folk Art Form	1. Film/Electronic Media Campaign
2. Pamphlet/Brochure/ Newsletter/Presentation	2. Public Meeting/Campaign/ Public Briefing	2. Display/Poster
3. Website/Blog/Group Email/Social Network Sites	3. Groups Discussion/ Stakeholder Interview/ Personal Contact	3. Exhibition/Fair
4. Study Report/Case Study/ Working Paper/Technical Note	4. Workshop/Seminar/ Consultation Meeting	
5. Press Release/	5. Media Briefing	

⁷ For more details, please refer a separate report on Communication work Plan

Written	Verbal	Visual
Advertisement		
6. Site Announcement Sign	6. Site Visit/Exposure Visit	
7. Letter/Email/Memo/Notice		
8. Policy/Legislation/ Government Order		
9. Agreement/Contract/ Memoranda of Understanding		
Action Taken Report		

J. 9. Reporting Arrangement

27. Consultation and participation process will be well documented for each and every consultations and consultation methods like workshops, FGDs, open meetings, and any other procedures) including records of participation, disclosure issues, concerns raised by the participants, options provided, and opinions referred to by the stakeholders. The records and recommendations will be reported in the periodical monitoring reports, mid-term reports and all evaluation reports.

CONTRIBUTION TO THE ADB RESULTS FRAMEWORK

Level 2 Results Framework Indicators			
No.	(Outputs and Outcomes)	Targets	Methods / Comments
1	Land improved through irrigation, drainage, and/or flood management (hectares)	11,150	Rural land area improved in the Vijayanagara Channels irrigation subproject through any new or improved: (i) irrigation and (ii) drainage services.

Source: Asian Development Bank.

ECONOMIC AND FINANCIAL ANALYSIS

A. Introduction

1. The Karnataka Integrated and Sustainable Water Resources Management Investment Program aims to manage and sustain selected water scarce river basins facing acute scarcity of water resources in the State of Karnataka in India. The investment program proposes a multipronged approach for integrated water resources management and water use efficiency aiming at quality irrigation service delivery systems through modernized irrigation infrastructure on Gondi and Vijayanagara medium subprojects and Tungabhadra Left Bank Canal sub project.

2. The Vijayanagara Channels (VNC) originally are run-of-the river irrigation canals spread across four talukas of three districts, namely, Bellary, Raichur and Koppal. Of the sixteen channels commencing from off-takes along the Tungabhadra river, ten are on the right-side and the river and six are on the left-side of the river. The cultivable command area (CCA) of these channels and the length of the channels vary, which are characterized by an uneven and inefficient distribution of water among the stakeholders and sub optimal water-use efficiency. Farmers in the upper and middle reaches get water for two seasons (kharif and rabi) whereas the tail end farmers received minimal water even during the kharif season. The beneficiaries of this project are mostly the small and marginal farmers along the 16 channels growing paddy, sugarcane and banana.

3. The analysis of the VNC Project (the project) was undertaken in accordance with ADB's guidelines for economic¹ and financial analysis². The main objectives of the study are to: (i) estimate the economic costs and benefits of carrying out rehabilitation works on each of the canals; (ii) assess the economic viability of the proposed interventions; (iii) estimate annual operation and maintenance costs; and (iv) to determine whether the proposed program will be economically viable and financially sustainable in the future.

B. Rationale

4. Justification for the project was provided by the Detailed Project Report (DPR, 2017) on the basis of: (i) a reduction in VNC irrigation water consumption from 12.5 thousand million cubic feet (TMC) to 5.80 TMC, a saving of 6.7 TMC; and (ii) to improve water delivery service to allow farmers to increase cropping intensities. During the focus group discussions conducted in the project area, farmers revealed that there is an unequal distribution of water along the length of the channels, especially at the tail ends. The tail-end farmers do not get their due share of water for their farmlands. Further, agricultural practices are still traditional with flood irrigation for paddy, the major crop.

5. The rationale for public intervention is that the private sector is unwilling to invest in irrigation infrastructure rehabilitation owing to the public good nature of irrigation water in India. Through a public-sector investment approach, the project will not only improve agricultural productivity but also generate resource use efficiencies.

6. The project will introduce river basin planning and volumetric monitoring of water flows. This will contribute not only to a better understanding of the resource availability, efficient utilization and improving agriculture water productivity but also result in efficient resource management, thus increasing net agriculture benefits per unit of water.

¹ ADB. 2017. *Guidelines for the Economic Analysis of Projects*. Manila.

² ADB. 2005. *Financial Management and Analysis of Projects*. Manila.

7. The project will strengthen water user cooperative societies (WUCS) through a capacity building program. Flow measuring devices will be introduced to monitor the flow regime at various hierarchies to ensure equitable distribution of water towards sustainable irrigation water management.

8. The specific objectives of the project include rehabilitation of anicuts, lining of channels, rehabilitation of structures on main canals, distributaries and field irrigation channels along with structures. The WUCS will be awarded directly the lining work and construction of structures on the field irrigation channels. The 16 main irrigation canals of the VNC will be rehabilitated in order to realize the maximum potential of CCA.

9. A GIS-based asset management and operation and maintenance plan will be developed to assist WUCS to monitor, manage and sustain their assets in good condition with timely and regular maintenance works with oversight by WUCS and WUCS federations.

10. The project will install measuring devices in main canals, branch canals and minors to improve operations and to monitor the distribution and water use efficiency. The project also proposes to strengthen the WUCS technically and financially by capacity building.

11. Thus, the economic rationale of the project is based upon the economic benefits from the improved water use and conveyance efficiency, increase in irrigated area, increase in equity and reliability of water supply combined with improved agronomic and irrigation practices at field level. As a consequence, the project seeks to create welfare, and eliminate poverty by enhancing sustainable income levels of the farmers with focus on marginal and small farmers.

C. Demand Analysis

12. Agriculture is the major occupation for a majority of the rural population in Karnataka. As per Census 2011, agriculture supports 13.74 million workers, of which 23.6% are cultivators and 25.7% agricultural workers. A total of 123,100 km² of land is cultivated in Karnataka constituting 64.6% of the total geographical area of the state. The agricultural sector of Karnataka is characterized by vast steppes of drought prone region and sporadic patches of irrigated area. Thus, a large portion of agricultural land in the state is exposed to the vagaries of monsoon with severe agro-climatic and resource constraints. Agriculture employs more than 60% of Karnataka's workforce. Agriculture in Karnataka is heavily dependent on the south-west monsoon. While only 26.5% of the sown area (30,900 km²) is under irrigation, 64.6% of the total geographical area is under cultivation.

13. Karnataka has 41.86 million hectares of irrigated area accounting for 34% of gross command area. Ground water is equally in demand for irrigation, industry and households. At present, gross annual supply of irrigation water does not meet the gross annual water requirements. As per the DPR, the project shall save water of around 6.7 TMC that has been reallocated for using in a new upstream irrigation system. The projected water requirement for the modernized VNC is in the range of 5.8 TMC.

14. The project will also be of importance to the country's major demand for agriculture production. Effective and sustainable development and management of the irrigation sector is vital to Karnataka's economic and social development and has a significant bearing on the economic sustenance and sustainable livelihoods of a large sector of population owning small and marginal farms and the potential for further development.

D. Alternative Analysis

15. Alternative approach comprises a without project scenario, that is, without carrying out the rehabilitation of anicuts and canals, crop diversification or introduction of new technology, or institutional strengthening of WUCS, a key component for sustainability. Further, it cannot bring any change in the cropping intensity, irrigation intensity, productivity of crops, water use optimization and farmer-management of structures and water distribution as implementation of new technology becomes impossible in the current poor condition of the 16 canals, and absence of flow control. Such a scenario brings in minimal change towards betterment in the livelihood of small and marginal farmers. Other problems encountered in the VNC with respect to operation and maintenance are: (i) existing canals have lost their section, thereby having difficulty in ensuring adequacy of flow; (ii) absence of measuring devices; (iii) damaged inlets and outlets; and (iv) unlined canals experiencing heavy seepage and weed growth (water hyacinth).

16. The project design aims to address all the problems mentioned above with many more intangible benefits accruing. The viable alternative is the planned rehabilitation of anicuts and canals as part of the investment program.

E. Cost-Benefit Analysis

1. Methodology

17. The study used a combination of economic surplus analysis and quantitative risk analysis to undertake the cost-benefit analysis (CBA) of the project. Economic surplus analysis was used to measure the agricultural benefits of the project through shifts in the supply of commodities (paddy, sugarcane, maize, sorghum, groundnuts, banana and cotton). This resulted in estimation of changes to producer surplus and consumer surplus within the project region and the rest of India. Given that a welfare economics approach to estimating project benefits, both non-incremental and incremental benefits are endogenously calculated and included in the producer surplus and consumer surplus results, and hence cannot be reported separately.

18. The quantitative risk model was adopted to measure the effect of uncertainty in key inputs upon the project results. The following inputs were defined as random variables and specified in terms of triangular probability distributions (represented by minimum, median, maximum values); (i) the gain in irrigation intensity; (ii) gain in crop yields; (iii) capital cost; and (iv) operation and maintenance (O&M) costs.

19. A Monte Carlo simulation model was used to measure the changes in economic surplus and measure the key economic results from the CBA: (i) economic internal rate of return (EIRR); and (ii) economic net present value (ENPV). The simulation model was solved for 5,000 iterations and the resulting EIRR and ENPV were reported in terms of summary statistics; mean, standard deviation, and percentiles. Given that a quantitative risk model is adopted to measure project benefits, and key project variability is captured within the input probability distributions, no further sensitivity analysis is undertaken.

2. Assumptions and data

20. The project evaluation period is 30 years, which includes a 3-year development period during which no project benefits occur. The discount rate was 9% in accordance with ADB guidelines. Investment costs were based on the latest cost estimates (June 2018). Prices were reported in terms of world price numeraire, with an exchange rate of ₹68.04=\$1.

21. To determine the economic values, financial prices were adjusted for transfer payments and price distortions in traded items and non-traded items that distort the real value to the society. Transfer payments include payments between the people and the government.

22. To adjust for price distortions, the economic value of crops produced is its export or import parity price (parity is the equivalence in value expressed in the currency of another country). These values are derived by adjusting the cost, insurance and freight price or the free on-board price, converted to economic values. Border prices were calculated for tradeable commodities paddy, sorghum, maize, cotton, banana, and traded inputs of urea, phosphorus, steel and cement. Other values were adjusted through application of standard conversion factor of 0.90, a shadow wage rate factor of 0.90, and a conversion factor of 0.82 for civil works.

3. Benefits

23. The quantified project benefits relate to improved crop production from: (i) increased yields in the head, middle and tail regions, and (ii) changes in cropped area and cropping intensity. Farmer surveys in each main canal irrigation system and published state statistics were used to estimate cropping patterns, crop area, yields, and crop inputs. Crop budgets were then developed for each main canal irrigation system for both with-project and without-project scenarios. For VNC, the aggregate changes in crop areas and cropping intensity and the weighted average change in crop yields are given in Table 1. Crop yields were obtained via survey and the with-project gains were due to response from the increased number of irrigations following system rehabilitation.

24. The project will also result in water savings from the project area which has formally been reallocated to the Upper Bhadra Project, which is a new major irrigation project that will irrigate 225,000 ha in an adjacent dryland area. Of the total 29.9 TMC allocated to the Upper Bhadra Project, 6.7 TMC will be provided from the project's water savings. The incremental production benefits from the water reallocation is treated as a non-quantified benefit.

Table 1: Crop area, yield and crop intensity for with-project and without-project scenarios

Crop	Without-project		With-project	
	Area (ha)	Yield (kg/ha)	Area (ha)	Yield (kg/ha)
Total area irrigated	11,154		11,154	
Paddy (kharif)	4,136	6,123	5,093	6,881
Sorghum (kharif)	145	1,200	130	1,290
Maize (kharif)	34	3,800	117	4,940
Cotton (kharif)	32	1,470	55	1,600
Paddy (rabi)	1,583	6,107	4,515	6,801
Groundnut (rabi)	849	7,104	664	8,086
Sugarcane	3,635	89,746	3,300	98,409
Banana	2,013	18,124	2,353	20,074
Crop area	12,427	13,593	16,227	
Cropping intensity (%)	111		145	

Source: Consultant estimates.

4. Costs

25. The project investment cost is estimated at ₹7,729 million including taxes and duties to be financed by the Government and financing charges during implementation. The total project cost in economic terms is ₹5,391 million (Table 2). Operation and maintenance cost requirements were calculated for each irrigation system based upon engineer's estimates. The annual O&M cost was calculated as ₹15.2 million per annum.

Table 2: Project costs (₹ million)

Category	CF	Financial	Economic
Civil works	0.82	4,636	3,793
Environmental management	0.90	116	104
Water user cooperative societies	0.90	1,264	1,138
Vehicles	0.90	55	50
On-farm extension works	0.90	341	307
Total		6,412	5,391

CF = conversion factor

Source: Asian Development Bank estimates.

5. Probability distributions

26. Triangular probability distribution parameters specified for each of the project's random variables are given in Table 3.

Table 3: Triangular probability distribution parameters for with-project scenario

Variable		Minimum	Median	Maximum
Gain in cropping intensity	%	30	34	38
Water savings area factor	%	20	50	60
Change in crop yield	scalar	0.9	1.0	1.1
Capital cost	scalar	0.9	1.0	1.2
Operation and maintenance cost	scalar	0.9	1.0	1.2

Source: Asian Development Bank estimates.

6. Results

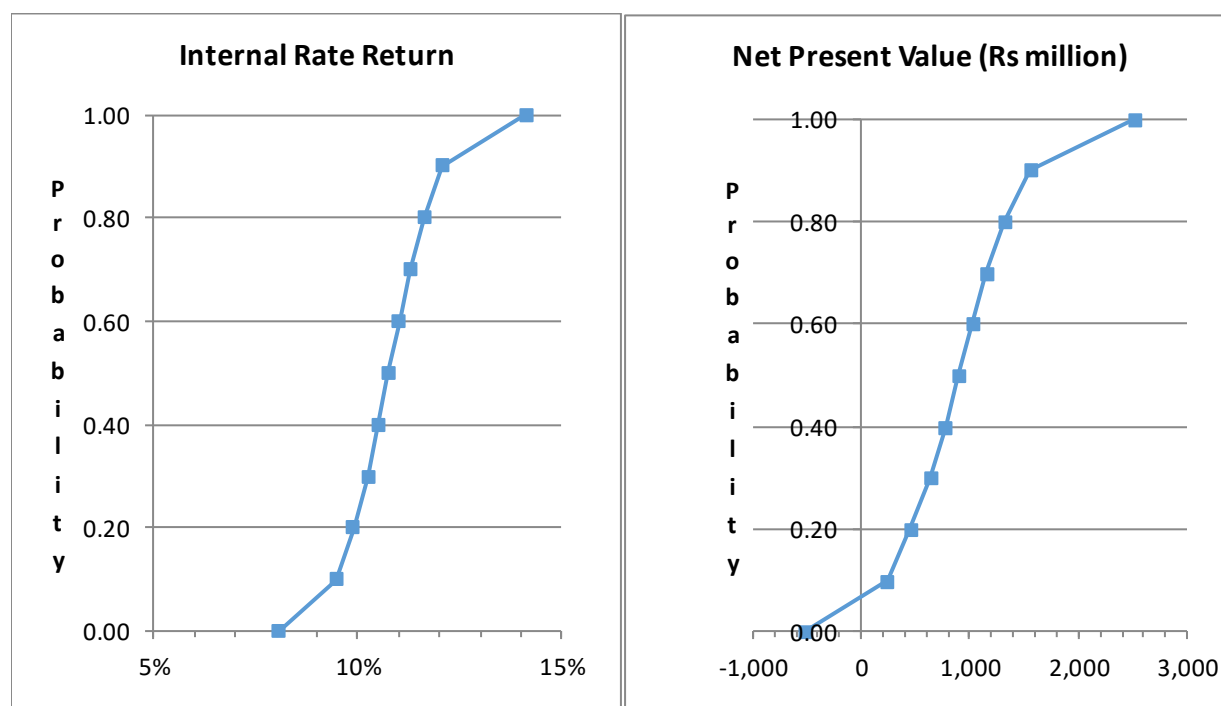
27. The project impacts were segregated into the gains and losses to producers and consumers of agricultural commodities in (i) the project area and (ii) the rest of India. Consumers within the region and rest of India will be expected to gain due to the increased production and resulting lower market prices. Producers in the region will gain due to their increased production, however producers in the rest of India will lose (marginally) due to the lower market prices. Simulation of the risk model derived statistical mean, standard deviation and percentiles of EIRR and ENPV (Table 4). The mean EIRR of 10.8% and mean NPV of ₹897 million indicates the project is economic. The simulated EIRRs ranged from 8.8% (5th percentile) to 13.1% (95th percentile). The simulation model indicated there was less than a 10% probability that the project would not achieve ADB's economic threshold target of 9% (Figure 1).

Table 4: Mean Annual Cash Flows (₹ million)

Year	Benefits				Costs		Net Benefits
	Consumers (region)	Producers (region)	Consumers (ROI)	Producers (ROI)	Capital	O&M	
1	0	0	0	0	1,617.3	15.2	-1,632.5
2	0	0	0	0	2,156.4	15.2	-2,171.6
3	0	0	0	0	1,617.3	15.2	-1,632.5
4	0.1	158.1	165.3	-164.9	0	15.2	143.4
5	0.3	412.9	475.3	-474.9	0	15.2	397.5
6	0.5	624.7	754.8	-754.1	0	15.2	610.7
7	0.7	838.7	1047.8	-1046.9	0	15.2	825.1
8	0.7	862.4	1073.1	-1072.2	0	15.2	848.8
9-30	0.7	867.5	1078.7	-1077.8	0	15.2	853.9
EIRR (%)							10.8
ENPV							897

EIRR=economic internal rate of return; ENPV=economic net present value; O&M= operation and maintenance; ROI=rest of India.

Source: Asian Development Bank estimates.

Figure 1: Cumulative probability distribution of EIRR and ENPV

Source: Asian Development Bank estimates.

F. Distribution and Poverty Analysis

28. The project beneficiary and stakeholder groups are identified into the following groups, i.e., marginal, small, semi-medium, medium and large farmer groups. The analysis seeks to identify the extent to which gain from benefits or bear costs associated with a project. The present study finds that the project benefits the marginal and small farmers groups. There are about 224,241 farming households in Siruguppa, Gangavathi, Koppal and Manvi talukas of Bellary,

Koppal and Raichur districts and out of which 70% are small and marginal farmers, which according to agriculture census data fall under the “poor” category.

29. It can be configured that the benefits enjoyed by poor will have a share of 70% of the total benefits accruing from the implementation of the project apart from the positive social impacts on the poor. An attempt has been made to find the poverty impact ratio (PIR) worked out by measuring the ratio between ENPV poor to ENPV total. The ENPV total was considered as a base and the ENPV poor was 70% of ENPV total as these households are small and marginal farmers who account for 70% of the total farmers in the project region holding 50% of the cultivated land. The poverty impact ratio was calculated to be 0.7.

G. Financial Sustainability

30. Rehabilitation of the channels and the structures on main canals, distributaries, and the FICs, erosion control and flood risk management services lead to improvement in the economic condition of the small and marginal farmers. This is achieved by way of improved supply of irrigation water through a regulated mechanism and ensuring sustainable availability of water resources through participatory management of water distribution led by WUCS. However, this approach does not generate revenue. Therefore, an analysis to determine: (i) incremental project net cash flows; and (ii) financial viability indicators such as the financial internal rate of return and financial net present value was not undertaken.

31. In order to ensure sustainability of O&M of the main irrigation canal systems, the project will put in place specific measures, namely, (i) state-wide measures and (ii) VNC project specific measures. At the state level Karnataka Neeravari Nigam Ltd. (KNNL), through the individual command area development authorities, will: (i) establish a scheme-wise management information system for O&M monitoring and planning; (ii) establish a mechanism of monitoring WUCS performance of water tax collection; (iii) put into operation participatory decision-making for fund allocation and utilization of O&M in each main irrigation canal system, and (d) introduce needs- and performance-based fund allocation system (for grant-in-aid followed by maintenance fund) by establishing direct linkage between the water tax collected and funds allocated. On implementation of the above systems, the VNC main irrigation canal systems are expected to function as per design standards of service delivery leading to its financial sustainability.

32. The annual O&M cost estimate for the project is compared to the a 5-year budget allocation of the KNNL irrigation system in Table 5. This indicates that the budget for O&M has mostly been increasing, and that the incremental O&M requirements from the project are very small in terms of current and future budget allocations. Hence, it is concluded that the government has sufficient financial capacity to maintain the project from existing resources, however the financial sustainably risk is considered *substantial* to raise awareness that O&M expenditure remains important.

Table 5: Project incremental O&M requirements against KNNL O&M budget

Year	Project O&M (₹ million)	KNNL O&M budget (₹ million)	Proportion of budget (%)
2020 -	15.2		
2012-13		924.8	1.64
2013-14		883.7	1.72
2014-15		1,743.3	0.87
2015-16		1,981.0	0.77
2016-17		2,041.2	0.74

Source: Karnataka Neeravari Nigam Ltd.

SUMMARY POVERTY REDUCTION AND SOCIAL STRATEGY

Country:	India	Project Title:	Karnataka Integrated and Sustainable Water Resources Investment Program – Tranche 2
Lending/Financing Modality:	Multitranche Financing Facility	Department/ Division:	South Asia Department Environment, Natural Resources and Agriculture Division

I. POVERTY AND SOCIAL ANALYSIS AND STRATEGY

Poverty targeting: General Intervention

A. Links to the National Poverty Reduction and Inclusive Growth Strategy and Country Partnership Strategy

The Karnataka Integrated and Sustainable Water Resources Management Investment Program (KISWRMIP) aims to improve water availability in selected river basins by implementing integrated water resources management. KISWRMIP will improve water use efficiency in irrigated agriculture and provide economic opportunities to improve rural incomes. The state and national government recognizes the need to reduce poverty as its primary goal and aims to reinvigorate the rural economy, strengthen agriculture and natural resources sectors, create employment, and bridge the divide between regions, sectors, and genders. ADB's overarching goal of poverty reduction (Strategy 2020) supports Indian government's priority to pursue rural poverty reduction through strengthening water resources management and irrigated agriculture. The government's 15-year vision document aims to achieve inclusive development and poverty reduction. The vision document comes into effect during 2017–2018 along with a seven-year National Development Agenda which will lay down the schemes, programs and strategies to achieve the vision. The poverty reduction and social strategy of the program is aligned with the ADB's Poverty and Social Analysis strategy (2012)¹ and ADB 2030 Strategy.²

B. Results from Poverty and Social Analysis during PPTA or Due Diligence

1. Key poverty and social issues. As per Census 2011, Karnataka State has 61.3% of its population in rural areas; thus, the majority of poor people reside in the rural areas. Population Census 2011 results and poverty head count ratio (2011–2012) point out that, 9.3 million of the total poor are from rural and 3.7 million from urban Karnataka.³ Karnataka accounts for 23.6 per cent population, with a rural and urban poverty figures of 26.1 and 19.6 per cent respectively. The overall poverty rate in the state is lower than the all-India average of 29.8 percent.

There has been a significant decline in Poverty Head Count Ratio in both rural and urban areas in Karnataka during the last decade.⁴ The percentage of population below poverty line has already declined by more than half of its position in 1993–1994, in 2011–2012 itself, at the State level as well as in rural and urban areas. Despite the constitutional privileges given to the minority groups, a large section continues to remain in severe poverty. Group dominance plays an important role in participation in growth and distribution and improving the standard of living. It has been seen that the benefits of economic growth have always taken time to reach the marginalized groups.

2. Beneficiaries. The population of the project area is scattered over the rural and peri-urban belt and the majority belongs to the rural areas. The rural and traditional character of the population has been the hallmark of villages coming under the project. Sex ratio is highest in Manvi taluk (1013) followed by Hospet (1006), Siruguppa (1005), Gangavathi (1003) and Koppal taluk (979).

In Koppal and Gangavathi, out of total workers, 24.1% are cultivators, 42.8% agricultural labor, 29.8% other workers; 3.2% are involved in household level activities.⁵ It is worth noting that women comprise 21% of cultivators and 60% agricultural laborers whereas men comprise 79% cultivators and 40% agricultural laborers. Special focus will be given to tail-end farmers, marginal, small, and landless farmers, and the poor and vulnerable women.

3. Impact channels. Project interventions will facilitate impact channels: (a) strengthened grass root level institutions and enhanced participation in governance and decision making processes by women, vulnerable groups and tail end farmers; (b) improved irrigation management practices on water productivity; (c) improved and equitable access to water by small, marginal and landless farmers; (d) enhanced off farm livelihood opportunities; (e) innovative and sustainable irrigation and agriculture practices, including drip and micro irrigation and crop diversification;

¹ ADB. 2012. *Handbook on Poverty and Social Analysis*. Philippines.

² ADB. 2018. *Strategy 2030: Achieving a Prosperous, Inclusive, Resilient, and Sustainable Asia and the Pacific*. Philippines

³ Govt. of India. 2011. *Population Census Report*. Govt. of India

⁴ Govt. of India. 2014. *Million Development Goals, Report of Karnataka*. Government of India

⁵ Govt. of India. 2011. *Population Census Report*. Government of India

(f) enhancing agricultural productivity, and (g) strengthening agricultural infrastructure, research and extension. All these are expected to lead to increase in agricultural productivity.

4. Other social and poverty issues. Along with water user cooperative societies (WUCS), the civil society organizations (CSOs) and women groups, allied departments will associate with the implementation of improved irrigation management and livelihood activities. Government of Karnataka has launched several programs and schemes for poverty alleviation and women empowerment programs over the period.

5. Design features. Pro-poor and gender inclusive components are incorporated within the overall program design framework as explained under point 3. The Project Implementation Office (PIO) at Irrigation Central Zone Munirabad supported by the program support consultant (PSC) and other stakeholders will be responsible for addressing these components as part of implementation.

Design and estimates of command area development (CAD) field irrigation channels will be prepared in consultation with WUCS, farmers associations and other grass root level institutions.

II. PARTICIPATION AND EMPOWERING THE POOR

1. Participatory approaches and project activities

Grassroots level institutions or community-based organizations are intended to establish based on local requirement. In most of the water resources projects, the concerned departments have usually determined the design, technology, service level and mode of scheme operation, typically based on the government rules/norms and procedures. For sustainability of systems and services, it is important that grassroot level institutions such as WUCS and farmers associations will be involved in the identification (and design) of solutions and play a key role in monitoring, cost recovery, and operation and maintenance of facilities constructed.

Poor, landless and other vulnerable groups participated and expressed their opinions in these events. Overall the participants showed great interest and confidence about the project as they are well aware of the benefits of this project that accrue post completion, and its effect on enhancing the living standards of farmers. It is important to note that around 50% of the participants who attended the meeting and the subsequent discussions belonged to the poor and vulnerable groups.

2. If civil society has a specific role in the project, summarize the actions taken to ensure their participation.

Along with the WUCS, the CSOs and women groups, allied departments are proposed to be associated with the implementation of improved irrigation management and livelihood activities. The Karnataka Neeravari Nigam Limited (KNNL, the executing agency) is encouraging the use of such institutions in implementation of CAD packages and operation and management of such facilities. The role of CSOs and community-based organizations are being listed in the terms of reference which will be published on KNNL website, newspapers and relevant media.

3. Civil society organizations. KNNL shall put in places support service teams (SSTs) to cover the VNC command area. The SST will work with the WUCS daily with effective community mobilization as its core objective. Each SST comprises two community organizers (one male and one female), one participatory on-farm water management agent, and an agricultural extension agent per 10 WUCS, to promote on farm trainings, women's participation, agricultural extension and related activities. The PSC experts will also be constantly carrying out community mobilization to ensure as many members from each WUCS jurisdiction as possible will be participating in the project. Similarly, networks will be established with existing civil society organizations, self-help groups and community-based organizations in the subproject areas.

1. The following forms of civil society organization participation are envisaged during project implementation, rated as high (H), medium (M), low (L), or not applicable (NA):

☒ Information gathering and sharing ☒ Consultation ☒ Collaboration ☒ Partnership

2. Participation plan.

☒ Yes. ☐ No. Participation is an integral part of the program and this will be considered as a priority while carrying out social mapping and assessment of the specific requirement and subsequent interventions. Civil society involvement in the project is included in the Stakeholder Communications Strategy.

III. GENDER AND DEVELOPMENT

Gender mainstreaming category: EGM

B. Key actions.

☒ Gender action plan ☐ Other actions or measures ☐ No action or measure

Key issues. Agriculture is the main livelihood activity of men and women in the project area. Nonetheless, women have limited access to productive resources (land, credit, inputs, transport, extension services, storage and technical assistance) which prevents them from adopting new technologies or encoring them economies of scale. Women farmers perceive that their opportunities in agriculture are reducing due to mechanization of the sector. The challenge

is further exacerbated by women's triple burden and 'second class status'.⁶ Women in the project area are frequently excluded from agricultural governance bodies and continue to be paid less than their male counterparts for the same activities. In the project area women's daily wages vary from INR 150 to 200 yet men receive between INR 250 to 350 for the same agricultural work. While Tranche 1 GAP activities are ongoing, monitoring data highlights that the project is on track to meet gender quotas for training and representation in key decision-making bodies. Furthermore, project interventions have increased women's income and livelihood standards through improved agricultural opportunities including crop diversification. Tranche 2 will continue and expand the scope of activities.

Key actions. The GAP envisages strengthening women's participation in grassroots governance and leadership with a planned 30 percent women's representation in the executive board. It will also focus on expanding women's knowledge of improved crop diversification and management, on-farm water management, operation and maintenance and systems management. The GAP aims to enable women to gain easier access to water for agricultural, domestic use, and health and hygiene purposes. Furthermore, the GAP will increase gender awareness among Water User Association members and staff in the Executing Agency.

IV. ADDRESSING SOCIAL SAFEGUARD ISSUES

A. Involuntary Resettlement

Safeguard Category: ☐ A ☐ B ☒ C ☐ FI

1. Key impacts. There will be no involuntary land acquisition or displacement as the project will be constructed within the government's preexisting right of way. Contractors can access the canal system from multiple vantage points and will avoid involuntary displacement to any non-title holders.

2. Strategy to address the impacts. A social due diligence report was compiled outlining the construction methodology that contractors will use to avoid involuntary resettlement.

3. Plan or other Actions.

- | | |
|---|--|
| ☐ Resettlement plan | ☐ Combined resettlement and indigenous peoples plan |
| ☒ Resettlement framework | ☐ Combined resettlement framework and indigenous peoples planning framework |
| ☐ Environmental and social management system arrangement | ☐ Social impact matrix |
| ☐ No action | |

B. Indigenous Peoples

Safeguard Category: ☐ A ☐ B ☒ C ☐ FI

1. Key impacts. The SPS 2009 Indigenous Peoples safeguard is not triggered despite that the project screening confirms the presence of different ethnic groups in the project footprint. The project implementation focuses exclusively on pre-existing governance and water resource infrastructure. The project will not affect i) the territories or natural or cultural resources of Indigenous Peoples, or ii) the dignity, human rights, livelihood systems, or culture of Indigenous Peoples.

Is broad community support triggered? ☒ Yes ☐ No

2. Strategy to address the impacts. The interest of IP shall be included as part of social development strategy and under no circumstances the STs/SCs will be deprived of the benefits from the program. This will be achieved by dovetailing government schemes for the IP into the project such as those of Departments of Social Justice, Women and Child Development and other allied sectors.

3. Plan or other actions.

- | | |
|--|--|
| ☐ Indigenous peoples plan | ☐ Combined resettlement plan and indigenous peoples plan |
| ☐ Indigenous peoples planning framework | ☐ Combined resettlement framework and indigenous peoples planning framework |
| ☒ Environmental and social management system arrangement | ☐ Indigenous peoples plan elements integrated in project with a summary |
| ☐ Social impact matrix | |
| ☐ No action | |

V. ADDRESSING OTHER SOCIAL RISKS

A. Risks in the Labor Market

1. Relevance of the project for the country's or region's or sector's labor market, indicated as high (H), medium (M), and low or not significant (L).

☒ unemployment ☒ underemployment ☐ retrenchment ☒ core labor standards

The project is expected to address the dual issues of unemployment and underemployment in the agriculture sector by providing alternative opportunities, especially the landless, women and the poor, with skilled farm and off farm job opportunities in subproject areas. The project will also address core labor standards, especially in the context of irrigation infrastructure development (e.g. canal modernization/lining, CAD works).

⁶ FAO, 2015. *Running out of time: The reduction of women's work burden in agricultural production*. Rome.

2. Labor market impact. The project will positively impact the agriculture labor market by providing unskilled and skilled employment opportunities, especially for the landless and vulnerable farming households, focusing on higher-value agricultural and irrigation practices.
B. Affordability Improved irrigation management, agricultural practices and farm and off farm income generating activities will be promoted by the project at no cost to the target populations. Improved market linkages and informed pricing mechanisms will be established to prevent exploitation of target populations. This is expected to positively influence an increased number of socially excluded and marginal populations, including women, to participate in project activities. This is also expected to protect them from becoming vulnerable to volatile market forces and fluctuating pricing mechanisms.
C. Communicable Diseases and Other Social Risks - Not Applicable 1. The impact of the following risks are rated as high (H), medium (M), low (L), or not applicable (NA): ☐ Communicable diseases (NA) ☐ Human trafficking ☐ Others (please specify) _____ 2. Risks to people in project area. None.
VI. MONITORING AND EVALUATION
Are social indicators included in the design and monitoring framework to facilitate monitoring of social development activities and/or social impacts during project implementation? ☒ Yes

GENDER ACTION PLAN FOR PROJECT 2

Activity	Indicators/Targets	Responsibility	Time Schedule
Output 1. State and Basin Institutions Strengthened for IWRM			
Develop the capacities of WRD staff including women in IWRM	At least 600 WRD staff (of which 200 are women staff) are certified in IWRM (2018 baseline: 150 staff are certified of which 40 are women)	WRD/PMU/PIOs	Year 2
Output 2. Irrigation system infrastructure and management modernized			
Build awareness of women's equal rights and participation in water resources management and livelihood improvement	30 sensitization workshops focusing on gender equality and social inclusion (GAP) approaches and women's role and participation in water resources management and farm/off farm livelihood activities conducted for all WUCS in the VNC subproject area (T: 30 workshops conducted with at least 60% men's participation)	PMU/PIOs	Year 2–3
	At least 30% of WUCS management board members are women (T: M– 150; F–50)	PMU/PIOs	Year 1–2
Improve the knowledge of relevant government departments on GAP approaches in agriculture, irrigation and water resources management	5 gender sensitization workshops provided to WRD and KNNL staff on GAP concepts and in the design and implementation of water resources management, irrigation and agriculture programs for vulnerable populations, including women and use of sex-disaggregated data for decision making (T: F–30; M–70)	WRD/PMU/PIOs	Year 1–2
Output 3: Program and management systems operational			
Monitor progress of gender and socially inclusive focused activities in water resources management	Project management information system with gender indicators	PMU/PIOs	Year 1–4
	Sex–disaggregated data collected, analyzed and reported in quarterly progress reports and safeguard monitoring reports	PMU/PIOs	Year 1–4

F = female, GAP = gender action plan, IWRM = integrated water resources management, M = male, NGO = nongovernment organization, PIM = participatory irrigation management, PIO = project implementation office, PIU = project implementation unit, PMU = project management unit, T = target, WRD = Water Resources Department, WUCS = Water Users Cooperative Society

Note: The Gender Action Plan is subject to further revisions post consultations with Asian Development Bank and State Government of Karnataka

RISK ASSESSMENT AND RISK MANAGEMENT PLAN

Risk Description	Rating	Mitigation Measures	Responsibility
Technical			
1. Extreme weather events during limited canal closure periods cause delay in project implementation	M	Contractor to mobilize additional resources and possible extension of contract and loan period.	PIO
Economic and financial			
2. Climate change impact on farm productivity.	M	Establishment and strengthening of WUCS. Upgrading of irrigation infrastructure including farmer irrigation canals. Support of on-farm water management and agrarian extension services.	KNNL, PIO, CADA
3. Poor collection by WUCS of irrigation service fees from farmers leading to poor maintenance and deterioration of farmer irrigation canals.	M	Establishment and strengthening of WUCS. Upgrading of irrigation infrastructure including farmer irrigation canals by the project prior to handover to WUCS. Farmers are willing to pay irrigation service fees once they experience the project's benefit of improved water delivery services.	PIO, CADA, WUCS
4. Financial management	M	Provide additional financial management staff, reduce staff transfers and provide regular training. Address weaknesses as highlighted in audit reports and improve reporting systems to include financial and disbursement reports.	KNNL, ACIWRM
5. Low O&M expenditure impacts long-term sustainability and performance of the irrigation system.	S	Develop an asset management system to assist with annual budget planning and workflow for O&M and integrate into the WUCS's, WRD's and KNNL's annual budget planning.	WRD, KNNL
Governance			
6. Loss of political, community and stakeholder support at the state level to sustain and proceed with reforms.	S	Continue supporting ACIWRM to strengthen IWRM knowledge and practices within Karnataka, including advocacy with key stakeholders.	SGOK
7. Trained WRD staff do not remain in office.	S	Ensure WRD maintain a succession staffing plan and young officers are engaged under the project to strengthen their knowledge about the project and its themes.	WRD, KNNL
8. State government does not operationalize policy for irrigation management transfer and WUCS collection of irrigation service fees.	S	Establishment and strengthening of WUCS to operate and maintain the tertiary irrigation systems in financial sustainable manner, including advocacy for payment of irrigation service fees. Modernize the irrigation systems to ensure the infrastructure is capable of quality water delivery services. Farmers are willing to pay if water delivery services meet their expectations.	WRD, KNNL, CADA
9. Delayed procurement due to multiple approval layers and irregular committee meetings and poor contract administration leading to delays in project completion.	M	Advance procurement actions will minimize need for KNNL Board approvals during implementation. One of two works packages have been awarded. ADB to provide regular procurement and contract administration trainings to ACIWRM and KNNL. ACIWRM to recruit procurement specialist with ADB procurement experience.	SGOK, WRD, KNNL, ACIWRM, ADB
10. Poor performance of consultants and low standard of outputs.	M	Administer to PMU and ACIWRM training programs to provide specific guidance on consultant performance evaluation and discuss case studies on consultant performance evaluation. A dedicated person in PMU and ACIWRM to be devoted to evaluation and monitoring of consultant's performance.	KNNL, ACIWRM

H = high, S = substantial, M = moderate, L = low.

ACIWRM = Advanced Center for Integrated Water Resources Management, ADB = Asian Development Bank, CADA = Command Area Development Authorities, IWRM = integrated water resources management, KNNL = Karnataka Neeravari Nigam Limited, O&M = operation and maintenance, PIO = project implementation office, SGOK = State Government of Karnataka, WRD = Water Resources Department, WUCS = water user cooperative societies

Source: Asian Development Bank.

Safeguards Reports

1. Environmental Assessment and Review Framework for the Karnataka Integrated and Sustainable Water Resources Management Investment Program.
Updated and disclosed on ADB's website in June 2019
<https://www.adb.org/sites/default/files/project-documents/43253/43253-013-earf-en.pdf>
2. Environmental Impact Assessment for Karnataka Integrated and Sustainable Water Resources Management Investment Program.
Updated and disclosed on ADB's website in June 2019.
<https://www.adb.org/projects/documents/ind-43253-026-eia-0>
3. Resettlement Framework for Karnataka Integrated and Sustainable Water Resources Management Investment Program.
Updated and disclosed on ADB's website in October 2019
<https://www.adb.org/projects/documents/ind-43253-026-rf>
4. Social Safeguards Due Diligence Report for Karnataka Integrated and Sustainable Water Resources Management Investment Program.
Updated and disclosed on ADB's website in October 2019.
<https://www.adb.org/projects/documents/ind-43253-026-sddr-0>